

Executable Residual-Return Alphas in Corporate Bondslarge Technical Specification of 20 Signals

Research specification

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Abstract

Implementation-level specifications for twenty corporate-bond microstructure and relative-value signals targeting five- and ten-trading-day residualised returns. The production object is executable net return conditional on fill, with point-in-time data, state classification, transaction costs and abstention. No signal is claimed to guarantee transaction-cost coverage.

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1 Common target and notation

Let i index bonds, $g(i)$ issuers, t decision timestamps and $H \in \{5, 10\}$ trading days. Let $P_{i,t}^*$ be a causal efficient dirty (or consistently total-return) price. Define the diagnostic residual return

$$y_{i,t,H}^{mid} = R_{i,t:H}^* - \widehat{\beta}_{i,t}^\top \Delta F_{t:t+H} - CarryRoll_{i,t,H}^{model},$$

where R^* includes coupon/accrual exactly once under the selected return convention. The production target is the signed executable package return

$$y_{i,t,H}^{exec} = R_{i,t,H}^{bond,exec} - R_{i,t,H}^{hedge,exec} - C_{i,t,H}^{fees} - C_{i,t,H}^{fin} - C_{i,t,H}^{impact} - C_{i,t,H}^{borrow}.$$

All terms use the same return denominator. Define price confidence $c \in [0, 1]$, information probability p^{info} , pressure-completion probability $p^{complete}$ and fill probability p^{fill} . Every changed hedge leg is costed. Historical fields unavailable at decision time are forbidden. Use chronological purged validation, held-out issuers, point-in-time terms/ratings/index membership, realistic quote latency and a daily position ledger. Unless stated otherwise, s denotes OAS or another consistently defined spread in basis points; positive spread residual means cheap in spread space, while positive price residual means rich in price space. Convert spread forecasts to price returns with the local spread-duration sign, $\Delta P/P \approx -SD \Delta s$, before combining with price-space signals.

2 Signal 01: Leave-one-out issuer-curve cheapness

2.1 Mathematics and feature construction

The canonical raw score is

$$z_{i,t}^{iss} = \frac{s_{i,t}^{fresh} - \hat{s}_{g,t}^{(-i)}(T_i) - \gamma^\top q_i}{\hat{\sigma}_{i,t} + \epsilon}. \quad (1)$$

Retain level, one- and three-day change, persistence, acceleration, trailing percentile and robust z-score. Add interactions with $c_{i,t}$, $p_{i,t}^{info}$, $p_{i,t}^{complete}$, liquidity regime, volatility and requested size. Thresholds and normalisation are learned on training data only. Missingness is explicit.

2.2 Point-in-time implementation

Maintain an immutable as-of ledger with economic, publication and receipt timestamps, corrections/cancels, security-master version and entitlements. At each decision time freeze the universe; resolve terms and corporate actions; construct fresh prices; compute peers/factors from earlier data; calculate the feature; obtain or simulate a fresh executable entry; and persist the feature snapshot. A confidence-weighted price can use

$$w_q = e^{-age_q/\tau_a} \min(1, size_q/v_0) w_d e^{-dispersion_q/\tau_\sigma}, \quad P^* = \frac{\sum_q w_q P_q}{\sum_q w_q}.$$

Never use future prints through symmetric smoothing.

2.3 Five- and ten-day forecast

Benchmark each horizon with

$$y_{i,t,H}^{exec} = a_H + b_H z + d_H z c + e_H z p^{info} + f_H z p^{complete} + u_{i,t,H}.$$

For dislocations estimate $D_{t+h} = D_t e^{-\kappa h} + \epsilon$ and expected capture $D_t(1 - e^{-\kappa H})$; for information diffusion use a bounded catch-up fraction. This gives an economic mapping from state to the 5d/10d horizon.

2.4 Execution and costs

Estimate jointly

$$p^{fill} = \mathbb{P}(fill \mid X, size, protocol), \quad (2)$$

$$C = g(size, spread, ADV, depth, vol, flow, toxicity, protocol), \quad (3)$$

$$\mu_H = \mathbb{E}[y_H^{exec} \mid fill, X]. \quad (4)$$

Use a fresh executable entry, never the historical print that revealed the signal. Model quote withdrawal and adverse selection. Relative-value packages pay every leg. Capacity is evaluated by rerunning net P&L at increasing requested size and participation constraints.

2.5 Falsification and monitoring

Kill the feature if it disappears with fresh independent prices, relies on unavailable data, is concentrated in a few issuers/events, or loses its edge under realistic fills and costs. Run leave-one-issuer-out tests, IG/HY and liquidity splits, calm/stress regimes, protocol splits and alternative residualisations. Monitor drift, price confidence, fill calibration, realised versus predicted costs, mark-outs and residual factor exposure.

2.6 Signal-specific mechanism and implementation

Data: Trades + fresh quotes + analytics | Target: Base and issuer-neutral | 5 and 10 days gap_i = OAS_i - curve_issuer,-i(tenor_i) - liquidity_adjustment_i Mechanism and evidence. A bond can be cheap relative to economically comparable liabilities of the same obligor without the issuer itself being cheap. The research hypothesis is convergence of the bond-specific gap, not compensation for broad credit risk. Liquidity and OTC-friction evidence motivates controlling persistent differences; no cited paper validates this exact spline signal at these horizons. Feature and implementation. Fit a robust, regularised issuer spread curve using only contemporaneously available peers, excluding the target bond. Weight observations by freshness, executable size and inverse estimated measurement variance. Use a sector/rating/tenor prior when issuer support is thin, but keep the prior uncertainty. As an initial eligibility rule require at least four useful peer observations across maturity, not four stale dealer copies. Control issue size, age, bid-ask cost and contractual differences. Keep legal obligor, guarantee and seniority consistent. Fit spread differences in basis points, retain the raw gap and its standard error, and standardise by a past-only robust residual scale. A positive adjusted gap is a candidate long-price signal. Cap extrapolation beyond the peer maturity range. Five-/ten-day forecast and hedge. Estimate conditional gap dynamics from previously observed, mature outcomes. Convert the expected narrowing into price-return basis points using spread duration; fit separate five- and ten-day coefficients. A ten-day position is not automatically twice a five-day position. Prefer an issuer-neutral label and hedge against liquid sibling bonds; remove remaining Treasury key-rate exposures separately. Condition expected convergence on a recent technical dislocation or improving flow. A static wide spread with no demonstrated short-horizon decay is not sufficient. Executable entry and costs. Compute the gap again at the actually available ask for a purchase and at the hedge bonds' bids, with valid size and timestamps. Admit only after the common all-leg cost and uncertainty gate. Exit at the chosen horizon, or earlier under a pretested convergence/invalidation policy. Use a no-trade band to avoid repeatedly crossing for tiny model changes. Falsification / kill test. Reject gaps explained by a stale target quote, different recovery claim, embedded option, tender, tax treatment or genuine liquidity premium. Refit without the single largest-weight peer. Kill the model if out-of-sample incremental net performance disappears against the near-twin signal or when both endpoints require fresh executable observations. A fair-value model can be precise and still wrong about convergence speed.

2.7 Agent implementation runbook: data discovery to safe production

Operating rule. The agent must advance through auditable gates rather than implement the entire alpha in one pass. If a required field is unavailable, ambiguous, delayed, or not historically point-in-time, stop that branch and record the limitation; never silently substitute a convenient proxy.

- A1. Data discovery.** Inventory candidate sources/tables/files; definitions; keys; units; time zones; economic, publication and receipt timestamps; corrections/cancels; historical coverage; entitlements; delayed-history rules; and whether each field was genuinely observable live. Output schema and coverage diagnostics only.
- A2. Raw profiling.** Report rows, bonds, issuers, days, missingness, duplicates, timestamp inversions, corrections, size distributions and join success by IG/HY, liquidity, sector and period. Do not impute yet.
- A3. Canonical identifiers.** Version the security master and map security IDs to legal obligor, parent, guarantee, seniority, currency, coupon, maturity, option schedule and corporate actions. Quarantine ambiguous/time-varying mappings.
- A4. As-of layer.** Require every query to accept a decision timestamp and return only records then available. Unit-test late records, corrections and cancels. Persist available-at and provenance fields.
- A5. Price layer.** Normalize clean/dirty price, accrued, spread convention, side, size and quote status. Create freshness, executable-size, dispersion and dealer-breadth metadata. Any forward fill must explicitly age and lose confidence.
- A6. Analytics layer.** Produce cash flows, yield/OAS as appropriate, spread duration, convexity, KRD, CS01/DV01, carry and roll from point-in-time terms; reconcile a sample against an independent implementation.
- A7. Targets.** Build 5d/10d efficient-price diagnostic and executable-return labels. Estimate residualisation only on prior data; cost every changed hedge leg, fees, financing and impact. Persist label maturity for purging.
- A8. Signal-specific prerequisites.** Execute the following in order, emitting exclusions, distributions and hand-audited examples after each sub-step:

- a) Discover issuer hierarchy, guarantee/seniority, OAS analytics, terms, trades and executable quotes.
- b) Build same-economic-claim sibling eligibility and point-in-time peer sets.
- c) Construct confidence-weighted fresh target/peer spreads; require adequate maturity support.
- d) Fit leave-one-out issuer curve, add contractual/liquidity adjustments and retain prediction uncertainty.
- e) Recompute cheapness at target executable side and hedge executable sides before admission.
- f) Refit without largest-weight peer and reject extrapolation/stale-price-driven gaps.

- A9. Literal raw feature.** Translate the documented equation exactly. Assert units, sign, finite values, peer counts, timestamp ordering and boundaries. Hand-reconcile at least 20 events.
- A10. Causal transforms.** Add rolling median/MAD z-score, changes, persistence, acceleration and percentile with past-only windows. Fit curves/betas/normalizers only inside training windows and version them.
- A11. State variables.** Join price confidence, information/toxicity, pressure completion, liquidity regime, volatility, event age and requested size as separate columns before tuning interactions.
- A12. Leakage tests.** Append records arriving after decision time and prove historical feature values do not change. Compare live-available with enhanced-history fields and test shifted timestamps. Any failure blocks modelling.
- A13. Measurement tests.** Recompute with fresh quotes only, trades only, independent prices where possible, alternative stale thresholds and stricter confidence. Reject effects caused only by asynchronous updating.
- A14. Transparent baseline.** Fit horizon-specific linear/robust models in walk-forward folds before nonlinear models; record sign, stability, sample size and incremental information.
- A15. State/convergence models.** Estimate information versus temporary-pressure probabilities and convergence speed where meaningful. Cross-fit state probabilities used as features.
- A16. Execution simulator.** Request a new executable entry after the signal timestamp; apply latency, size, withdrawal, partial-fill and legging rules. No executable observation means unfilled, not a midpoint fill.
- A17. Fill/cost models.** Estimate fill probability and conditional spread/impact/fees/financing by size, protocol and state. Calibrate on held-out fills and cost every changed leg.
- A18. Net outcomes.** Store expected executable return conditional on fill, conservative lower quantile and capacity. Keep gross, mark, executable and fully-net outcomes as distinct audit columns.
- A19. Adversarial validation.** Use chronological purged folds, issuer holdouts, liquidity/rating/regime splits, event exclusions, alternative residualisations, placebo dates and within-family ablations.
- A20. Shadow mode.** Freeze versions and generate live predictions without trading. Log source timestamps, features, quotes, hypothetical orders, predicted fills/costs and 1d/5d/10d mark-outs.
- A21. Canary.** Start with strict size, concentration, liquidity and turnover limits. Kill automatically on timestamp anomalies, data gaps, feature drift, cost underprediction, unexplained residual beta or repeated adverse selection.
- A22. Promotion/rollback.** Promote only a fully reproducible version. Default to no-trade when health checks fail and retain one-command rollback to the prior approved model.

Incremental safe prompts. Use these as separate prompts/commits; each prompt may inspect prior artifacts but must not expand scope silently.

- P1.** “Inventory only the required data, schemas, keys, timestamps, live-history availability, entitlements, coverage and unresolved questions. Do not build a feature.”
- P2.** “Build and unit-test canonical identifiers and as-of joins. Return diagnostics and quarantined ambiguities. Do not calculate alpha.”
- P3.** “Build the price/analytics layer with freshness/confidence and reconcile a hand-audited sample. Stop on unexplained discrepancies.”

- P4.** “Implement only the literal raw signal with assertions and 20 hand-reconciled examples. Do not optimize parameters.”
- P5.** “Add past-only transforms and state variables. Run leakage and measurement-artifact tests before fitting any predictive model.”
- P6.** “Fit the transparent walk-forward baseline for 5d/10d. Report coefficients, uncertainty, ablations and failures; do not tune on the final test period.”
- P7.** “Add execution, fill and transaction-cost simulation. Recompute conditional executable net outcomes and mark unavailable opportunities unfilled.”
- P8.** “Run adversarial validation and within-family ablations. Produce a go/no-go research recommendation based only on predeclared gates.”
- P9.** “Run shadow mode with frozen versions. Do not send orders. Compare predicted versus observable fills, costs and mark-outs.”
- P10.** “Only after explicit human approval, create a tightly limited canary configuration with hard risk limits, health checks and rollback; do not broaden limits automatically.”

2.8 Production row and acceptance gate

Persist raw score, z-score, event age, quote age, peer count/effective sample size, confidence, state probabilities, size/ADV, expected spread/impact, issuer/sector/rating/tenor, KRD/CS01 and source timestamps. The feature graduates only if it adds incremental out-of-sample information after other members of its economic family, survives alternative residualisations and retains a positive conservative executable-net-return statistic at realistic capacity.

3 Signal 02: Local issuer-curve kink / butterfly

3.1 Mathematics and feature construction

The canonical raw score is

$$B_{i,t} = s_{i,t} - w_{L,i,t}s_{L,t} - w_{R,i,t}s_{R,t}, \text{quad} w_{L,i,t} + w_{R,i,t} = 1, \quad (5)$$

Retain level, one- and three-day change, persistence, acceleration, trailing percentile and robust z-score. Add interactions with $c_{i,t}$, $p_{i,t}^{info}$, $p_{i,t}^{complete}$, liquidity regime, volatility and requested size. Thresholds and normalisation are learned on training data only. Missingness is explicit.

3.2 Point-in-time implementation

Maintain an immutable as-of ledger with economic, publication and receipt timestamps, corrections/cancels, security-master version and entitlements. At each decision time freeze the universe; resolve terms and corporate actions; construct fresh prices; compute peers/factors from earlier data; calculate the feature; obtain or simulate a fresh executable entry; and persist the feature snapshot. A confidence-weighted price can use

$$w_q = e^{-age_q/\tau_a} \min(1, size_q/v_0) w_d e^{-dispersion_q/\tau_\sigma}, \quad P^* = \frac{\sum_q w_q P_q}{\sum_q w_q}.$$

Never use future prints through symmetric smoothing.

3.3 Five- and ten-day forecast

Benchmark each horizon with

$$y_{i,t,H}^{exec} = a_H + b_H z + d_H zc + e_H z p^{info} + f_H z p^{complete} + u_{i,t,H}.$$

For dislocations estimate $D_{t+h} = D_t e^{-\kappa h} + \epsilon$ and expected capture $D_t(1 - e^{-\kappa H})$; for information diffusion use a bounded catch-up fraction. This gives an economic mapping from state to the 5d/10d horizon.

3.4 Execution and costs

Estimate jointly

$$p^{fill} = \mathbb{P}(fill \mid X, size, protocol), \quad (6)$$

$$C = g(size, spread, ADV, depth, vol, flow, toxicity, protocol), \quad (7)$$

$$\mu_H = \mathbb{E}[y_H^{exec} \mid fill, X]. \quad (8)$$

Use a fresh executable entry, never the historical print that revealed the signal. Model quote withdrawal and adverse selection. Relative-value packages pay every leg. Capacity is evaluated by rerunning net P&L at increasing requested size and participation constraints.

3.5 Falsification and monitoring

Kill the feature if it disappears with fresh independent prices, relies on unavailable data, is concentrated in a few issuers/events, or loses its edge under realistic fills and costs. Run leave-one-issuer-out tests, IG/HY and liquidity splits, calm/stress regimes, protocol splits and alternative residualisations. Monitor drift, price confidence, fill calibration, realised versus predicted costs, mark-outs and residual factor exposure.

3.6 Signal-specific mechanism and implementation

Data: Synchronous sibling quotes + cash-flow risk | Target: Issuer-neutral preferred | 5 and 10 days kink = $s_{\text{middle}} - [w*s_{\text{short}} + (1-w)*s_{\text{long}}]$; then remove signal 01 Mechanism and evidence. This isolates a local distortion in curve shape rather than an overall cheap bond or broad issuer spread move. Its distinct hypothesis is that a temporary concentration of demand at one maturity creates curvature that subsequently relaxes. This is a proposed implementation based on relative pricing; OTC-friction research supplies background, not a proven butterfly alpha. Feature and implementation. Choose short-, middle- and long-maturity bonds with the same economic credit claim and reliable current quotes. Interpolate the two wings to the middle bond's tenor to construct a spread-space curvature feature; w is the interpolation weight, not the trading notional. Remove the broad leave-one-out value score and persistent local curve shape using training-only regressions. A positive remaining kink suggests the middle bond is cheap. Record wing width, coupon mismatch, liquidity asymmetry, the age of all quotes and the condition number of the hedge system. Start with adjacent liquid maturities; do not optimise thousands of arbitrary triplets on the test set. Five-/ten-day forecast and hedge. Fit five- and ten-day mean curvature changes separately, conditioned on a transient-flow trigger and issuer event calendar. Estimate results against the issuer-neutral target. Size the tradable package with cash-flow/key-rate and spread-sensitivity calculations, not equal face values. A three-bond package cannot generally neutralise every Treasury key rate, credit level, credit slope and cash constraint simultaneously: add Treasury hedges or relax a documented constraint. Record the resulting residual slope/convexity risk. Executable entry and costs. Evaluate all three bonds and any rate hedges at contemporaneous executable prices. Count two-way costs on every changed leg. A small apparent curvature edge is particularly vulnerable to multi-leg friction and legging risk. Set a maximum time between leg executions, reserve for failed hedges, and cancel the package when one required leg is not available. Prefer rebalancing existing inventory to opening a completely new butterfly. Falsification / kill test. Exclude maturity points with real structural segmentation, unusual call schedules, concentrated redemptions or covenant/recovery differences. Run leave-one-wing-out stability tests and compare with a simpler target-versus-one-sibling trade. This is low priority because much of its information may already be in signal 01 while costs are higher. Retain it only if its net contribution survives that direct ablation.

3.7 Agent implementation runbook: data discovery to safe production

Operating rule. The agent must advance through auditable gates rather than implement the entire alpha in one pass. If a required field is unavailable, ambiguous, delayed, or not historically point-in-time, stop that branch and record the limitation; never silently substitute a convenient proxy.

- A1. Data discovery.** Inventory candidate sources/tables/files; definitions; keys; units; time zones; economic, publication and receipt timestamps; corrections/cancels; historical coverage; entitlements; delayed-history rules; and whether each field was genuinely observable live. Output schema and coverage diagnostics only.
- A2. Raw profiling.** Report rows, bonds, issuers, days, missingness, duplicates, timestamp inversions, corrections, size distributions and join success by IG/HY, liquidity, sector and period. Do not impute yet.
- A3. Canonical identifiers.** Version the security master and map security IDs to legal obligor, parent, guarantee, seniority, currency, coupon, maturity, option schedule and corporate actions. Quarantine ambiguous/time-varying mappings.
- A4. As-of layer.** Require every query to accept a decision timestamp and return only records then available. Unit-test late records, corrections and cancels. Persist available-at and provenance fields.
- A5. Price layer.** Normalize clean/dirty price, accrued, spread convention, side, size and quote status. Create freshness, executable-size, dispersion and dealer-breadth metadata. Any forward fill must explicitly age and lose confidence.
- A6. Analytics layer.** Produce cash flows, yield/OAS as appropriate, spread duration, convexity, KRD, CS01/DV01, carry and roll from point-in-time terms; reconcile a sample against an independent implementation.
- A7. Targets.** Build 5d/10d efficient-price diagnostic and executable-return labels. Estimate residualisation only on prior data; cost every changed hedge leg, fees, financing and impact. Persist label maturity for purging.
- A8. Signal-specific prerequisites.** Execute the following in order, emitting exclusions, distributions and hand-audited examples after each sub-step:
 - a) Discover same-claim siblings, fresh executable quotes, spread analytics and KRD/CS01.

- b) Enumerate adjacent maturity triplets with identical obligor, currency, guarantee and seniority; exclude optionality mismatches.
- c) Choose interpolation weights from tenor/duration coordinates, but solve hedge notionals separately from KRD/CS01 constraints.
- d) Compute curvature and residualise against signal 01 and persistent local shape using training-only fits.
- e) Reprice all cash and rate-hedge legs at synchronized executable sides; reject packages outside the legging window.
- f) Stress wing choice, quote freshness, hedge conditioning and one-wing removal; require incremental net value over simpler sibling RV.

- A9. Literal raw feature.** Translate the documented equation exactly. Assert units, sign, finite values, peer counts, timestamp ordering and boundaries. Hand-reconcile at least 20 events.
- A10. Causal transforms.** Add rolling median/MAD z-score, changes, persistence, acceleration and percentile with past-only windows. Fit curves/betas/normalizers only inside training windows and version them.
- A11. State variables.** Join price confidence, information/toxicity, pressure completion, liquidity regime, volatility, event age and requested size as separate columns before tuning interactions.
- A12. Leakage tests.** Append records arriving after decision time and prove historical feature values do not change. Compare live-available with enhanced-history fields and test shifted timestamps. Any failure blocks modelling.
- A13. Measurement tests.** Recompute with fresh quotes only, trades only, independent prices where possible, alternative stale thresholds and stricter confidence. Reject effects caused only by asynchronous updating.
- A14. Transparent baseline.** Fit horizon-specific linear/robust models in walk-forward folds before nonlinear models; record sign, stability, sample size and incremental information.
- A15. State/convergence models.** Estimate information versus temporary-pressure probabilities and convergence speed where meaningful. Cross-fit state probabilities used as features.
- A16. Execution simulator.** Request a new executable entry after the signal timestamp; apply latency, size, withdrawal, partial-fill and legging rules. No executable observation means unfilled, not a midpoint fill.
- A17. Fill/cost models.** Estimate fill probability and conditional spread/impact/fees/financing by size, protocol and state. Calibrate on held-out fills and cost every changed leg.
- A18. Net outcomes.** Store expected executable return conditional on fill, conservative lower quantile and capacity. Keep gross, mark, executable and fully-net outcomes as distinct audit columns.
- A19. Adversarial validation.** Use chronological purged folds, issuer holdouts, liquidity/rating/regime splits, event exclusions, alternative residualisations, placebo dates and within-family ablations.
- A20. Shadow mode.** Freeze versions and generate live predictions without trading. Log source timestamps, features, quotes, hypothetical orders, predicted fills/costs and 1d/5d/10d mark-outs.
- A21. Canary.** Start with strict size, concentration, liquidity and turnover limits. Kill automatically on timestamp anomalies, data gaps, feature drift, cost underprediction, unexplained residual beta or repeated adverse selection.
- A22. Promotion/rollback.** Promote only a fully reproducible version. Default to no-trade when health checks fail and retain one-command rollback to the prior approved model.

Incremental safe prompts. Use these as separate prompts/commits; each prompt may inspect prior artifacts but must not expand scope silently.

- P1.** “Inventory only the required data, schemas, keys, timestamps, live-history availability, entitlements, coverage and unresolved questions. Do not build a feature.”
- P2.** “Build and unit-test canonical identifiers and as-of joins. Return diagnostics and quarantined ambiguities. Do not calculate alpha.”

- P3.** “Build the price/analytics layer with freshness/confidence and reconcile a hand-audited sample. Stop on unexplained discrepancies.”
- P4.** “Implement only the literal raw signal with assertions and 20 hand-reconciled examples. Do not optimize parameters.”
- P5.** “Add past-only transforms and state variables. Run leakage and measurement-artifact tests before fitting any predictive model.”
- P6.** “Fit the transparent walk-forward baseline for 5d/10d. Report coefficients, uncertainty, ablations and failures; do not tune on the final test period.”
- P7.** “Add execution, fill and transaction-cost simulation. Recompute conditional executable net outcomes and mark unavailable opportunities unfilled.”
- P8.** “Run adversarial validation and within-family ablations. Produce a go/no-go research recommendation based only on predeclared gates.”
- P9.** “Run shadow mode with frozen versions. Do not send orders. Compare predicted versus observable fills, costs and mark-outs.”
- P10.** “Only after explicit human approval, create a tightly limited canary configuration with hard risk limits, health checks and rollback; do not broaden limits automatically.”

3.8 Production row and acceptance gate

Persist raw score, z-score, event age, quote age, peer count/effective sample size, confidence, state probabilities, size/ADV, expected spread/impact, issuer/sector/rating/tenor, KRD/CS01 and source timestamps. The feature graduates only if it adds incremental out-of-sample information after other members of its economic family, survives alternative residualisations and retains a positive conservative executable-net-return statistic at realistic capacity.

4 Signal 03: Near-twin bond divergence

4.1 Mathematics and feature construction

The canonical raw score is

$$D_{ij,t} = s_{i,t} - \beta_{ij,t} s_{j,t} - \theta^\top x_{ij,t}. \quad (9)$$

Retain level, one- and three-day change, persistence, acceleration, trailing percentile and robust z-score. Add interactions with $c_{i,t}$, $p_{i,t}^{info}$, $p_{i,t}^{complete}$, liquidity regime, volatility and requested size. Thresholds and normalisation are learned on training data only. Missingness is explicit.

4.2 Point-in-time implementation

Maintain an immutable as-of ledger with economic, publication and receipt timestamps, corrections/cancels, security-master version and entitlements. At each decision time freeze the universe; resolve terms and corporate actions; construct fresh prices; compute peers/factors from earlier data; calculate the feature; obtain or simulate a fresh executable entry; and persist the feature snapshot. A confidence-weighted price can use

$$w_q = e^{-age_q/\tau_a} \min(1, size_q/v_0) w_d e^{-dispersion_q/\tau_\sigma}, \quad P^* = \frac{\sum_q w_q P_q}{\sum_q w_q}.$$

Never use future prints through symmetric smoothing.

4.3 Five- and ten-day forecast

Benchmark each horizon with

$$y_{i,t,H}^{exec} = a_H + b_H z + d_H z c + e_H z p^{info} + f_H z p^{complete} + u_{i,t,H}.$$

For dislocations estimate $D_{t+h} = D_t e^{-\kappa h} + \epsilon$ and expected capture $D_t(1 - e^{-\kappa H})$; for information diffusion use a bounded catch-up fraction. This gives an economic mapping from state to the 5d/10d horizon.

4.4 Execution and costs

Estimate jointly

$$p^{fill} = \mathbb{P}(fill \mid X, size, protocol), \quad (10)$$

$$C = g(size, spread, ADV, depth, vol, flow, toxicity, protocol), \quad (11)$$

$$\mu_H = \mathbb{E}[y_H^{exec} \mid fill, X]. \quad (12)$$

Use a fresh executable entry, never the historical print that revealed the signal. Model quote withdrawal and adverse selection. Relative-value packages pay every leg. Capacity is evaluated by rerunning net P&L at increasing requested size and participation constraints.

4.5 Falsification and monitoring

Kill the feature if it disappears with fresh independent prices, relies on unavailable data, is concentrated in a few issuers/events, or loses its edge under realistic fills and costs. Run leave-one-issuer-out tests, IG/HY and liquidity splits, calm/stress regimes, protocol splits and alternative residualisations. Monitor drift, price confidence, fill calibration, realised versus predicted costs, mark-outs and residual factor exposure.

4.6 Signal-specific mechanism and implementation

Data: Bond master + paired trades / quotes | Target: Issuer-neutral preferred | 5 and 10 days pair_gap = (s_i - s_j) - expected_persistent_spread_difference(i,j) Mechanism and evidence. Two close substitutes can temporarily diverge,

but a persistent liquidity or contractual difference is not mispricing. This signal tests the unexpected change in their relationship, not whether spreads should be identical. Corporate-bond illiquidity evidence motivates the adjustment. The exact matching, decay and trading rule are research proposals. Feature and implementation. Build a point-in-time nearest-neighbour graph within legal obligor, currency, seniority and guarantee. Match on effective maturity, cash-flow shape, coupon, issue age, size and trading activity; explicitly exclude optionality or structural mismatches. Require current prices for both members. Estimate their normal spread difference with a robust trailing model using, for example, 60–120 available trading sessions, while preserving uncertainty for young bonds. Include changing liquidity differentials rather than subtracting an unconditional mean forever. The score is this adjusted gap divided by a robust historical scale. Positive means long i versus j, subject to costs. Five-/ten-day forecast and hedge. Estimate pair-specific dynamics with shrinkage to an issuer/sector pool; most pairs have too few independent five-/ten-day observations for unrestricted estimation. Calculate a duration-consistent price-return forecast rather than treating spread basis points as price basis points. Test both horizons and require that the forecasted convergence has not already occurred between signal calculation and execution. A paired hedge generally makes this more compatible with a strict issuer-neutral target than cross-issuer value. Executable entry and costs. Use a paired executable gap, including the short bond's borrow cost and availability where the strategy actually shorts. For an existing long-only portfolio, compare swapping the expensive bond into the cheap one against holding the original; that is a different payoff and cost denominator from a financed long–short pair. Cap concentration in one issuer and count correlation among multiple pairs sharing the same bond. Falsification / kill test. Watch for benchmark/on-the-run liquidity changes, issue reopenings, merger guarantees and maturity-dependent recovery expectations. Pairs need not remain cointegrated after a capital-structure event. Demand incremental performance over signal 01 and over simply buying the more liquid sibling. Delete duplicated graph edges from the portfolio-level risk calculation, not just from a statistical significance table.

4.7 Agent implementation runbook: data discovery to safe production

Operating rule. The agent must advance through auditable gates rather than implement the entire alpha in one pass. If a required field is unavailable, ambiguous, delayed, or not historically point-in-time, stop that branch and record the limitation; never silently substitute a convenient proxy.

- A1. Data discovery.** Inventory candidate sources/tables/files; definitions; keys; units; time zones; economic, publication and receipt timestamps; corrections/cancels; historical coverage; entitlements; delayed-history rules; and whether each field was genuinely observable live. Output schema and coverage diagnostics only.
- A2. Raw profiling.** Report rows, bonds, issuers, days, missingness, duplicates, timestamp inversions, corrections, size distributions and join success by IG/HY, liquidity, sector and period. Do not impute yet.
- A3. Canonical identifiers.** Version the security master and map security IDs to legal obligor, parent, guarantee, seniority, currency, coupon, maturity, option schedule and corporate actions. Quarantine ambiguous/time-varying mappings.
- A4. As-of layer.** Require every query to accept a decision timestamp and return only records then available. Unit-test late records, corrections and cancels. Persist available-at and provenance fields.
- A5. Price layer.** Normalize clean/dirty price, accrued, spread convention, side, size and quote status. Create freshness, executable-size, dispersion and dealer-breadth metadata. Any forward fill must explicitly age and lose confidence.
- A6. Analytics layer.** Produce cash flows, yield/OAS as appropriate, spread duration, convexity, KRD, CS01/DV01, carry and roll from point-in-time terms; reconcile a sample against an independent implementation.
- A7. Targets.** Build 5d/10d efficient-price diagnostic and executable-return labels. Estimate residualisation only on prior data; cost every changed hedge leg, fees, financing and impact. Persist label maturity for purging.
- A8. Signal-specific prerequisites.** Execute the following in order, emitting exclusions, distributions and hand-audited examples after each sub-step:
 - a) Discover candidate twins with terms, optionality, liquidity, fresh quotes and borrow/locate history.
 - b) Build a point-in-time nearest-neighbour graph with hard claim constraints and soft distance on maturity, coupon, age, size and liquidity.
 - c) Estimate the normal pair spread difference with a trailing robust/shrunk model and retain its uncertainty.

- d) Compute the unexpected gap and duration-consistent convergence; de-duplicate overlapping graph edges at portfolio level.
- e) Evaluate long/short legs at executable sides including borrow, or separately define a long-only swap implementation.
- f) Break pairs around reopenings, tenders, guarantee/capital-structure changes and liquidity regime shifts; compare against signal 01.

- A9. Literal raw feature.** Translate the documented equation exactly. Assert units, sign, finite values, peer counts, timestamp ordering and boundaries. Hand-reconcile at least 20 events.
- A10. Causal transforms.** Add rolling median/MAD z-score, changes, persistence, acceleration and percentile with past-only windows. Fit curves/betas/normalizers only inside training windows and version them.
- A11. State variables.** Join price confidence, information/toxicity, pressure completion, liquidity regime, volatility, event age and requested size as separate columns before tuning interactions.
- A12. Leakage tests.** Append records arriving after decision time and prove historical feature values do not change. Compare live-available with enhanced-history fields and test shifted timestamps. Any failure blocks modelling.
- A13. Measurement tests.** Recompute with fresh quotes only, trades only, independent prices where possible, alternative stale thresholds and stricter confidence. Reject effects caused only by asynchronous updating.
- A14. Transparent baseline.** Fit horizon-specific linear/robust models in walk-forward folds before nonlinear models; record sign, stability, sample size and incremental information.
- A15. State/convergence models.** Estimate information versus temporary-pressure probabilities and convergence speed where meaningful. Cross-fit state probabilities used as features.
- A16. Execution simulator.** Request a new executable entry after the signal timestamp; apply latency, size, withdrawal, partial-fill and legging rules. No executable observation means unfilled, not a midpoint fill.
- A17. Fill/cost models.** Estimate fill probability and conditional spread/impact/fees/financing by size, protocol and state. Calibrate on held-out fills and cost every changed leg.
- A18. Net outcomes.** Store expected executable return conditional on fill, conservative lower quantile and capacity. Keep gross, mark, executable and fully-net outcomes as distinct audit columns.
- A19. Adversarial validation.** Use chronological purged folds, issuer holdouts, liquidity/rating/regime splits, event exclusions, alternative residualisations, placebo dates and within-family ablations.
- A20. Shadow mode.** Freeze versions and generate live predictions without trading. Log source timestamps, features, quotes, hypothetical orders, predicted fills/costs and 1d/5d/10d mark-outs.
- A21. Canary.** Start with strict size, concentration, liquidity and turnover limits. Kill automatically on timestamp anomalies, data gaps, feature drift, cost underprediction, unexplained residual beta or repeated adverse selection.
- A22. Promotion/rollback.** Promote only a fully reproducible version. Default to no-trade when health checks fail and retain one-command rollback to the prior approved model.

Incremental safe prompts. Use these as separate prompts/commits; each prompt may inspect prior artifacts but must not expand scope silently.

- P1.** “Inventory only the required data, schemas, keys, timestamps, live-history availability, entitlements, coverage and unresolved questions. Do not build a feature.”
- P2.** “Build and unit-test canonical identifiers and as-of joins. Return diagnostics and quarantined ambiguities. Do not calculate alpha.”
- P3.** “Build the price/analytics layer with freshness/confidence and reconcile a hand-audited sample. Stop on unexplained discrepancies.”
- P4.** “Implement only the literal raw signal with assertions and 20 hand-reconciled examples. Do not optimize parameters.”

- P5.** “Add past-only transforms and state variables. Run leakage and measurement-artifact tests before fitting any predictive model.”
- P6.** “Fit the transparent walk-forward baseline for 5d/10d. Report coefficients, uncertainty, ablations and failures; do not tune on the final test period.”
- P7.** “Add execution, fill and transaction-cost simulation. Recompute conditional executable net outcomes and mark unavailable opportunities unfilled.”
- P8.** “Run adversarial validation and within-family ablations. Produce a go/no-go research recommendation based only on predeclared gates.”
- P9.** “Run shadow mode with frozen versions. Do not send orders. Compare predicted versus observable fills, costs and mark-outs.”
- P10.** “Only after explicit human approval, create a tightly limited canary configuration with hard risk limits, health checks and rollback; do not broaden limits automatically.”

4.8 Production row and acceptance gate

Persist raw score, z-score, event age, quote age, peer count/effective sample size, confidence, state probabilities, size/ADV, expected spread/impact, issuer/sector/rating/tenor, KRD/CS01 and source timestamps. The feature graduates only if it adds incremental out-of-sample information after other members of its economic family, survives alternative residualisations and retains a positive conservative executable-net-return statistic at realistic capacity.

5 Signal 04: Liquidity-adjusted cross-issuer fair value

5.1 Mathematics and feature construction

The canonical raw score is

$$RV_{i,t} = s_{i,t} - \hat{f}_t(X^{credit}, X^{liq}, X^{contract}). \quad (13)$$

Retain level, one- and three-day change, persistence, acceleration, trailing percentile and robust z-score. Add interactions with $c_{i,t}$, $p_{i,t}^{info}$, $p_{i,t}^{complete}$, liquidity regime, volatility and requested size. Thresholds and normalisation are learned on training data only. Missingness is explicit.

5.2 Point-in-time implementation

Maintain an immutable as-of ledger with economic, publication and receipt timestamps, corrections/cancels, security-master version and entitlements. At each decision time freeze the universe; resolve terms and corporate actions; construct fresh prices; compute peers/factors from earlier data; calculate the feature; obtain or simulate a fresh executable entry; and persist the feature snapshot. A confidence-weighted price can use

$$w_q = e^{-age_q/\tau_a} \min(1, size_q/v_0) w_d e^{-dispersion_q/\tau_\sigma}, \quad P^* = \frac{\sum_q w_q P_q}{\sum_q w_q}.$$

Never use future prints through symmetric smoothing.

5.3 Five- and ten-day forecast

Benchmark each horizon with

$$y_{i,t,H}^{exec} = a_H + b_H z + d_H z c + e_H z p^{info} + f_H z p^{complete} + u_{i,t,H}.$$

For dislocations estimate $D_{t+h} = D_t e^{-\kappa h} + \epsilon$ and expected capture $D_t(1 - e^{-\kappa H})$; for information diffusion use a bounded catch-up fraction. This gives an economic mapping from state to the 5d/10d horizon.

5.4 Execution and costs

Estimate jointly

$$p^{fill} = \mathbb{P}(fill \mid X, size, protocol), \quad (14)$$

$$C = g(size, spread, ADV, depth, vol, flow, toxicity, protocol), \quad (15)$$

$$\mu_H = \mathbb{E}[y_H^{exec} \mid fill, X]. \quad (16)$$

Use a fresh executable entry, never the historical print that revealed the signal. Model quote withdrawal and adverse selection. Relative-value packages pay every leg. Capacity is evaluated by rerunning net P&L at increasing requested size and participation constraints.

5.5 Falsification and monitoring

Kill the feature if it disappears with fresh independent prices, relies on unavailable data, is concentrated in a few issuers/events, or loses its edge under realistic fills and costs. Run leave-one-issuer-out tests, IG/HY and liquidity splits, calm/stress regimes, protocol splits and alternative residualisations. Monitor drift, price confidence, fill calibration, realised versus predicted costs, mark-outs and residual factor exposure.

5.6 Signal-specific mechanism and implementation

Data: Prices + point-in-time fundamentals | Target: Base; weak issuer-neutral fit | 10-day emphasis; test 5 value_gap_i = s_i - predicted_spread(credit, tenor, structure, liquidity) Mechanism and evidence. This asks whether a bond

is unusually wide after observable credit and liquidity differences. Valuation-related corporate-bond predictability has published support, but book-to-market evidence is not validation of this peer-spread model or of convergence in one or two weeks. Treat the short-horizon version as a higher-uncertainty extension. Feature and implementation. Use a robust additive model or a tightly regularised tree model to estimate spread from sector, rating, maturity, seniority, leverage, coverage, profitability, equity volatility, issue size/age and transaction-cost proxies. Join accounting inputs by actual publication time, not fiscal period end. Separate IG/HY and financial/nonfinancial regimes. Train without the target issuer when measuring cross-issuer generalisation; shrink issuer effects rather than allowing the model to memorise identities. Residualise the value score against size, liquidity, carry and broad credit exposures. Preserve model dispersion across reasonable specifications as an uncertainty feature. A positive residual spread is a tentative long signal. Five-/ten-day forecast and hedge. Require an additional short-horizon catalyst or technical confirmation: fading forced sales, quote recovery or a fresh issuer-curve discrepancy. Fit independent five- and ten-day residual-return calibrations. Do not multiply an annual valuation premium by $5/252$ and call the result a forecast. This signal is naturally tested against the base target; strict issuer-neutralisation can remove most of the issuer-wide valuation thesis, leaving only bond-specific deviations. Executable entry and costs. Hedge sector/rating/tenor credit exposures with genuinely available instruments and price all basis risk and hedge costs. Restrict position size using both liquidity and model extrapolation uncertainty. Enter only where the remaining expected move exceeds all-in costs, not simply in the widest-spread decile. Run a benchmark that holds a similar bond with lower costs; the proposed alpha must beat that feasible alternative. Falsification / kill test. A wide residual can be compensation for an omitted default, recovery, covenant or liquidity risk. Fresh adverse information can make stale fundamentals dangerous. Kill the short-horizon implementation if it works only on slow evaluated marks, only on distressed exclusions, or only before cost-neutralisation. This belongs behind local issuer value because identifying fair value and identifying a five-day catalyst are different problems.

5.7 Agent implementation runbook: data discovery to safe production

Operating rule. The agent must advance through auditable gates rather than implement the entire alpha in one pass. If a required field is unavailable, ambiguous, delayed, or not historically point-in-time, stop that branch and record the limitation; never silently substitute a convenient proxy.

- A1. Data discovery.** Inventory candidate sources/tables/files; definitions; keys; units; time zones; economic, publication and receipt timestamps; corrections/cancels; historical coverage; entitlements; delayed-history rules; and whether each field was genuinely observable live. Output schema and coverage diagnostics only.
- A2. Raw profiling.** Report rows, bonds, issuers, days, missingness, duplicates, timestamp inversions, corrections, size distributions and join success by IG/HY, liquidity, sector and period. Do not impute yet.
- A3. Canonical identifiers.** Version the security master and map security IDs to legal obligor, parent, guarantee, seniority, currency, coupon, maturity, option schedule and corporate actions. Quarantine ambiguous/time-varying mappings.
- A4. As-of layer.** Require every query to accept a decision timestamp and return only records then available. Unit-test late records, corrections and cancels. Persist available-at and provenance fields.
- A5. Price layer.** Normalize clean/dirty price, accrued, spread convention, side, size and quote status. Create freshness, executable-size, dispersion and dealer-breadth metadata. Any forward fill must explicitly age and lose confidence.
- A6. Analytics layer.** Produce cash flows, yield/OAS as appropriate, spread duration, convexity, KRD, CS01/DV01, carry and roll from point-in-time terms; reconcile a sample against an independent implementation.
- A7. Targets.** Build 5d/10d efficient-price diagnostic and executable-return labels. Estimate residualisation only on prior data; cost every changed hedge leg, fees, financing and impact. Persist label maturity for purging.
- A8. Signal-specific prerequisites.** Execute the following in order, emitting exclusions, distributions and hand-audited examples after each sub-step:
 - a) Discover point-in-time terms, ratings, fundamentals by publication time, liquidity measures and executable prices.
 - b) Define train-only peer universes and prevent issuer identity or future ratings from leaking into fair value.

- c) Fit regularised fair-spread models by economic regime and generate strictly out-of-fold fair values for training rows.
- d) Form residual value and orthogonalise against broad carry, liquidity, size and credit-beta exposures; retain model dispersion.
- e) Translate spread residual to executable expected return using spread duration and actual entry/hedge sides.
- f) Use issuer-held-out/calendar-held-out tests and alternative fair-value models; reject compensation for omitted credit/liquidity risk.

- A9. Literal raw feature.** Translate the documented equation exactly. Assert units, sign, finite values, peer counts, timestamp ordering and boundaries. Hand-reconcile at least 20 events.
- A10. Causal transforms.** Add rolling median/MAD z-score, changes, persistence, acceleration and percentile with past-only windows. Fit curves/betas/normalizers only inside training windows and version them.
- A11. State variables.** Join price confidence, information/toxicity, pressure completion, liquidity regime, volatility, event age and requested size as separate columns before tuning interactions.
- A12. Leakage tests.** Append records arriving after decision time and prove historical feature values do not change. Compare live-available with enhanced-history fields and test shifted timestamps. Any failure blocks modelling.
- A13. Measurement tests.** Recompute with fresh quotes only, trades only, independent prices where possible, alternative stale thresholds and stricter confidence. Reject effects caused only by asynchronous updating.
- A14. Transparent baseline.** Fit horizon-specific linear/robust models in walk-forward folds before nonlinear models; record sign, stability, sample size and incremental information.
- A15. State/convergence models.** Estimate information versus temporary-pressure probabilities and convergence speed where meaningful. Cross-fit state probabilities used as features.
- A16. Execution simulator.** Request a new executable entry after the signal timestamp; apply latency, size, withdrawal, partial-fill and legging rules. No executable observation means unfilled, not a midpoint fill.
- A17. Fill/cost models.** Estimate fill probability and conditional spread/impact/fees/financing by size, protocol and state. Calibrate on held-out fills and cost every changed leg.
- A18. Net outcomes.** Store expected executable return conditional on fill, conservative lower quantile and capacity. Keep gross, mark, executable and fully-net outcomes as distinct audit columns.
- A19. Adversarial validation.** Use chronological purged folds, issuer holdouts, liquidity/rating/regime splits, event exclusions, alternative residualisations, placebo dates and within-family ablations.
- A20. Shadow mode.** Freeze versions and generate live predictions without trading. Log source timestamps, features, quotes, hypothetical orders, predicted fills/costs and 1d/5d/10d mark-outs.
- A21. Canary.** Start with strict size, concentration, liquidity and turnover limits. Kill automatically on timestamp anomalies, data gaps, feature drift, cost underprediction, unexplained residual beta or repeated adverse selection.
- A22. Promotion/rollback.** Promote only a fully reproducible version. Default to no-trade when health checks fail and retain one-command rollback to the prior approved model.

Incremental safe prompts. Use these as separate prompts/commits; each prompt may inspect prior artifacts but must not expand scope silently.

- P1.** “Inventory only the required data, schemas, keys, timestamps, live-history availability, entitlements, coverage and unresolved questions. Do not build a feature.”
- P2.** “Build and unit-test canonical identifiers and as-of joins. Return diagnostics and quarantined ambiguities. Do not calculate alpha.”
- P3.** “Build the price/analytics layer with freshness/confidence and reconcile a hand-audited sample. Stop on unexplained discrepancies.”

- P4.** “Implement only the literal raw signal with assertions and 20 hand-reconciled examples. Do not optimize parameters.”
- P5.** “Add past-only transforms and state variables. Run leakage and measurement-artifact tests before fitting any predictive model.”
- P6.** “Fit the transparent walk-forward baseline for 5d/10d. Report coefficients, uncertainty, ablations and failures; do not tune on the final test period.”
- P7.** “Add execution, fill and transaction-cost simulation. Recompute conditional executable net outcomes and mark unavailable opportunities unfilled.”
- P8.** “Run adversarial validation and within-family ablations. Produce a go/no-go research recommendation based only on predeclared gates.”
- P9.** “Run shadow mode with frozen versions. Do not send orders. Compare predicted versus observable fills, costs and mark-outs.”
- P10.** “Only after explicit human approval, create a tightly limited canary configuration with hard risk limits, health checks and rollback; do not broaden limits automatically.”

5.8 Production row and acceptance gate

Persist raw score, z-score, event age, quote age, peer count/effective sample size, confidence, state probabilities, size/ADV, expected spread/impact, issuer/sector/rating/tenor, KRD/CS01 and source timestamps. The feature graduates only if it adds incremental out-of-sample information after other members of its economic family, survives alternative residualisations and retains a positive conservative executable-net-return statistic at realistic capacity.

6 Signal 05: Issuer leader–laggard catch-up

6.1 Mathematics and feature construction

The canonical raw score is

$$C_{i,t} = \hat{\beta}_i I_{g,t}^{leader} - I_{i,t}^{bond}. \quad (17)$$

Retain level, one- and three-day change, persistence, acceleration, trailing percentile and robust z-score. Add interactions with $c_{i,t}$, $p_{i,t}^{info}$, $p_{i,t}^{complete}$, liquidity regime, volatility and requested size. Thresholds and normalisation are learned on training data only. Missingness is explicit.

6.2 Point-in-time implementation

Maintain an immutable as-of ledger with economic, publication and receipt timestamps, corrections/cancels, security-master version and entitlements. At each decision time freeze the universe; resolve terms and corporate actions; construct fresh prices; compute peers/factors from earlier data; calculate the feature; obtain or simulate a fresh executable entry; and persist the feature snapshot. A confidence-weighted price can use

$$w_q = e^{-age_q/\tau_a} \min(1, size_q/v_0) w_d e^{-dispersion_q/\tau_\sigma}, \quad P^* = \frac{\sum_q w_q P_q}{\sum_q w_q}.$$

Never use future prints through symmetric smoothing.

6.3 Five- and ten-day forecast

Benchmark each horizon with

$$y_{i,t,H}^{exec} = a_H + b_H z + d_H z c + e_H z p^{info} + f_H z p^{complete} + u_{i,t,H}.$$

For dislocations estimate $D_{t+h} = D_t e^{-\kappa h} + \epsilon$ and expected capture $D_t(1 - e^{-\kappa H})$; for information diffusion use a bounded catch-up fraction. This gives an economic mapping from state to the 5d/10d horizon.

6.4 Execution and costs

Estimate jointly

$$p^{fill} = \mathbb{P}(fill \mid X, size, protocol), \quad (18)$$

$$C = g(size, spread, ADV, depth, vol, flow, toxicity, protocol), \quad (19)$$

$$\mu_H = \mathbb{E}[y_H^{exec} \mid fill, X]. \quad (20)$$

Use a fresh executable entry, never the historical print that revealed the signal. Model quote withdrawal and adverse selection. Relative-value packages pay every leg. Capacity is evaluated by rerunning net P&L at increasing requested size and participation constraints.

6.5 Falsification and monitoring

Kill the feature if it disappears with fresh independent prices, relies on unavailable data, is concentrated in a few issuers/events, or loses its edge under realistic fills and costs. Run leave-one-issuer-out tests, IG/HY and liquidity splits, calm/stress regimes, protocol splits and alternative residualisations. Monitor drift, price confidence, fill calibration, realised versus predicted costs, mark-outs and residual factor exposure.

6.6 Signal-specific mechanism and implementation

Data: Fresh sibling prices + timestamped quotes | Target: Base and relative issuer-neutral | 5-day emphasis; test 10 catchup_i = beta_i * issuer_leader_move_since_refresh_i - observed_move_i Mechanism and evidence. A liquid

issuer bond may incorporate issuer information before a less actively updated sibling. The tradable hypothesis is delayed adjustment after common rate and market moves are removed. It is an engineering extension of credit price-discovery research, not a claim that historical CDS leadership proves bond-to-bond lagged profits. Feature and implementation. Define leaders ex ante using trailing trade frequency, quote freshness, executable size and historically measured information leadership. Estimate a sparse directed graph, forbidding future information in edge selection. Measure leaders' residual price changes only since the target's last reliable refresh; subtract the target's actual subsequent movement. Use a robust weighted issuer innovation excluding the target itself. Positive catch-up means the target has underperformed positive issuer information and is a candidate long. Control for cash-flow sensitivity and remove Treasury/market moves before constructing the feature. Retain information age and the number of independently refreshed leaders. Five-/ten-day forecast and hedge. Start with innovations accumulated over the previous one to three trading days and estimate their remaining five- and ten-day effect after actual entry. A historical lead of a few minutes does not establish a daily strategy. The most direct issuer-neutral label compares target returns with a leave-one-out sibling basket: the common shock may already be in the leaders, while the relative catch-up remains in the target. Test the same feature on the base label to show how much is simply issuer exposure. Executable entry and costs. The target must still be available at a genuine, fresh quote that embeds the lag. If contacting a dealer immediately updates the price, there is no executable alpha even though the database label catches up tomorrow. Simulate decision-to-quote latency, failed RFQs and hedge fills. Use a liquid sibling hedge and recalculate edge at the final quote, not the stale last trade. Falsification / kill test. The decisive falsification is to replace stale/evaluated endpoints with fresh independently sourced executable prices. Kill the signal if its IC then vanishes or if the quoted edge is routinely withdrawn. Exclude differences caused by recovery/seniority, maturity-specific refinancing risk or a bond-specific tender. Compare directly against signals 01, 06, 07 and 14 to avoid counting one common information event four times.

6.7 Agent implementation runbook: data discovery to safe production

Operating rule. The agent must advance through auditable gates rather than implement the entire alpha in one pass. If a required field is unavailable, ambiguous, delayed, or not historically point-in-time, stop that branch and record the limitation; never silently substitute a convenient proxy.

- A1. Data discovery.** Inventory candidate sources/tables/files; definitions; keys; units; time zones; economic, publication and receipt timestamps; corrections/cancels; historical coverage; entitlements; delayed-history rules; and whether each field was genuinely observable live. Output schema and coverage diagnostics only.
- A2. Raw profiling.** Report rows, bonds, issuers, days, missingness, duplicates, timestamp inversions, corrections, size distributions and join success by IG/HY, liquidity, sector and period. Do not impute yet.
- A3. Canonical identifiers.** Version the security master and map security IDs to legal obligor, parent, guarantee, seniority, currency, coupon, maturity, option schedule and corporate actions. Quarantine ambiguous/time-varying mappings.
- A4. As-of layer.** Require every query to accept a decision timestamp and return only records then available. Unit-test late records, corrections and cancels. Persist available-at and provenance fields.
- A5. Price layer.** Normalize clean/dirty price, accrued, spread convention, side, size and quote status. Create freshness, executable-size, dispersion and dealer-breadth metadata. Any forward fill must explicitly age and lose confidence.
- A6. Analytics layer.** Produce cash flows, yield/OAS as appropriate, spread duration, convexity, KRD, CS01/DV01, carry and roll from point-in-time terms; reconcile a sample against an independent implementation.
- A7. Targets.** Build 5d/10d efficient-price diagnostic and executable-return labels. Estimate residualisation only on prior data; cost every changed hedge leg, fees, financing and impact. Persist label maturity for purging.
- A8. Signal-specific prerequisites.** Execute the following in order, emitting exclusions, distributions and hand-audited examples after each sub-step:
 - a) Discover same-issuer bonds and identify reliable price leaders from historical freshness and response speed.
 - b) Estimate issuer common shocks from leaders without allowing the laggard target into shock construction.
 - c) Measure target catch-up shortfall using past-only beta and explicit event age.
 - d) Classify the leader move as information-bearing versus transient using sibling/CDS/equity/quote confirmation.

- e) Refresh the target executable quote after the leader shock and reject already-adjusted targets.
- f) Estimate catch-up speed separately at 5d/10d and ablate against issuer-curve cheapness.

- A9. Literal raw feature.** Translate the documented equation exactly. Assert units, sign, finite values, peer counts, timestamp ordering and boundaries. Hand-reconcile at least 20 events.
- A10. Causal transforms.** Add rolling median/MAD z-score, changes, persistence, acceleration and percentile with past-only windows. Fit curves/betas/normalizers only inside training windows and version them.
- A11. State variables.** Join price confidence, information/toxicity, pressure completion, liquidity regime, volatility, event age and requested size as separate columns before tuning interactions.
- A12. Leakage tests.** Append records arriving after decision time and prove historical feature values do not change. Compare live-available with enhanced-history fields and test shifted timestamps. Any failure blocks modelling.
- A13. Measurement tests.** Recompute with fresh quotes only, trades only, independent prices where possible, alternative stale thresholds and stricter confidence. Reject effects caused only by asynchronous updating.
- A14. Transparent baseline.** Fit horizon-specific linear/robust models in walk-forward folds before nonlinear models; record sign, stability, sample size and incremental information.
- A15. State/convergence models.** Estimate information versus temporary-pressure probabilities and convergence speed where meaningful. Cross-fit state probabilities used as features.
- A16. Execution simulator.** Request a new executable entry after the signal timestamp; apply latency, size, withdrawal, partial-fill and legging rules. No executable observation means unfilled, not a midpoint fill.
- A17. Fill/cost models.** Estimate fill probability and conditional spread/impact/fees/financing by size, protocol and state. Calibrate on held-out fills and cost every changed leg.
- A18. Net outcomes.** Store expected executable return conditional on fill, conservative lower quantile and capacity. Keep gross, mark, executable and fully-net outcomes as distinct audit columns.
- A19. Adversarial validation.** Use chronological purged folds, issuer holdouts, liquidity/rating/regime splits, event exclusions, alternative residualisations, placebo dates and within-family ablations.
- A20. Shadow mode.** Freeze versions and generate live predictions without trading. Log source timestamps, features, quotes, hypothetical orders, predicted fills/costs and 1d/5d/10d mark-outs.
- A21. Canary.** Start with strict size, concentration, liquidity and turnover limits. Kill automatically on timestamp anomalies, data gaps, feature drift, cost underprediction, unexplained residual beta or repeated adverse selection.
- A22. Promotion/rollback.** Promote only a fully reproducible version. Default to no-trade when health checks fail and retain one-command rollback to the prior approved model.

Incremental safe prompts. Use these as separate prompts/commits; each prompt may inspect prior artifacts but must not expand scope silently.

- P1.** “Inventory only the required data, schemas, keys, timestamps, live-history availability, entitlements, coverage and unresolved questions. Do not build a feature.”
- P2.** “Build and unit-test canonical identifiers and as-of joins. Return diagnostics and quarantined ambiguities. Do not calculate alpha.”
- P3.** “Build the price/analytics layer with freshness/confidence and reconcile a hand-audited sample. Stop on unexplained discrepancies.”
- P4.** “Implement only the literal raw signal with assertions and 20 hand-reconciled examples. Do not optimize parameters.”
- P5.** “Add past-only transforms and state variables. Run leakage and measurement-artifact tests before fitting any predictive model.”

- P6.** “Fit the transparent walk-forward baseline for 5d/10d. Report coefficients, uncertainty, ablations and failures; do not tune on the final test period.”
- P7.** “Add execution, fill and transaction-cost simulation. Recompute conditional executable net outcomes and mark unavailable opportunities unfilled.”
- P8.** “Run adversarial validation and within-family ablations. Produce a go/no-go research recommendation based only on predeclared gates.”
- P9.** “Run shadow mode with frozen versions. Do not send orders. Compare predicted versus observable fills, costs and mark-outs.”
- P10.** “Only after explicit human approval, create a tightly limited canary configuration with hard risk limits, health checks and rollback; do not broaden limits automatically.”

6.8 Production row and acceptance gate

Persist raw score, z-score, event age, quote age, peer count/effective sample size, confidence, state probabilities, size/ADV, expected spread/impact, issuer/sector/rating/tenor, KRD/CS01 and source timestamps. The feature graduates only if it adds incremental out-of-sample information after other members of its economic family, survives alternative residualisations and retains a positive conservative executable-net-return statistic at realistic capacity.

7 Signal 06: Equity-to-credit delayed adjustment

7.1 Mathematics and feature construction

The canonical raw score is

$$E_{i,t} = \hat{\beta}_i^{EC} \varepsilon_{g,t}^{eq} - \varepsilon_{i,t}^{credit}. \quad (21)$$

Retain level, one- and three-day change, persistence, acceleration, trailing percentile and robust z-score. Add interactions with $c_{i,t}$, $p_{i,t}^{info}$, $p_{i,t}^{complete}$, liquidity regime, volatility and requested size. Thresholds and normalisation are learned on training data only. Missingness is explicit.

7.2 Point-in-time implementation

Maintain an immutable as-of ledger with economic, publication and receipt timestamps, corrections/cancels, security-master version and entitlements. At each decision time freeze the universe; resolve terms and corporate actions; construct fresh prices; compute peers/factors from earlier data; calculate the feature; obtain or simulate a fresh executable entry; and persist the feature snapshot. A confidence-weighted price can use

$$w_q = e^{-age_q/\tau_a} \min(1, size_q/v_0) w_d e^{-dispersion_q/\tau_\sigma}, \quad P^* = \frac{\sum_q w_q P_q}{\sum_q w_q}.$$

Never use future prints through symmetric smoothing.

7.3 Five- and ten-day forecast

Benchmark each horizon with

$$y_{i,t,H}^{exec} = a_H + b_H z + d_H zc + e_H z p^{info} + f_H z p^{complete} + u_{i,t,H}.$$

For dislocations estimate $D_{t+h} = D_t e^{-\kappa h} + \epsilon$ and expected capture $D_t(1 - e^{-\kappa H})$; for information diffusion use a bounded catch-up fraction. This gives an economic mapping from state to the 5d/10d horizon.

7.4 Execution and costs

Estimate jointly

$$p^{fill} = \mathbb{P}(fill \mid X, size, protocol), \quad (22)$$

$$C = g(size, spread, ADV, depth, vol, flow, toxicity, protocol), \quad (23)$$

$$\mu_H = \mathbb{E}[y_H^{exec} \mid fill, X]. \quad (24)$$

Use a fresh executable entry, never the historical print that revealed the signal. Model quote withdrawal and adverse selection. Relative-value packages pay every leg. Capacity is evaluated by rerunning net P&L at increasing requested size and participation constraints.

7.5 Falsification and monitoring

Kill the feature if it disappears with fresh independent prices, relies on unavailable data, is concentrated in a few issuers/events, or loses its edge under realistic fills and costs. Run leave-one-issuer-out tests, IG/HY and liquidity splits, calm/stress regimes, protocol splits and alternative residualisations. Monitor drift, price confidence, fill calibration, realised versus predicted costs, mark-outs and residual factor exposure.

7.6 Signal-specific mechanism and implementation

Data: Equity / volatility + credit prices | Target: Base; issuer-neutral may erase it | 5 and 10 days equity_credit_gap = implied_credit_price_innovation - realised_bond_innovation Mechanism and evidence. Issuer equity and credit

can react differently to firm-specific information. The proposal is to isolate an equity-implied credit change not yet incorporated in the bond, rather than buy bonds after any positive stock return. Historical cross-market research supports examining information transmission, but leadership is sample-dependent and does not establish present-day net alpha. Feature and implementation. Compute stock returns net of equity market and industry exposure and changes in option-implied or realised equity volatility. Map these innovations into expected bond spread changes using a past-only issuer/sector model with leverage, rating and spread level interactions. Convert the implied spread change to a positive-for-price-return innovation, then subtract the bond's fresh observed response. Treat equity returns and volatility separately because a positive stock reaction to a leveraged acquisition can be bad for creditors. Use only event information publicly released before the decision, with matching exchange and bond-market clocks. Five-/ten-day forecast and hedge. Fit remaining five- and ten-day bond residual returns after entry, not simultaneous stock/bond correlation. A simple restricted starting model uses one- and three-day equity innovations with interactions for earnings, financing announcements and HY/IG. Most issuer-wide information should disappear under strict issuer-neutralisation; report that honestly instead of changing the label to preserve the signal. Where it survives, show that different bonds have different adjustment lags, not just different measured betas. Executable entry and costs. Require a fresh quote confirming that the predicted credit movement is still uncaptured. Bond costs alone can exceed a modest equity-implied move. Hedging broad credit is different from selling the stock: an equity hedge creates equity/credit basis and event risk and must be separately tested, priced and constrained. Do not assume that any shareholder-good news is bondholder-good news. Falsification / kill test. Exclude or model buybacks, recapitalisations, acquisitions, capital distributions and distress where the relationship changes sign. Kill a model driven by stale bond closes or unrestricted nonlinear fitting to a handful of crises. Require incremental net improvement beyond fresh CDS and sibling-bond innovations; otherwise equity is a useful state variable or veto, not an independent trading alpha.

7.7 Agent implementation runbook: data discovery to safe production

Operating rule. The agent must advance through auditable gates rather than implement the entire alpha in one pass. If a required field is unavailable, ambiguous, delayed, or not historically point-in-time, stop that branch and record the limitation; never silently substitute a convenient proxy.

- A1. Data discovery.** Inventory candidate sources/tables/files; definitions; keys; units; time zones; economic, publication and receipt timestamps; corrections/cancels; historical coverage; entitlements; delayed-history rules; and whether each field was genuinely observable live. Output schema and coverage diagnostics only.
- A2. Raw profiling.** Report rows, bonds, issuers, days, missingness, duplicates, timestamp inversions, corrections, size distributions and join success by IG/HY, liquidity, sector and period. Do not impute yet.
- A3. Canonical identifiers.** Version the security master and map security IDs to legal obligor, parent, guarantee, seniority, currency, coupon, maturity, option schedule and corporate actions. Quarantine ambiguous/time-varying mappings.
- A4. As-of layer.** Require every query to accept a decision timestamp and return only records then available. Unit-test late records, corrections and cancels. Persist available-at and provenance fields.
- A5. Price layer.** Normalize clean/dirty price, accrued, spread convention, side, size and quote status. Create freshness, executable-size, dispersion and dealer-breadth metadata. Any forward fill must explicitly age and lose confidence.
- A6. Analytics layer.** Produce cash flows, yield/OAS as appropriate, spread duration, convexity, KRD, CS01/DV01, carry and roll from point-in-time terms; reconcile a sample against an independent implementation.
- A7. Targets.** Build 5d/10d efficient-price diagnostic and executable-return labels. Estimate residualisation only on prior data; cost every changed hedge leg, fees, financing and impact. Persist label maturity for purging.
- A8. Signal-specific prerequisites.** Execute the following in order, emitting exclusions, distributions and hand-audited examples after each sub-step:
 - a) Discover issuer equity, bond quotes/trades, corporate-action calendars and point-in-time equity-to-obligor mappings.
 - b) Construct equity innovations after market/sector removal, using only information received before the bond decision time.
 - c) Estimate prior-data equity-to-credit beta with shrinkage and allow rating/leverage/stress interactions.

- d) Measure missing bond response and cross-check siblings/CDS so equity-specific noise is not called credit information.
- e) Refresh executable bond price after the equity shock and reject already-adjusted or ambiguously mapped targets.
- f) Ablate earnings/M&A days, beta windows and sibling confirmation; require incremental value beyond issuer leader-lag.

- A9. Literal raw feature.** Translate the documented equation exactly. Assert units, sign, finite values, peer counts, timestamp ordering and boundaries. Hand-reconcile at least 20 events.
- A10. Causal transforms.** Add rolling median/MAD z-score, changes, persistence, acceleration and percentile with past-only windows. Fit curves/betas/normalizers only inside training windows and version them.
- A11. State variables.** Join price confidence, information/toxicity, pressure completion, liquidity regime, volatility, event age and requested size as separate columns before tuning interactions.
- A12. Leakage tests.** Append records arriving after decision time and prove historical feature values do not change. Compare live-available with enhanced-history fields and test shifted timestamps. Any failure blocks modelling.
- A13. Measurement tests.** Recompute with fresh quotes only, trades only, independent prices where possible, alternative stale thresholds and stricter confidence. Reject effects caused only by asynchronous updating.
- A14. Transparent baseline.** Fit horizon-specific linear/robust models in walk-forward folds before nonlinear models; record sign, stability, sample size and incremental information.
- A15. State/convergence models.** Estimate information versus temporary-pressure probabilities and convergence speed where meaningful. Cross-fit state probabilities used as features.
- A16. Execution simulator.** Request a new executable entry after the signal timestamp; apply latency, size, withdrawal, partial-fill and legging rules. No executable observation means unfilled, not a midpoint fill.
- A17. Fill/cost models.** Estimate fill probability and conditional spread/impact/fees/financing by size, protocol and state. Calibrate on held-out fills and cost every changed leg.
- A18. Net outcomes.** Store expected executable return conditional on fill, conservative lower quantile and capacity. Keep gross, mark, executable and fully-net outcomes as distinct audit columns.
- A19. Adversarial validation.** Use chronological purged folds, issuer holdouts, liquidity/rating/regime splits, event exclusions, alternative residualisations, placebo dates and within-family ablations.
- A20. Shadow mode.** Freeze versions and generate live predictions without trading. Log source timestamps, features, quotes, hypothetical orders, predicted fills/costs and 1d/5d/10d mark-outs.
- A21. Canary.** Start with strict size, concentration, liquidity and turnover limits. Kill automatically on timestamp anomalies, data gaps, feature drift, cost underprediction, unexplained residual beta or repeated adverse selection.
- A22. Promotion/rollback.** Promote only a fully reproducible version. Default to no-trade when health checks fail and retain one-command rollback to the prior approved model.

Incremental safe prompts. Use these as separate prompts/commits; each prompt may inspect prior artifacts but must not expand scope silently.

- P1.** “Inventory only the required data, schemas, keys, timestamps, live-history availability, entitlements, coverage and unresolved questions. Do not build a feature.”
- P2.** “Build and unit-test canonical identifiers and as-of joins. Return diagnostics and quarantined ambiguities. Do not calculate alpha.”
- P3.** “Build the price/analytics layer with freshness/confidence and reconcile a hand-audited sample. Stop on unexplained discrepancies.”
- P4.** “Implement only the literal raw signal with assertions and 20 hand-reconciled examples. Do not optimize parameters.”

- P5.** “Add past-only transforms and state variables. Run leakage and measurement-artifact tests before fitting any predictive model.”
- P6.** “Fit the transparent walk-forward baseline for 5d/10d. Report coefficients, uncertainty, ablations and failures; do not tune on the final test period.”
- P7.** “Add execution, fill and transaction-cost simulation. Recompute conditional executable net outcomes and mark unavailable opportunities unfilled.”
- P8.** “Run adversarial validation and within-family ablations. Produce a go/no-go research recommendation based only on predeclared gates.”
- P9.** “Run shadow mode with frozen versions. Do not send orders. Compare predicted versus observable fills, costs and mark-outs.”
- P10.** “Only after explicit human approval, create a tightly limited canary configuration with hard risk limits, health checks and rollback; do not broaden limits automatically.”

7.8 Production row and acceptance gate

Persist raw score, z-score, event age, quote age, peer count/effective sample size, confidence, state probabilities, size/ADV, expected spread/impact, issuer/sector/rating/tenor, KRD/CS01 and source timestamps. The feature graduates only if it adds incremental out-of-sample information after other members of its economic family, survives alternative residualisations and retains a positive conservative executable-net-return statistic at realistic capacity.

8 Signal 07: CDS-to-cash catch-up, adjusted for basis

8.1 Mathematics and feature construction

The canonical raw score is

$$G_{i,t} = s_{i,t}^{cash} - \phi(CDS_{g,t}, T_i, R) - \hat{b}_{i,t}. \quad (25)$$

Retain level, one- and three-day change, persistence, acceleration, trailing percentile and robust z-score. Add interactions with $c_{i,t}$, $p_{i,t}^{info}$, $p_{i,t}^{complete}$, liquidity regime, volatility and requested size. Thresholds and normalisation are learned on training data only. Missingness is explicit.

8.2 Point-in-time implementation

Maintain an immutable as-of ledger with economic, publication and receipt timestamps, corrections/cancels, security-master version and entitlements. At each decision time freeze the universe; resolve terms and corporate actions; construct fresh prices; compute peers/factors from earlier data; calculate the feature; obtain or simulate a fresh executable entry; and persist the feature snapshot. A confidence-weighted price can use

$$w_q = e^{-age_q/\tau_a} \min(1, size_q/v_0) w_d e^{-dispersion_q/\tau_\sigma}, \quad P^* = \frac{\sum_q w_q P_q}{\sum_q w_q}.$$

Never use future prints through symmetric smoothing.

8.3 Five- and ten-day forecast

Benchmark each horizon with

$$y_{i,t,H}^{exec} = a_H + b_H z + d_H z c + e_H z p^{info} + f_H z p^{complete} + u_{i,t,H}.$$

For dislocations estimate $D_{t+h} = D_t e^{-\kappa h} + \epsilon$ and expected capture $D_t(1 - e^{-\kappa H})$; for information diffusion use a bounded catch-up fraction. This gives an economic mapping from state to the 5d/10d horizon.

8.4 Execution and costs

Estimate jointly

$$p^{fill} = \mathbb{P}(fill \mid X, size, protocol), \quad (26)$$

$$C = g(size, spread, ADV, depth, vol, flow, toxicity, protocol), \quad (27)$$

$$\mu_H = \mathbb{E}[y_H^{exec} \mid fill, X]. \quad (28)$$

Use a fresh executable entry, never the historical print that revealed the signal. Model quote withdrawal and adverse selection. Relative-value packages pay every leg. Capacity is evaluated by rerunning net P&L at increasing requested size and participation constraints.

8.5 Falsification and monitoring

Kill the feature if it disappears with fresh independent prices, relies on unavailable data, is concentrated in a few issuers/events, or loses its edge under realistic fills and costs. Run leave-one-issuer-out tests, IG/HY and liquidity splits, calm/stress regimes, protocol splits and alternative residualisations. Monitor drift, price confidence, fill calibration, realised versus predicted costs, mark-outs and residual factor exposure.

8.6 Signal-specific mechanism and implementation

Data: Licensed CDS curve + cash bond quotes | Target: Base; relative version possible | 5-day emphasis; test 10 gap_i = cash_spread_i - comparable_CDS_spread_i - expected_basis_i Mechanism and evidence. Historical

investment-grade evidence finds an important CDS price-discovery role. The implementable hypothesis is delayed cash adjustment after a new CDS innovation, not guaranteed convergence of the entire cash–CDS basis in ten days. The source also highlights frictions that separate the instruments economically. Feature and implementation. Map each bond to the correct reference entity and contract seniority, currency, restructuring terms and maturity. Convert CDS quotes through a consistent hazard/recovery and cash-flow framework, or use a carefully calibrated comparable-spread mapping. Track coupon, discounting, deliverability and known restructuring differences. Estimate the persistent basis conditional on financing, liquidity and bond characteristics. Construct both the adjusted level gap and the recent CDS innovation not yet reflected in fresh cash prices. Positive cash spread relative to a valid comparison indicates a potential long-price catch-up, but only if the dynamic coefficient supports narrowing. Five-/ten-day forecast and hedge. Fit error-correction and lagged-innovation terms with separate five-/ten-day outcomes. Estimate which market leads by issuer and regime rather than imposing the 2004/2005 sample’s result forever. A large stable basis with no estimated short-horizon adjustment receives little forecast. Test base residual returns first; a sibling-relative implementation can isolate different cash-bond adjustment speeds after removing issuer-wide credit moves. Executable entry and costs. A cash-bond purchase hedged with purchased CDS protection is not costless or perfectly neutral. Include CDS bid–ask/upfront costs, premium accrual, financing, roll and cash/derivative settlement or jump-to-default mismatch. Price an index hedge alternative and its basis risk where single-name protection is not practical. Confirm quotes are executable at relevant size rather than dealer indications. Both legs must pass the common uncertainty and liquidity gate. Falsification / kill test. Do not equate an OAS number directly with a CDS quote without a validated convention mapping. Veto deliverability changes, restructuring disputes, stale single-name CDS quotes and balance-sheet stress that can sustain a basis. Require incremental information beyond signal 05. Kill the five-/ten-day thesis if all apparent gains require holding the basis for months or financing terms unavailable to the desk.

8.7 Agent implementation runbook: data discovery to safe production

Operating rule. The agent must advance through auditable gates rather than implement the entire alpha in one pass. If a required field is unavailable, ambiguous, delayed, or not historically point-in-time, stop that branch and record the limitation; never silently substitute a convenient proxy.

- A1. Data discovery.** Inventory candidate sources/tables/files; definitions; keys; units; time zones; economic, publication and receipt timestamps; corrections/cancels; historical coverage; entitlements; delayed-history rules; and whether each field was genuinely observable live. Output schema and coverage diagnostics only.
- A2. Raw profiling.** Report rows, bonds, issuers, days, missingness, duplicates, timestamp inversions, corrections, size distributions and join success by IG/HY, liquidity, sector and period. Do not impute yet.
- A3. Canonical identifiers.** Version the security master and map security IDs to legal obligor, parent, guarantee, seniority, currency, coupon, maturity, option schedule and corporate actions. Quarantine ambiguous/time-varying mappings.
- A4. As-of layer.** Require every query to accept a decision timestamp and return only records then available. Unit-test late records, corrections and cancels. Persist available-at and provenance fields.
- A5. Price layer.** Normalize clean/dirty price, accrued, spread convention, side, size and quote status. Create freshness, executable-size, dispersion and dealer-breadth metadata. Any forward fill must explicitly age and lose confidence.
- A6. Analytics layer.** Produce cash flows, yield/OAS as appropriate, spread duration, convexity, KRD, CS01/DV01, carry and roll from point-in-time terms; reconcile a sample against an independent implementation.
- A7. Targets.** Build 5d/10d efficient-price diagnostic and executable-return labels. Estimate residualisation only on prior data; cost every changed hedge leg, fees, financing and impact. Persist label maturity for purging.
- A8. Signal-specific prerequisites.** Execute the following in order, emitting exclusions, distributions and hand-audited examples after each sub-step:
 - a) Discover CDS curves/quotes, contract/restructuring convention, reference-entity mapping, recovery assumptions and fresh cash prices.
 - b) Map CDS reference entity/deliverable obligations to the bond legal exposure; quarantine successor/guarantee ambiguities.

- c) Convert CDS to a maturity-matched credit measure under explicit recovery/convention assumptions and estimate slow basis on prior data.
- d) Form unexpected basis innovation, separating funding/liquidity/contract effects from information-driven CDS moves.
- e) Reprice cash and CDS hedge at executable sides with upfront/running conversion, accrual, financing and bid/offer handled consistently.
- f) Stress recovery, interpolation and basis windows; require transient catch-up rather than permanent basis to explain returns.

- A9. Literal raw feature.** Translate the documented equation exactly. Assert units, sign, finite values, peer counts, timestamp ordering and boundaries. Hand-reconcile at least 20 events.
- A10. Causal transforms.** Add rolling median/MAD z-score, changes, persistence, acceleration and percentile with past-only windows. Fit curves/betas/normalizers only inside training windows and version them.
- A11. State variables.** Join price confidence, information/toxicity, pressure completion, liquidity regime, volatility, event age and requested size as separate columns before tuning interactions.
- A12. Leakage tests.** Append records arriving after decision time and prove historical feature values do not change. Compare live-available with enhanced-history fields and test shifted timestamps. Any failure blocks modelling.
- A13. Measurement tests.** Recompute with fresh quotes only, trades only, independent prices where possible, alternative stale thresholds and stricter confidence. Reject effects caused only by asynchronous updating.
- A14. Transparent baseline.** Fit horizon-specific linear/robust models in walk-forward folds before nonlinear models; record sign, stability, sample size and incremental information.
- A15. State/convergence models.** Estimate information versus temporary-pressure probabilities and convergence speed where meaningful. Cross-fit state probabilities used as features.
- A16. Execution simulator.** Request a new executable entry after the signal timestamp; apply latency, size, withdrawal, partial-fill and legging rules. No executable observation means unfilled, not a midpoint fill.
- A17. Fill/cost models.** Estimate fill probability and conditional spread/impact/fees/financing by size, protocol and state. Calibrate on held-out fills and cost every changed leg.
- A18. Net outcomes.** Store expected executable return conditional on fill, conservative lower quantile and capacity. Keep gross, mark, executable and fully-net outcomes as distinct audit columns.
- A19. Adversarial validation.** Use chronological purged folds, issuer holdouts, liquidity/rating/regime splits, event exclusions, alternative residualisations, placebo dates and within-family ablations.
- A20. Shadow mode.** Freeze versions and generate live predictions without trading. Log source timestamps, features, quotes, hypothetical orders, predicted fills/costs and 1d/5d/10d mark-outs.
- A21. Canary.** Start with strict size, concentration, liquidity and turnover limits. Kill automatically on timestamp anomalies, data gaps, feature drift, cost underprediction, unexplained residual beta or repeated adverse selection.
- A22. Promotion/rollback.** Promote only a fully reproducible version. Default to no-trade when health checks fail and retain one-command rollback to the prior approved model.

Incremental safe prompts. Use these as separate prompts/commits; each prompt may inspect prior artifacts but must not expand scope silently.

- P1.** “Inventory only the required data, schemas, keys, timestamps, live-history availability, entitlements, coverage and unresolved questions. Do not build a feature.”
- P2.** “Build and unit-test canonical identifiers and as-of joins. Return diagnostics and quarantined ambiguities. Do not calculate alpha.”
- P3.** “Build the price/analytics layer with freshness/confidence and reconcile a hand-audited sample. Stop on unexplained discrepancies.”

- P4.** “Implement only the literal raw signal with assertions and 20 hand-reconciled examples. Do not optimize parameters.”
- P5.** “Add past-only transforms and state variables. Run leakage and measurement-artifact tests before fitting any predictive model.”
- P6.** “Fit the transparent walk-forward baseline for 5d/10d. Report coefficients, uncertainty, ablations and failures; do not tune on the final test period.”
- P7.** “Add execution, fill and transaction-cost simulation. Recompute conditional executable net outcomes and mark unavailable opportunities unfilled.”
- P8.** “Run adversarial validation and within-family ablations. Produce a go/no-go research recommendation based only on predeclared gates.”
- P9.** “Run shadow mode with frozen versions. Do not send orders. Compare predicted versus observable fills, costs and mark-outs.”
- P10.** “Only after explicit human approval, create a tightly limited canary configuration with hard risk limits, health checks and rollback; do not broaden limits automatically.”

8.8 Production row and acceptance gate

Persist raw score, z-score, event age, quote age, peer count/effective sample size, confidence, state probabilities, size/ADV, expected spread/impact, issuer/sector/rating/tenor, KRD/CS01 and source timestamps. The feature graduates only if it adds incremental out-of-sample information after other members of its economic family, survives alternative residualisations and retains a positive conservative executable-net-return statistic at realistic capacity.

9 Signal 08: ETF credit-price discovery lead-lag

9.1 Mathematics and feature construction

The canonical raw score is

$$G_{i,t}^{ETF} = \lambda_i \varepsilon_t^{ETF} - \varepsilon_{i,t}^{bond}. \quad (29)$$

Retain level, one- and three-day change, persistence, acceleration, trailing percentile and robust z-score. Add interactions with $c_{i,t}$, $p_{i,t}^{info}$, $p_{i,t}^{complete}$, liquidity regime, volatility and requested size. Thresholds and normalisation are learned on training data only. Missingness is explicit.

9.2 Point-in-time implementation

Maintain an immutable as-of ledger with economic, publication and receipt timestamps, corrections/cancels, security-master version and entitlements. At each decision time freeze the universe; resolve terms and corporate actions; construct fresh prices; compute peers/factors from earlier data; calculate the feature; obtain or simulate a fresh executable entry; and persist the feature snapshot. A confidence-weighted price can use

$$w_q = e^{-age_q/\tau_a} \min(1, size_q/v_0) w_d e^{-dispersion_q/\tau_\sigma}, \quad P^* = \frac{\sum_q w_q P_q}{\sum_q w_q}.$$

Never use future prints through symmetric smoothing.

9.3 Five- and ten-day forecast

Benchmark each horizon with

$$y_{i,t,H}^{exec} = a_H + b_H z + d_H z c + e_H z p^{info} + f_H z p^{complete} + u_{i,t,H}.$$

For dislocations estimate $D_{t+h} = D_t e^{-\kappa h} + \epsilon$ and expected capture $D_t(1 - e^{-\kappa H})$; for information diffusion use a bounded catch-up fraction. This gives an economic mapping from state to the 5d/10d horizon.

9.4 Execution and costs

Estimate jointly

$$p^{fill} = \mathbb{P}(fill \mid X, size, protocol), \quad (30)$$

$$C = g(size, spread, ADV, depth, vol, flow, toxicity, protocol), \quad (31)$$

$$\mu_H = \mathbb{E}[y_H^{exec} \mid fill, X]. \quad (32)$$

Use a fresh executable entry, never the historical print that revealed the signal. Model quote withdrawal and adverse selection. Relative-value packages pay every leg. Capacity is evaluated by rerunning net P&L at increasing requested size and participation constraints.

9.5 Falsification and monitoring

Kill the feature if it disappears with fresh independent prices, relies on unavailable data, is concentrated in a few issuers/events, or loses its edge under realistic fills and costs. Run leave-one-issuer-out tests, IG/HY and liquidity splits, calm/stress regimes, protocol splits and alternative residualisations. Monitor drift, price confidence, fill calibration, realised versus predicted costs, mark-outs and residual factor exposure.

9.6 Signal-specific mechanism and implementation

Data: ETF prices + holdings/risk + bond quotes | Target: Base; much may be factor exposure | 5-day test; 10-day secondary ETF_gap_i = exposure_i * ETF_credit_innovation - bond_credit_innovation_i Mechanism and evidence.

An ETF can update faster than the valuation marks of the bonds it holds. That observation does not make a discount to reported NAV an arbitrage signal: bond baskets, liquidity and valuation timing differ. The proposed alpha is a remaining bond-level information lag, not a mechanical ETF premium trade. Feature and implementation. Remove Treasury, cash-distribution and broad-market effects from timestamp-matched ETF returns. Map the remaining credit innovation to a bond using a holdings-aware issuer/sector/rating/tenor exposure model. Compare that implied movement with the bond's actually refreshed price movement. Prefer several relevant ETFs and penalise inconsistent signals. Do not use an end-of-day NAV or holdings file before publication. A positive ETF-led price innovation not incorporated in the bond is a candidate long, subject to fresh confirmation. Keep the ETF's own flow-driven pricing component separate from information about underlying credit. Five-/ten-day forecast and hedge. Evaluate the residual effect over five and ten days after the executable quote is obtained. Much price discovery may complete inside the day, leaving no weekly opportunity. Broad credit-factor residualisation can remove most of this feature by design; a remaining score must reflect bond-specific lag or mismatched investor demand, not an omitted ETF/index beta. Consider the feature first as a stale-mark detector and risk veto before treating it as a standalone alpha. Executable entry and costs. Hedge using a liquid credit instrument only after checking tracking error and costs. Never book profit from buying a stale cash mark against a live ETF price. Simulate ETF hedge slippage, bond RFQ latency and the possibility that the bond quote updates before a fill. Show net results conditional on successful fills and report the opportunities that disappear upon inquiry. Falsification / kill test. Kill the signal if it works only against stale NAV/evaluated prices, in broad risk-on periods, or before removing rate/credit beta. Test incremental value against sibling and CDS leadership. Stress periods can generate persistent ETF/bond valuation differences, and authorised-participant basket economics need not be accessible to a generic trading strategy. Distinguish this information feature from basket-specific flow pressure in signal 16.

9.7 Agent implementation runbook: data discovery to safe production

Operating rule. The agent must advance through auditable gates rather than implement the entire alpha in one pass. If a required field is unavailable, ambiguous, delayed, or not historically point-in-time, stop that branch and record the limitation; never silently substitute a convenient proxy.

- A1. Data discovery.** Inventory candidate sources/tables/files; definitions; keys; units; time zones; economic, publication and receipt timestamps; corrections/cancels; historical coverage; entitlements; delayed-history rules; and whether each field was genuinely observable live. Output schema and coverage diagnostics only.
- A2. Raw profiling.** Report rows, bonds, issuers, days, missingness, duplicates, timestamp inversions, corrections, size distributions and join success by IG/HY, liquidity, sector and period. Do not impute yet.
- A3. Canonical identifiers.** Version the security master and map security IDs to legal obligor, parent, guarantee, seniority, currency, coupon, maturity, option schedule and corporate actions. Quarantine ambiguous/time-varying mappings.
- A4. As-of layer.** Require every query to accept a decision timestamp and return only records then available. Unit-test late records, corrections and cancels. Persist available-at and provenance fields.
- A5. Price layer.** Normalize clean/dirty price, accrued, spread convention, side, size and quote status. Create freshness, executable-size, dispersion and dealer-breadth metadata. Any forward fill must explicitly age and lose confidence.
- A6. Analytics layer.** Produce cash flows, yield/OAS as appropriate, spread duration, convexity, KRD, CS01/DV01, carry and roll from point-in-time terms; reconcile a sample against an independent implementation.
- A7. Targets.** Build 5d/10d efficient-price diagnostic and executable-return labels. Estimate residualisation only on prior data; cost every changed hedge leg, fees, financing and impact. Persist label maturity for purging.
- A8. Signal-specific prerequisites.** Execute the following in order, emitting exclusions, distributions and hand-audited examples after each sub-step:
 - a) Discover bond ETFs, intraday ETF prices/NAV proxies, creation-redemption baskets, holdings, flows and bond mappings with publication times.
 - b) Use actual basket membership when available; do not equate fund holdings with the creation/redemption basket.

- c) Extract ETF credit innovations after rate/equity removal and estimate bond sensitivity on prior data with shrinkage.
- d) Measure missing bond response only when ETF information precedes the cash-bond update and exposure is economically relevant.
- e) Refresh cash executable quote and include any ETF/hedge costs; reject stale-NAV and asynchronous-close artifacts.
- f) Ablate ETF choice, basket membership, premium/discount state and timing; require incremental value beyond CDS/equity/issuer leaders.

- A9. Literal raw feature.** Translate the documented equation exactly. Assert units, sign, finite values, peer counts, timestamp ordering and boundaries. Hand-reconcile at least 20 events.
- A10. Causal transforms.** Add rolling median/MAD z-score, changes, persistence, acceleration and percentile with past-only windows. Fit curves/betas/normalizers only inside training windows and version them.
- A11. State variables.** Join price confidence, information/toxicity, pressure completion, liquidity regime, volatility, event age and requested size as separate columns before tuning interactions.
- A12. Leakage tests.** Append records arriving after decision time and prove historical feature values do not change. Compare live-available with enhanced-history fields and test shifted timestamps. Any failure blocks modelling.
- A13. Measurement tests.** Recompute with fresh quotes only, trades only, independent prices where possible, alternative stale thresholds and stricter confidence. Reject effects caused only by asynchronous updating.
- A14. Transparent baseline.** Fit horizon-specific linear/robust models in walk-forward folds before nonlinear models; record sign, stability, sample size and incremental information.
- A15. State/convergence models.** Estimate information versus temporary-pressure probabilities and convergence speed where meaningful. Cross-fit state probabilities used as features.
- A16. Execution simulator.** Request a new executable entry after the signal timestamp; apply latency, size, withdrawal, partial-fill and legging rules. No executable observation means unfilled, not a midpoint fill.
- A17. Fill/cost models.** Estimate fill probability and conditional spread/impact/fees/financing by size, protocol and state. Calibrate on held-out fills and cost every changed leg.
- A18. Net outcomes.** Store expected executable return conditional on fill, conservative lower quantile and capacity. Keep gross, mark, executable and fully-net outcomes as distinct audit columns.
- A19. Adversarial validation.** Use chronological purged folds, issuer holdouts, liquidity/rating/regime splits, event exclusions, alternative residualisations, placebo dates and within-family ablations.
- A20. Shadow mode.** Freeze versions and generate live predictions without trading. Log source timestamps, features, quotes, hypothetical orders, predicted fills/costs and 1d/5d/10d mark-outs.
- A21. Canary.** Start with strict size, concentration, liquidity and turnover limits. Kill automatically on timestamp anomalies, data gaps, feature drift, cost underprediction, unexplained residual beta or repeated adverse selection.
- A22. Promotion/rollback.** Promote only a fully reproducible version. Default to no-trade when health checks fail and retain one-command rollback to the prior approved model.

Incremental safe prompts. Use these as separate prompts/commits; each prompt may inspect prior artifacts but must not expand scope silently.

- P1.** “Inventory only the required data, schemas, keys, timestamps, live-history availability, entitlements, coverage and unresolved questions. Do not build a feature.”
- P2.** “Build and unit-test canonical identifiers and as-of joins. Return diagnostics and quarantined ambiguities. Do not calculate alpha.”
- P3.** “Build the price/analytics layer with freshness/confidence and reconcile a hand-audited sample. Stop on unexplained discrepancies.”

- P4.** “Implement only the literal raw signal with assertions and 20 hand-reconciled examples. Do not optimize parameters.”
- P5.** “Add past-only transforms and state variables. Run leakage and measurement-artifact tests before fitting any predictive model.”
- P6.** “Fit the transparent walk-forward baseline for 5d/10d. Report coefficients, uncertainty, ablations and failures; do not tune on the final test period.”
- P7.** “Add execution, fill and transaction-cost simulation. Recompute conditional executable net outcomes and mark unavailable opportunities unfilled.”
- P8.** “Run adversarial validation and within-family ablations. Produce a go/no-go research recommendation based only on predeclared gates.”
- P9.** “Run shadow mode with frozen versions. Do not send orders. Compare predicted versus observable fills, costs and mark-outs.”
- P10.** “Only after explicit human approval, create a tightly limited canary configuration with hard risk limits, health checks and rollback; do not broaden limits automatically.”

9.8 Production row and acceptance gate

Persist raw score, z-score, event age, quote age, peer count/effective sample size, confidence, state probabilities, size/ADV, expected spread/impact, issuer/sector/rating/tenor, KRD/CS01 and source timestamps. The feature graduates only if it adds incremental out-of-sample information after other members of its economic family, survives alternative residualisations and retains a positive conservative executable-net-return statistic at realistic capacity.

10 Signal 09: De-noised short-horizon reversal

10.1 Mathematics and feature construction

The canonical raw score is

$$REV_{i,t} = -\hat{u}_{i,t-k:t}^{efficient}(1 - \hat{p}_{i,t}^{info}). \quad (33)$$

Retain level, one- and three-day change, persistence, acceleration, trailing percentile and robust z-score. Add interactions with $c_{i,t}$, $p_{i,t}^{info}$, $p_{i,t}^{complete}$, liquidity regime, volatility and requested size. Thresholds and normalisation are learned on training data only. Missingness is explicit.

10.2 Point-in-time implementation

Maintain an immutable as-of ledger with economic, publication and receipt timestamps, corrections/cancels, security-master version and entitlements. At each decision time freeze the universe; resolve terms and corporate actions; construct fresh prices; compute peers/factors from earlier data; calculate the feature; obtain or simulate a fresh executable entry; and persist the feature snapshot. A confidence-weighted price can use

$$w_q = e^{-age_q/\tau_a} \min(1, size_q/v_0) w_d e^{-dispersion_q/\tau_\sigma}, \quad P^* = \frac{\sum_q w_q P_q}{\sum_q w_q}.$$

Never use future prints through symmetric smoothing.

10.3 Five- and ten-day forecast

Benchmark each horizon with

$$y_{i,t,H}^{exec} = a_H + b_H z + d_H zc + e_H z p^{info} + f_H z p^{complete} + u_{i,t,H}.$$

For dislocations estimate $D_{t+h} = D_t e^{-\kappa h} + \epsilon$ and expected capture $D_t(1 - e^{-\kappa H})$; for information diffusion use a bounded catch-up fraction. This gives an economic mapping from state to the 5d/10d horizon.

10.4 Execution and costs

Estimate jointly

$$p^{fill} = \mathbb{P}(fill \mid X, size, protocol), \quad (34)$$

$$C = g(size, spread, ADV, depth, vol, flow, toxicity, protocol), \quad (35)$$

$$\mu_H = \mathbb{E}[y_H^{exec} \mid fill, X]. \quad (36)$$

Use a fresh executable entry, never the historical print that revealed the signal. Model quote withdrawal and adverse selection. Relative-value packages pay every leg. Capacity is evaluated by rerunning net P&L at increasing requested size and participation constraints.

10.5 Falsification and monitoring

Kill the feature if it disappears with fresh independent prices, relies on unavailable data, is concentrated in a few issuers/events, or loses its edge under realistic fills and costs. Run leave-one-issuer-out tests, IG/HY and liquidity splits, calm/stress regimes, protocol splits and alternative residualisations. Monitor drift, price confidence, fill calibration, realised versus predicted costs, mark-outs and residual factor exposure.

10.6 Signal-specific mechanism and implementation

Data: Trades + side inference + fresh quotes | Target: Base and issuer-neutral | 5-day emphasis; test 10 reversal_i = -sum(past_1_to_5_day_latent_residual_returns_i) Mechanism and evidence. Observed short-term reversals can mix

genuine temporary impact with bid–ask bounce and bad marks. Published illiquidity and transient-impact research motivates separating those components. Neither negative autocovariance nor a fitted impact kernel proves a tradable multi-day return. Feature and implementation. Construct a causal estimate of the efficient price using side-adjusted trades and fresh quotes, with transaction-size, age and venue-dependent measurement noise. Use a one-sided filter, not a full-sample smoother that incorporates future trades. Remove carry, rates and chosen credit factors before aggregating one-, three- and five-day residual moves. Keep a separate observation-noise score. A negative latent residual move supplies a positive reversal feature only when independent prices confirm the dislocation and public issuer news does not explain it. Estimate buy- and sell-pressure responses separately rather than assuming symmetry. Five-/ten-day forecast and hedge. Fit five- and ten-day response functions, with state interactions for quote depth, volatility and persistent-flow probability. Use a restricted baseline first: a single negative lagged residual return and a measurement-quality gate. Estimate how much of the move has already reversed by actual entry. Longer horizons can dilute a short-lived bounce while accumulating more risk; do not add the same predicted rebound twice to the ten-day target. Executable entry and costs. A low customer-sale print is not an available ask. Recompute the signal from the price at which the strategy can buy and from a realistic future exit side. Apply the common cost gate, including the hedge and conditional market impact. Prefer situations where the concession is still quoted and liquidity has begun to recover. Compare with the economics of improving an existing portfolio’s rebalance rather than assuming a new financed round trip is always necessary. Falsification / kill test. Run the strongest negative control: calculate the score using same-side matched trades or independent fresh quote midpoints. Kill it if the effect disappears once bid–ask bounce is removed, if it depends on forward smoothing, or if losses concentrate in genuine issuer deterioration. Ablate against order-flow reversal, block concessions and curve value; these may be different measurements of the same temporary shock.

10.7 Agent implementation runbook: data discovery to safe production

Operating rule. The agent must advance through auditable gates rather than implement the entire alpha in one pass. If a required field is unavailable, ambiguous, delayed, or not historically point-in-time, stop that branch and record the limitation; never silently substitute a convenient proxy.

- A1. Data discovery.** Inventory candidate sources/tables/files; definitions; keys; units; time zones; economic, publication and receipt timestamps; corrections/cancels; historical coverage; entitlements; delayed-history rules; and whether each field was genuinely observable live. Output schema and coverage diagnostics only.
- A2. Raw profiling.** Report rows, bonds, issuers, days, missingness, duplicates, timestamp inversions, corrections, size distributions and join success by IG/HY, liquidity, sector and period. Do not impute yet.
- A3. Canonical identifiers.** Version the security master and map security IDs to legal obligor, parent, guarantee, seniority, currency, coupon, maturity, option schedule and corporate actions. Quarantine ambiguous/time-varying mappings.
- A4. As-of layer.** Require every query to accept a decision timestamp and return only records then available. Unit-test late records, corrections and cancels. Persist available-at and provenance fields.
- A5. Price layer.** Normalize clean/dirty price, accrued, spread convention, side, size and quote status. Create freshness, executable-size, dispersion and dealer-breadth metadata. Any forward fill must explicitly age and lose confidence.
- A6. Analytics layer.** Produce cash flows, yield/OAS as appropriate, spread duration, convexity, KRD, CS01/DV01, carry and roll from point-in-time terms; reconcile a sample against an independent implementation.
- A7. Targets.** Build 5d/10d efficient-price diagnostic and executable-return labels. Estimate residualisation only on prior data; cost every changed hedge leg, fees, financing and impact. Persist label maturity for purging.
- A8. Signal-specific prerequisites.** Execute the following in order, emitting exclusions, distributions and hand-audited examples after each sub-step:
 - a) Discover prints and executable quote histories with timestamps sufficient to separate bid–ask bounce from efficient moves.
 - b) Build a causal efficient-price filter and validate against later independent quotes without using them in construction.
 - c) Compute recent residual efficient-price innovation rather than last-trade return.

- d) Apply reversal only in low-information states after controlling issuer/market shocks.
- e) Require a new executable entry after the move; never assume access to the initiating print.
- f) Run stale-price/placebo reversals; the effect must strengthen, not weaken, under high-confidence observations.

- A9. Literal raw feature.** Translate the documented equation exactly. Assert units, sign, finite values, peer counts, timestamp ordering and boundaries. Hand-reconcile at least 20 events.
- A10. Causal transforms.** Add rolling median/MAD z-score, changes, persistence, acceleration and percentile with past-only windows. Fit curves/betas/normalizers only inside training windows and version them.
- A11. State variables.** Join price confidence, information/toxicity, pressure completion, liquidity regime, volatility, event age and requested size as separate columns before tuning interactions.
- A12. Leakage tests.** Append records arriving after decision time and prove historical feature values do not change. Compare live-available with enhanced-history fields and test shifted timestamps. Any failure blocks modelling.
- A13. Measurement tests.** Recompute with fresh quotes only, trades only, independent prices where possible, alternative stale thresholds and stricter confidence. Reject effects caused only by asynchronous updating.
- A14. Transparent baseline.** Fit horizon-specific linear/robust models in walk-forward folds before nonlinear models; record sign, stability, sample size and incremental information.
- A15. State/convergence models.** Estimate information versus temporary-pressure probabilities and convergence speed where meaningful. Cross-fit state probabilities used as features.
- A16. Execution simulator.** Request a new executable entry after the signal timestamp; apply latency, size, withdrawal, partial-fill and legging rules. No executable observation means unfilled, not a midpoint fill.
- A17. Fill/cost models.** Estimate fill probability and conditional spread/impact/fees/financing by size, protocol and state. Calibrate on held-out fills and cost every changed leg.
- A18. Net outcomes.** Store expected executable return conditional on fill, conservative lower quantile and capacity. Keep gross, mark, executable and fully-net outcomes as distinct audit columns.
- A19. Adversarial validation.** Use chronological purged folds, issuer holdouts, liquidity/rating/regime splits, event exclusions, alternative residualisations, placebo dates and within-family ablations.
- A20. Shadow mode.** Freeze versions and generate live predictions without trading. Log source timestamps, features, quotes, hypothetical orders, predicted fills/costs and 1d/5d/10d mark-outs.
- A21. Canary.** Start with strict size, concentration, liquidity and turnover limits. Kill automatically on timestamp anomalies, data gaps, feature drift, cost underprediction, unexplained residual beta or repeated adverse selection.
- A22. Promotion/rollback.** Promote only a fully reproducible version. Default to no-trade when health checks fail and retain one-command rollback to the prior approved model.

Incremental safe prompts. Use these as separate prompts/commits; each prompt may inspect prior artifacts but must not expand scope silently.

- P1.** “Inventory only the required data, schemas, keys, timestamps, live-history availability, entitlements, coverage and unresolved questions. Do not build a feature.”
- P2.** “Build and unit-test canonical identifiers and as-of joins. Return diagnostics and quarantined ambiguities. Do not calculate alpha.”
- P3.** “Build the price/analytics layer with freshness/confidence and reconcile a hand-audited sample. Stop on unexplained discrepancies.”
- P4.** “Implement only the literal raw signal with assertions and 20 hand-reconciled examples. Do not optimize parameters.”
- P5.** “Add past-only transforms and state variables. Run leakage and measurement-artifact tests before fitting any predictive model.”

- P6.** “Fit the transparent walk-forward baseline for 5d/10d. Report coefficients, uncertainty, ablations and failures; do not tune on the final test period.”
- P7.** “Add execution, fill and transaction-cost simulation. Recompute conditional executable net outcomes and mark unavailable opportunities unfilled.”
- P8.** “Run adversarial validation and within-family ablations. Produce a go/no-go research recommendation based only on predeclared gates.”
- P9.** “Run shadow mode with frozen versions. Do not send orders. Compare predicted versus observable fills, costs and mark-outs.”
- P10.** “Only after explicit human approval, create a tightly limited canary configuration with hard risk limits, health checks and rollback; do not broaden limits automatically.”

10.8 Production row and acceptance gate

Persist raw score, z-score, event age, quote age, peer count/effective sample size, confidence, state probabilities, size/ADV, expected spread/impact, issuer/sector/rating/tenor, KRD/CS01 and source timestamps. The feature graduates only if it adds incremental out-of-sample information after other members of its economic family, survives alternative residualisations and retains a positive conservative executable-net-return statistic at realistic capacity.

11 Signal 10: Toxicity-gated order-flow reversal

11.1 Mathematics and feature construction

The canonical raw score is

$$OF_{i,t}^{rev} = -FlowShock_{i,t}(1 - p_{i,t}^{info})p_{i,t}^{complete}. \quad (37)$$

Retain level, one- and three-day change, persistence, acceleration, trailing percentile and robust z-score. Add interactions with $c_{i,t}$, $p_{i,t}^{info}$, $p_{i,t}^{complete}$, liquidity regime, volatility and requested size. Thresholds and normalisation are learned on training data only. Missingness is explicit.

11.2 Point-in-time implementation

Maintain an immutable as-of ledger with economic, publication and receipt timestamps, corrections/cancels, security-master version and entitlements. At each decision time freeze the universe; resolve terms and corporate actions; construct fresh prices; compute peers/factors from earlier data; calculate the feature; obtain or simulate a fresh executable entry; and persist the feature snapshot. A confidence-weighted price can use

$$w_q = e^{-age_q/\tau_a} \min(1, size_q/v_0) w_d e^{-dispersion_q/\tau_\sigma}, \quad P^* = \frac{\sum_q w_q P_q}{\sum_q w_q}.$$

Never use future prints through symmetric smoothing.

11.3 Five- and ten-day forecast

Benchmark each horizon with

$$y_{i,t,H}^{exec} = a_H + b_H z + d_H zc + e_H z p^{info} + f_H z p^{complete} + u_{i,t,H}.$$

For dislocations estimate $D_{t+h} = D_t e^{-\kappa h} + \epsilon$ and expected capture $D_t(1 - e^{-\kappa H})$; for information diffusion use a bounded catch-up fraction. This gives an economic mapping from state to the 5d/10d horizon.

11.4 Execution and costs

Estimate jointly

$$p^{fill} = \mathbb{P}(fill \mid X, size, protocol), \quad (38)$$

$$C = g(size, spread, ADV, depth, vol, flow, toxicity, protocol), \quad (39)$$

$$\mu_H = \mathbb{E}[y_H^{exec} \mid fill, X]. \quad (40)$$

Use a fresh executable entry, never the historical print that revealed the signal. Model quote withdrawal and adverse selection. Relative-value packages pay every leg. Capacity is evaluated by rerunning net P&L at increasing requested size and participation constraints.

11.5 Falsification and monitoring

Kill the feature if it disappears with fresh independent prices, relies on unavailable data, is concentrated in a few issuers/events, or loses its edge under realistic fills and costs. Run leave-one-issuer-out tests, IG/HY and liquidity splits, calm/stress regimes, protocol splits and alternative residualisations. Monitor drift, price confidence, fill calibration, realised versus predicted costs, mark-outs and residual factor exposure.

11.6 Signal-specific mechanism and implementation

Data: Signed customer flow + quote/news state | Target: Base and issuer-neutral | 5 and 10 days flow = $\text{sum}(\text{sign_customer} * q_weight * age_weight)$; score = $-(1 - \text{toxicity}) * \text{flow}$ Mechanism and evidence. Liquidity-motivated

customer selling can leave dealers with inventory and depress prices temporarily. Customer selling can also carry genuine information. This signal reverses only flow classified, probabilistically and ex ante, as predominantly transitory. Inventory and impact research supports the distinction; the classifier and its thresholds are proposed. Feature and implementation. Define customer purchases as +1 and sales as -1, carefully inverting reporting-dealer side when required. Exclude or separately model interdealer activity and avoid double-counting riskless-principal chains. Aggregate one- to three-day signed notional with concave size weights, capped-print indicators and recency decay; divide by trailing typical volume or risk capacity. Build a toxicity estimate from previously matured outcomes, public event flags, cross-market agreement, quote revisions and whether selling persists across independent observations. A negative flow with low estimated toxicity gives a positive price-reversal score. Any chain classification must use only reports available at the decision. Five-/ten-day forecast and hedge. Calibrate separate five-/ten-day returns and an empirical completion variable: slowing sale intensity, stabilising bid depth or renewed two-sided quotes. Do not label a current flow using its future markout. Under issuer-neutralisation, measure the bond's imbalance relative to siblings to avoid merely buying issuer-wide deterioration. Fit continuation and reversal jointly as regime alternatives, not independently additive alphas with contradictory signs. Executable entry and costs. Buy only when an executable concession remains after the flow has begun to stabilise. The observed last sale and the available ask are different prices. Test passive and aggressive routes separately, conditioning expected return on fill; a passive fill can be disproportionately toxic. Require all-leg costs and uncertainty reserves to be covered at the chosen horizon and size. Falsification / kill test. Kill the feature if trade-side labels are unreliable, public caps are silently replaced with delayed true sizes, or a classifier uses future information. Treat broad persistent issuer selling plus negative CDS/equity news as a veto or continuation state. Show incremental net performance beyond simple reversal and dealer-inventory proxies. A volume imbalance without a price concession or a credible completion signal is not enough.

11.7 Agent implementation runbook: data discovery to safe production

Operating rule. The agent must advance through auditable gates rather than implement the entire alpha in one pass. If a required field is unavailable, ambiguous, delayed, or not historically point-in-time, stop that branch and record the limitation; never silently substitute a convenient proxy.

- A1. Data discovery.** Inventory candidate sources/tables/files; definitions; keys; units; time zones; economic, publication and receipt timestamps; corrections/cancels; historical coverage; entitlements; delayed-history rules; and whether each field was genuinely observable live. Output schema and coverage diagnostics only.
- A2. Raw profiling.** Report rows, bonds, issuers, days, missingness, duplicates, timestamp inversions, corrections, size distributions and join success by IG/HY, liquidity, sector and period. Do not impute yet.
- A3. Canonical identifiers.** Version the security master and map security IDs to legal obligor, parent, guarantee, seniority, currency, coupon, maturity, option schedule and corporate actions. Quarantine ambiguous/time-varying mappings.
- A4. As-of layer.** Require every query to accept a decision timestamp and return only records then available. Unit-test late records, corrections and cancels. Persist available-at and provenance fields.
- A5. Price layer.** Normalize clean/dirty price, accrued, spread convention, side, size and quote status. Create freshness, executable-size, dispersion and dealer-breadth metadata. Any forward fill must explicitly age and lose confidence.
- A6. Analytics layer.** Produce cash flows, yield/OAS as appropriate, spread duration, convexity, KRD, CS01/DV01, carry and roll from point-in-time terms; reconcile a sample against an independent implementation.
- A7. Targets.** Build 5d/10d efficient-price diagnostic and executable-return labels. Estimate residualisation only on prior data; cost every changed hedge leg, fees, financing and impact. Persist label maturity for purging.
- A8. Signal-specific prerequisites.** Execute the following in order, emitting exclusions, distributions and hand-audited examples after each sub-step:
 - a) Discover signed-flow/RFQ proxies, trade direction, price response and cross-market information sources.
 - b) Normalize flow shock by bond-specific historical activity, liquidity and requested size.
 - c) Train a cross-fitted information/toxicity classifier from sibling, CDS, equity and quote confirmation.
 - d) Estimate pressure completion/exhaustion separately from information toxicity.
 - e) Form reversal only when pressure is large, information probability low and completion probability high.

f) Model fill selection explicitly; easy fills after toxic flow are treated as adverse-selection evidence.

- A9. Literal raw feature.** Translate the documented equation exactly. Assert units, sign, finite values, peer counts, timestamp ordering and boundaries. Hand-reconcile at least 20 events.
- A10. Causal transforms.** Add rolling median/MAD z-score, changes, persistence, acceleration and percentile with past-only windows. Fit curves/betas/normalizers only inside training windows and version them.
- A11. State variables.** Join price confidence, information/toxicity, pressure completion, liquidity regime, volatility, event age and requested size as separate columns before tuning interactions.
- A12. Leakage tests.** Append records arriving after decision time and prove historical feature values do not change. Compare live-available with enhanced-history fields and test shifted timestamps. Any failure blocks modelling.
- A13. Measurement tests.** Recompute with fresh quotes only, trades only, independent prices where possible, alternative stale thresholds and stricter confidence. Reject effects caused only by asynchronous updating.
- A14. Transparent baseline.** Fit horizon-specific linear/robust models in walk-forward folds before nonlinear models; record sign, stability, sample size and incremental information.
- A15. State/convergence models.** Estimate information versus temporary-pressure probabilities and convergence speed where meaningful. Cross-fit state probabilities used as features.
- A16. Execution simulator.** Request a new executable entry after the signal timestamp; apply latency, size, withdrawal, partial-fill and legging rules. No executable observation means unfilled, not a midpoint fill.
- A17. Fill/cost models.** Estimate fill probability and conditional spread/impact/fees/financing by size, protocol and state. Calibrate on held-out fills and cost every changed leg.
- A18. Net outcomes.** Store expected executable return conditional on fill, conservative lower quantile and capacity. Keep gross, mark, executable and fully-net outcomes as distinct audit columns.
- A19. Adversarial validation.** Use chronological purged folds, issuer holdouts, liquidity/rating/regime splits, event exclusions, alternative residualisations, placebo dates and within-family ablations.
- A20. Shadow mode.** Freeze versions and generate live predictions without trading. Log source timestamps, features, quotes, hypothetical orders, predicted fills/costs and 1d/5d/10d mark-outs.
- A21. Canary.** Start with strict size, concentration, liquidity and turnover limits. Kill automatically on timestamp anomalies, data gaps, feature drift, cost underprediction, unexplained residual beta or repeated adverse selection.
- A22. Promotion/rollback.** Promote only a fully reproducible version. Default to no-trade when health checks fail and retain one-command rollback to the prior approved model.

Incremental safe prompts. Use these as separate prompts/commits; each prompt may inspect prior artifacts but must not expand scope silently.

- P1.** “Inventory only the required data, schemas, keys, timestamps, live-history availability, entitlements, coverage and unresolved questions. Do not build a feature.”
- P2.** “Build and unit-test canonical identifiers and as-of joins. Return diagnostics and quarantined ambiguities. Do not calculate alpha.”
- P3.** “Build the price/analytics layer with freshness/confidence and reconcile a hand-audited sample. Stop on unexplained discrepancies.”
- P4.** “Implement only the literal raw signal with assertions and 20 hand-reconciled examples. Do not optimize parameters.”
- P5.** “Add past-only transforms and state variables. Run leakage and measurement-artifact tests before fitting any predictive model.”
- P6.** “Fit the transparent walk-forward baseline for 5d/10d. Report coefficients, uncertainty, ablations and failures; do not tune on the final test period.”

- P7.** “Add execution, fill and transaction-cost simulation. Recompute conditional executable net outcomes and mark unavailable opportunities unfilled.”
- P8.** “Run adversarial validation and within-family ablations. Produce a go/no-go research recommendation based only on predeclared gates.”
- P9.** “Run shadow mode with frozen versions. Do not send orders. Compare predicted versus observable fills, costs and mark-outs.”
- P10.** “Only after explicit human approval, create a tightly limited canary configuration with hard risk limits, health checks and rollback; do not broaden limits automatically.”

11.8 Production row and acceptance gate

Persist raw score, z-score, event age, quote age, peer count/effective sample size, confidence, state probabilities, size/ADV, expected spread/impact, issuer/sector/rating/tenor, KRD/CS01 and source timestamps. The feature graduates only if it adds incremental out-of-sample information after other members of its economic family, survives alternative residualisations and retains a positive conservative executable-net-return statistic at realistic capacity.

12 Signal 11: Persistent informed-flow continuation

12.1 Mathematics and feature construction

The canonical raw score is

$$OF_{i,t}^{cont} = FlowShock_{i,t} p_{i,t}^{info} p_{i,t}^{persist}. \quad (41)$$

Retain level, one- and three-day change, persistence, acceleration, trailing percentile and robust z-score. Add interactions with $c_{i,t}$, $p_{i,t}^{info}$, $p_{i,t}^{complete}$, liquidity regime, volatility and requested size. Thresholds and normalisation are learned on training data only. Missingness is explicit.

12.2 Point-in-time implementation

Maintain an immutable as-of ledger with economic, publication and receipt timestamps, corrections/cancels, security-master version and entitlements. At each decision time freeze the universe; resolve terms and corporate actions; construct fresh prices; compute peers/factors from earlier data; calculate the feature; obtain or simulate a fresh executable entry; and persist the feature snapshot. A confidence-weighted price can use

$$w_q = e^{-age_q/\tau_a} \min(1, size_q/v_0) w_d e^{-dispersion_q/\tau_\sigma}, \quad P^* = \frac{\sum_q w_q P_q}{\sum_q w_q}.$$

Never use future prints through symmetric smoothing.

12.3 Five- and ten-day forecast

Benchmark each horizon with

$$y_{i,t,H}^{exec} = a_H + b_H z + d_H zc + e_H z p^{info} + f_H z p^{complete} + u_{i,t,H}.$$

For dislocations estimate $D_{t+h} = D_t e^{-\kappa h} + \epsilon$ and expected capture $D_t(1 - e^{-\kappa H})$; for information diffusion use a bounded catch-up fraction. This gives an economic mapping from state to the 5d/10d horizon.

12.4 Execution and costs

Estimate jointly

$$p^{fill} = \mathbb{P}(fill \mid X, size, protocol), \quad (42)$$

$$C = g(size, spread, ADV, depth, vol, flow, toxicity, protocol), \quad (43)$$

$$\mu_H = \mathbb{E}[y_H^{exec} \mid fill, X]. \quad (44)$$

Use a fresh executable entry, never the historical print that revealed the signal. Model quote withdrawal and adverse selection. Relative-value packages pay every leg. Capacity is evaluated by rerunning net P&L at increasing requested size and participation constraints.

12.5 Falsification and monitoring

Kill the feature if it disappears with fresh independent prices, relies on unavailable data, is concentrated in a few issuers/events, or loses its edge under realistic fills and costs. Run leave-one-issuer-out tests, IG/HY and liquidity splits, calm/stress regimes, protocol splits and alternative residualisations. Monitor drift, price confidence, fill calibration, realised versus predicted costs, mark-outs and residual factor exposure.

12.6 Signal-specific mechanism and implementation

Data: Permissioned completed flow + mature markouts | Target: Base; relative version possible | 5 and 10 days continuation = toxicity_probability * persistent_signed_completed_flow Mechanism and evidence. Some order flow

predicts continuing price adjustment rather than a bounce. Identity-rich research documents an information role for client-dealer relationships, but its confidential regulatory dataset is not a public trading feed. The proposal uses only lawful, permissioned, sufficiently aggregated completed activity and public corroboration. Feature and implementation. Start from the same signed-flow ledger as signal 10. Measure persistence across days and independent channels, not simply the number of repeated prints from one order. Estimate the probability that a flow episode reflects lasting price information using only historically matured markouts and contemporaneously available market state. Incorporate public CDS/equity/issuer-bond agreement and revisions in fresh dealer prices. Train client-related statistics only where contractual permissions, privacy controls and the desk's information barriers permit; otherwise use anonymous aggregate features. Positive customer purchase pressure with high persistent-information probability is a tentative long-price signal. Five-/ten-day forecast and hedge. Estimate residual continuation over five and ten trading days separately after a feasible fill. Track whether the signal's effect is a gradual price response or just compensation for taking broad issuer risk. Most truly issuer-wide information will be removed by the strict issuer-neutral target. A relative version must identify differing adjustment speeds across bonds, rather than relabel an issuer directional bet. Use a shared regime model with signal 10 so one episode is not simultaneously called informed continuation and uninformed reversal. Executable entry and costs. Do not use pending identifiable client instructions to trade ahead of that client. Research governance must approve the source, aggregation and time delay before feature engineering starts. Execution requires a fresh quote with an uncaptured expected movement large enough to pay the spread and hedges. Include fill-selection effects: dealers may offer attractive prices exactly when their information disagrees with the model. Falsification / kill test. A good historical customer markout is not proof of persistent skill and can reflect selection, market beta or changing relationships. Use shrinkage and leave-client/channel-out tests where permissible. Reject a feature that needs confidential identifiers unavailable in production, that dies after delayed publication is respected, or that merely duplicates signal 05 or 07. Compare continuation, reversal and abstention in the same locked out-of-sample experiment.

12.7 Agent implementation runbook: data discovery to safe production

Operating rule. The agent must advance through auditable gates rather than implement the entire alpha in one pass. If a required field is unavailable, ambiguous, delayed, or not historically point-in-time, stop that branch and record the limitation; never silently substitute a convenient proxy.

- A1. Data discovery.** Inventory candidate sources/tables/files; definitions; keys; units; time zones; economic, publication and receipt timestamps; corrections/cancels; historical coverage; entitlements; delayed-history rules; and whether each field was genuinely observable live. Output schema and coverage diagnostics only.
- A2. Raw profiling.** Report rows, bonds, issuers, days, missingness, duplicates, timestamp inversions, corrections, size distributions and join success by IG/HY, liquidity, sector and period. Do not impute yet.
- A3. Canonical identifiers.** Version the security master and map security IDs to legal obligor, parent, guarantee, seniority, currency, coupon, maturity, option schedule and corporate actions. Quarantine ambiguous/time-varying mappings.
- A4. As-of layer.** Require every query to accept a decision timestamp and return only records then available. Unit-test late records, corrections and cancels. Persist available-at and provenance fields.
- A5. Price layer.** Normalize clean/dirty price, accrued, spread convention, side, size and quote status. Create freshness, executable-size, dispersion and dealer-breadth metadata. Any forward fill must explicitly age and lose confidence.
- A6. Analytics layer.** Produce cash flows, yield/OAS as appropriate, spread duration, convexity, KRD, CS01/DV01, carry and roll from point-in-time terms; reconcile a sample against an independent implementation.
- A7. Targets.** Build 5d/10d efficient-price diagnostic and executable-return labels. Estimate residualisation only on prior data; cost every changed hedge leg, fees, financing and impact. Persist label maturity for purging.
- A8. Signal-specific prerequisites.** Execute the following in order, emitting exclusions, distributions and hand-audited examples after each sub-step:
 - a) Discover signed flow/RFQ data, price response, quote revisions and cross-market confirmations with exact receipt times.
 - b) Normalize flow by bond activity and separate repeated child prints from distinct parent demand where defensible.

- c) Cross-fit information and persistence probabilities; persistence may use only flow observed by the decision time.
- d) Construct continuation only in information-confirmed persistent states and exclude low-information exhaustion states used by signal 10.
- e) Request a new executable quote and model selective fills/adverse selection explicitly.
- f) Ablate events, aggregation windows and parent-order heuristics; require complementarity rather than duplication with signal 10.

- A9. Literal raw feature.** Translate the documented equation exactly. Assert units, sign, finite values, peer counts, timestamp ordering and boundaries. Hand-reconcile at least 20 events.
- A10. Causal transforms.** Add rolling median/MAD z-score, changes, persistence, acceleration and percentile with past-only windows. Fit curves/betas/normalizers only inside training windows and version them.
- A11. State variables.** Join price confidence, information/toxicity, pressure completion, liquidity regime, volatility, event age and requested size as separate columns before tuning interactions.
- A12. Leakage tests.** Append records arriving after decision time and prove historical feature values do not change. Compare live-available with enhanced-history fields and test shifted timestamps. Any failure blocks modelling.
- A13. Measurement tests.** Recompute with fresh quotes only, trades only, independent prices where possible, alternative stale thresholds and stricter confidence. Reject effects caused only by asynchronous updating.
- A14. Transparent baseline.** Fit horizon-specific linear/robust models in walk-forward folds before nonlinear models; record sign, stability, sample size and incremental information.
- A15. State/convergence models.** Estimate information versus temporary-pressure probabilities and convergence speed where meaningful. Cross-fit state probabilities used as features.
- A16. Execution simulator.** Request a new executable entry after the signal timestamp; apply latency, size, withdrawal, partial-fill and legging rules. No executable observation means unfilled, not a midpoint fill.
- A17. Fill/cost models.** Estimate fill probability and conditional spread/impact/fees/financing by size, protocol and state. Calibrate on held-out fills and cost every changed leg.
- A18. Net outcomes.** Store expected executable return conditional on fill, conservative lower quantile and capacity. Keep gross, mark, executable and fully-net outcomes as distinct audit columns.
- A19. Adversarial validation.** Use chronological purged folds, issuer holdouts, liquidity/rating/regime splits, event exclusions, alternative residualisations, placebo dates and within-family ablations.
- A20. Shadow mode.** Freeze versions and generate live predictions without trading. Log source timestamps, features, quotes, hypothetical orders, predicted fills/costs and 1d/5d/10d mark-outs.
- A21. Canary.** Start with strict size, concentration, liquidity and turnover limits. Kill automatically on timestamp anomalies, data gaps, feature drift, cost underprediction, unexplained residual beta or repeated adverse selection.
- A22. Promotion/rollback.** Promote only a fully reproducible version. Default to no-trade when health checks fail and retain one-command rollback to the prior approved model.

Incremental safe prompts. Use these as separate prompts/commits; each prompt may inspect prior artifacts but must not expand scope silently.

- P1.** “Inventory only the required data, schemas, keys, timestamps, live-history availability, entitlements, coverage and unresolved questions. Do not build a feature.”
- P2.** “Build and unit-test canonical identifiers and as-of joins. Return diagnostics and quarantined ambiguities. Do not calculate alpha.”
- P3.** “Build the price/analytics layer with freshness/confidence and reconcile a hand-audited sample. Stop on unexplained discrepancies.”

- P4.** “Implement only the literal raw signal with assertions and 20 hand-reconciled examples. Do not optimize parameters.”
- P5.** “Add past-only transforms and state variables. Run leakage and measurement-artifact tests before fitting any predictive model.”
- P6.** “Fit the transparent walk-forward baseline for 5d/10d. Report coefficients, uncertainty, ablations and failures; do not tune on the final test period.”
- P7.** “Add execution, fill and transaction-cost simulation. Recompute conditional executable net outcomes and mark unavailable opportunities unfilled.”
- P8.** “Run adversarial validation and within-family ablations. Produce a go/no-go research recommendation based only on predeclared gates.”
- P9.** “Run shadow mode with frozen versions. Do not send orders. Compare predicted versus observable fills, costs and mark-outs.”
- P10.** “Only after explicit human approval, create a tightly limited canary configuration with hard risk limits, health checks and rollback; do not broaden limits automatically.”

12.8 Production row and acceptance gate

Persist raw score, z-score, event age, quote age, peer count/effective sample size, confidence, state probabilities, size/ADV, expected spread/impact, issuer/sector/rating/tenor, KRD/CS01 and source timestamps. The feature graduates only if it adds incremental out-of-sample information after other members of its economic family, survives alternative residualisations and retains a positive conservative executable-net-return statistic at realistic capacity.

13 Signal 12: Remaining block concession / impact decay

13.1 Mathematics and feature construction

The canonical raw score is

$$BC_{i,t} = P_{i,t}^{fair} - P_{i,t}^{fresh,exec}. \quad (45)$$

Retain level, one- and three-day change, persistence, acceleration, trailing percentile and robust z-score. Add interactions with $c_{i,t}$, $p_{i,t}^{info}$, $p_{i,t}^{complete}$, liquidity regime, volatility and requested size. Thresholds and normalisation are learned on training data only. Missingness is explicit.

13.2 Point-in-time implementation

Maintain an immutable as-of ledger with economic, publication and receipt timestamps, corrections/cancels, security-master version and entitlements. At each decision time freeze the universe; resolve terms and corporate actions; construct fresh prices; compute peers/factors from earlier data; calculate the feature; obtain or simulate a fresh executable entry; and persist the feature snapshot. A confidence-weighted price can use

$$w_q = e^{-age_q/\tau_a} \min(1, size_q/v_0) w_d e^{-dispersion_q/\tau_\sigma}, \quad P^* = \frac{\sum_q w_q P_q}{\sum_q w_q}.$$

Never use future prints through symmetric smoothing.

13.3 Five- and ten-day forecast

Benchmark each horizon with

$$y_{i,t,H}^{exec} = a_H + b_H z + d_H z c + e_H z p^{info} + f_H z p^{complete} + u_{i,t,H}.$$

For dislocations estimate $D_{t+h} = D_t e^{-\kappa h} + \epsilon$ and expected capture $D_t(1 - e^{-\kappa H})$; for information diffusion use a bounded catch-up fraction. This gives an economic mapping from state to the 5d/10d horizon.

13.4 Execution and costs

Estimate jointly

$$p^{fill} = \mathbb{P}(fill \mid X, size, protocol), \quad (46)$$

$$C = g(size, spread, ADV, depth, vol, flow, toxicity, protocol), \quad (47)$$

$$\mu_H = \mathbb{E}[y_H^{exec} \mid fill, X]. \quad (48)$$

Use a fresh executable entry, never the historical print that revealed the signal. Model quote withdrawal and adverse selection. Relative-value packages pay every leg. Capacity is evaluated by rerunning net P&L at increasing requested size and participation constraints.

13.5 Falsification and monitoring

Kill the feature if it disappears with fresh independent prices, relies on unavailable data, is concentrated in a few issuers/events, or loses its edge under realistic fills and costs. Run leave-one-issuer-out tests, IG/HY and liquidity splits, calm/stress regimes, protocol splits and alternative residualisations. Monitor drift, price confidence, fill calibration, realised versus predicted costs, mark-outs and residual factor exposure.

13.6 Signal-specific mechanism and implementation

Data: Block prints + independent executable quotes | Target: Base and issuer-neutral | 5-day emphasis; test 10 remaining_concession = independent_fair_price_now - executable_ask_now Mechanism and evidence. A large

customer sale may involve a temporary price concession. The potential alpha is the part still obtainable after the print is observed, not the difference between a historical block price and a later mark. Transient-impact and immediacy studies provide the mechanism; remaining executable alpha must be measured independently. Feature and implementation. Detect unusually large completed customer trades relative to the bond's trailing size distribution, retaining a censored-size flag where public quantities are capped. Estimate a pre-event fair-price path from independent sibling/market references and contemporaneous quotes, then subtract the permanent information component supported by public cross-market moves. Measure the present concession at the ask for a long trade. Record elapsed clock time and number of independent price refreshes since the block. Treat multi-print parent orders and riskless-principal chains as one episode only when that relationship is known as of the decision. Five-/ten-day forecast and hedge. Fit a side-and regime-specific transient-impact decay model using matured historical episodes, then translate the remaining component into five- and ten-day price-return forecasts. Estimate in both event time and calendar trading days: a five-trade recovery need not equal a five-day recovery. Require evidence of a nonzero remaining concession at entry. Use issuer-relative controls so broad credit deterioration is not mistaken for temporary block impact. Executable entry and costs. Never assume the strategy can transact at the block's past price after seeing the report. Obtain a new executable quote and record the requested size, quote expiry and fill/no-fill outcome. The entry can already consume nearly all of the concession. Count hedge execution, impact from the strategy's own order and the exit spread; reserve for another block arriving before the inventory clears. Refuse to average into a deteriorating information state merely because the historical block price was lower. Falsification / kill test. Kill the strategy if profitability vanishes when report-arrival latency, capped sizes and quote refreshes are respected. Compare customer-buy and customer-sale episodes separately. Exclude unmodelled credit events and repeated forced-sale programs whose completion cannot be observed. Test incremental net value beyond signal 09, 10 and 13; a block flag alone may add no new information after price, flow and inventory state are known.

13.7 Agent implementation runbook: data discovery to safe production

Operating rule. The agent must advance through auditable gates rather than implement the entire alpha in one pass. If a required field is unavailable, ambiguous, delayed, or not historically point-in-time, stop that branch and record the limitation; never silently substitute a convenient proxy.

- A1. Data discovery.** Inventory candidate sources/tables/files; definitions; keys; units; time zones; economic, publication and receipt timestamps; corrections/cancels; historical coverage; entitlements; delayed-history rules; and whether each field was genuinely observable live. Output schema and coverage diagnostics only.
- A2. Raw profiling.** Report rows, bonds, issuers, days, missingness, duplicates, timestamp inversions, corrections, size distributions and join success by IG/HY, liquidity, sector and period. Do not impute yet.
- A3. Canonical identifiers.** Version the security master and map security IDs to legal obligor, parent, guarantee, seniority, currency, coupon, maturity, option schedule and corporate actions. Quarantine ambiguous/time-varying mappings.
- A4. As-of layer.** Require every query to accept a decision timestamp and return only records then available. Unit-test late records, corrections and cancels. Persist available-at and provenance fields.
- A5. Price layer.** Normalize clean/dirty price, accrued, spread convention, side, size and quote status. Create freshness, executable-size, dispersion and dealer-breadth metadata. Any forward fill must explicitly age and lose confidence.
- A6. Analytics layer.** Produce cash flows, yield/OAS as appropriate, spread duration, convexity, KRD, CS01/DV01, carry and roll from point-in-time terms; reconcile a sample against an independent implementation.
- A7. Targets.** Build 5d/10d efficient-price diagnostic and executable-return labels. Estimate residualisation only on prior data; cost every changed hedge leg, fees, financing and impact. Persist label maturity for purging.
- A8. Signal-specific prerequisites.** Execute the following in order, emitting exclusions, distributions and hand-audited examples after each sub-step:
 - a) Discover large-trade prints, size information, side classification and fresh post-trade quotes.
 - b) Define blocks relative to bond-specific normal size and liquidity using only prior history.
 - c) Estimate pre-block fair price from independent quotes/peers and measure initial concession.
 - d) Track causal post-block recovery and estimate concession remaining at the actual decision time.

- e) Model whether initiating pressure is complete; do not enter while seller pressure is accelerating.
- f) Requote before entry and include own impact at requested size; reject historical-print-only opportunities.

- A9. Literal raw feature.** Translate the documented equation exactly. Assert units, sign, finite values, peer counts, timestamp ordering and boundaries. Hand-reconcile at least 20 events.
- A10. Causal transforms.** Add rolling median/MAD z-score, changes, persistence, acceleration and percentile with past-only windows. Fit curves/betas/normalizers only inside training windows and version them.
- A11. State variables.** Join price confidence, information/toxicity, pressure completion, liquidity regime, volatility, event age and requested size as separate columns before tuning interactions.
- A12. Leakage tests.** Append records arriving after decision time and prove historical feature values do not change. Compare live-available with enhanced-history fields and test shifted timestamps. Any failure blocks modelling.
- A13. Measurement tests.** Recompute with fresh quotes only, trades only, independent prices where possible, alternative stale thresholds and stricter confidence. Reject effects caused only by asynchronous updating.
- A14. Transparent baseline.** Fit horizon-specific linear/robust models in walk-forward folds before nonlinear models; record sign, stability, sample size and incremental information.
- A15. State/convergence models.** Estimate information versus temporary-pressure probabilities and convergence speed where meaningful. Cross-fit state probabilities used as features.
- A16. Execution simulator.** Request a new executable entry after the signal timestamp; apply latency, size, withdrawal, partial-fill and legging rules. No executable observation means unfilled, not a midpoint fill.
- A17. Fill/cost models.** Estimate fill probability and conditional spread/impact/fees/financing by size, protocol and state. Calibrate on held-out fills and cost every changed leg.
- A18. Net outcomes.** Store expected executable return conditional on fill, conservative lower quantile and capacity. Keep gross, mark, executable and fully-net outcomes as distinct audit columns.
- A19. Adversarial validation.** Use chronological purged folds, issuer holdouts, liquidity/rating/regime splits, event exclusions, alternative residualisations, placebo dates and within-family ablations.
- A20. Shadow mode.** Freeze versions and generate live predictions without trading. Log source timestamps, features, quotes, hypothetical orders, predicted fills/costs and 1d/5d/10d mark-outs.
- A21. Canary.** Start with strict size, concentration, liquidity and turnover limits. Kill automatically on timestamp anomalies, data gaps, feature drift, cost underprediction, unexplained residual beta or repeated adverse selection.
- A22. Promotion/rollback.** Promote only a fully reproducible version. Default to no-trade when health checks fail and retain one-command rollback to the prior approved model.

Incremental safe prompts. Use these as separate prompts/commits; each prompt may inspect prior artifacts but must not expand scope silently.

- P1.** “Inventory only the required data, schemas, keys, timestamps, live-history availability, entitlements, coverage and unresolved questions. Do not build a feature.”
- P2.** “Build and unit-test canonical identifiers and as-of joins. Return diagnostics and quarantined ambiguities. Do not calculate alpha.”
- P3.** “Build the price/analytics layer with freshness/confidence and reconcile a hand-audited sample. Stop on unexplained discrepancies.”
- P4.** “Implement only the literal raw signal with assertions and 20 hand-reconciled examples. Do not optimize parameters.”
- P5.** “Add past-only transforms and state variables. Run leakage and measurement-artifact tests before fitting any predictive model.”

- P6.** “Fit the transparent walk-forward baseline for 5d/10d. Report coefficients, uncertainty, ablations and failures; do not tune on the final test period.”
- P7.** “Add execution, fill and transaction-cost simulation. Recompute conditional executable net outcomes and mark unavailable opportunities unfilled.”
- P8.** “Run adversarial validation and within-family ablations. Produce a go/no-go research recommendation based only on predeclared gates.”
- P9.** “Run shadow mode with frozen versions. Do not send orders. Compare predicted versus observable fills, costs and mark-outs.”
- P10.** “Only after explicit human approval, create a tightly limited canary configuration with hard risk limits, health checks and rollback; do not broaden limits automatically.”

13.8 Production row and acceptance gate

Persist raw score, z-score, event age, quote age, peer count/effective sample size, confidence, state probabilities, size/ADV, expected spread/impact, issuer/sector/rating/tenor, KRD/CS01 and source timestamps. The feature graduates only if it adds incremental out-of-sample information after other members of its economic family, survives alternative residualisations and retains a positive conservative executable-net-return statistic at realistic capacity.

14 Signal 13: Inventory / dealer-axe pressure exhaustion

14.1 Mathematics and feature construction

The canonical raw score is

$$INV_{i,t} = z(Burden_{d,i,t}), z(Cheapness_{i,t}), p_{d,i,t}^{exhaust}(1 - p_{i,t}^{info}), \quad (49)$$

Retain level, one- and three-day change, persistence, acceleration, trailing percentile and robust z-score. Add interactions with $c_{i,t}$, $p_{i,t}^{info}$, $p_{i,t}^{complete}$, liquidity regime, volatility and requested size. Thresholds and normalisation are learned on training data only. Missingness is explicit.

14.2 Point-in-time implementation

Maintain an immutable as-of ledger with economic, publication and receipt timestamps, corrections/cancels, security-master version and entitlements. At each decision time freeze the universe; resolve terms and corporate actions; construct fresh prices; compute peers/factors from earlier data; calculate the feature; obtain or simulate a fresh executable entry; and persist the feature snapshot. A confidence-weighted price can use

$$w_q = e^{-age_q/\tau_a} \min(1, size_q/v_0) w_d e^{-dispersion_q/\tau_\sigma}, \quad P^* = \frac{\sum_q w_q P_q}{\sum_q w_q}.$$

Never use future prints through symmetric smoothing.

14.3 Five- and ten-day forecast

Benchmark each horizon with

$$y_{i,t,H}^{exec} = a_H + b_H z + d_H zc + e_H z p^{info} + f_H z p^{complete} + u_{i,t,H}.$$

For dislocations estimate $D_{t+h} = D_t e^{-\kappa h} + \epsilon$ and expected capture $D_t(1 - e^{-\kappa H})$; for information diffusion use a bounded catch-up fraction. This gives an economic mapping from state to the 5d/10d horizon.

14.4 Execution and costs

Estimate jointly

$$p^{fill} = \mathbb{P}(fill \mid X, size, protocol), \quad (50)$$

$$C = g(size, spread, ADV, depth, vol, flow, toxicity, protocol), \quad (51)$$

$$\mu_H = \mathbb{E}[y_H^{exec} \mid fill, X]. \quad (52)$$

Use a fresh executable entry, never the historical print that revealed the signal. Model quote withdrawal and adverse selection. Relative-value packages pay every leg. Capacity is evaluated by rerunning net P&L at increasing requested size and participation constraints.

14.5 Falsification and monitoring

Kill the feature if it disappears with fresh independent prices, relies on unavailable data, is concentrated in a few issuers/events, or loses its edge under realistic fills and costs. Run leave-one-issuer-out tests, IG/HY and liquidity splits, calm/stress regimes, protocol splits and alternative residualisations. Monitor drift, price confidence, fill calibration, realised versus predicted costs, mark-outs and residual factor exposure.

14.6 Signal-specific mechanism and implementation

Data: Own inventory or licensed dealer axes | Target: Base and issuer-neutral | 5 and 10 days inventory_score = inventory_burden * remaining_cheapness * completion_probability Mechanism and evidence. Inventory-sensitive intermediaries may quote concessions to reduce unwanted positions. A 2015 working paper reports weekly inventory-related predictability using privileged dealer-identified transactions. That evidence motivates the mechanism, not a claim that public TRACE reconstructs all dealers' inventory or that the reported gross return is net profit. Feature and implementation. With authorised internal data, track bond risk inventory relative to a dealer/desk target and capacity, including principal versus agency trades and transfers. With licensed axes instead, build a separately named proxy from persistent sell-interest breadth, quoted size, price aggressiveness and recent withdrawals; an axe is an indication of interest, not verified inventory. Keep source identities and permissions straight. Positive excess inventory or sell-axis pressure must coincide with an executable cheap price and evidence that the burden is easing. Normalise within dealer/bond and control market-wide balance-sheet state. Do not sum dealer stocks inferred from an incomplete public flow history. Five-/ten-day forecast and hedge. Estimate five-/ten-day outcomes conditional on observed burden and its recent change. Inventory can remain elevated, so level alone is insufficient. Test whether the relevant signal is exhaustion, falling pressure, or a current concession that pays for waiting. In the strict issuer-neutral version, compare inventory/axis pressure across siblings and hedge the common issuer move. Own-desk inventory may mainly predict the desk's quote skew rather than market returns; measure those outcomes separately. Executable entry and costs. Entry must capture a present concession without exhausting the attractive dealer quote. Model counterparty-specific capacity, information leakage and the effect of consuming displayed interest. Cost all legs, financing and inventory holding risk. Record axe withdrawals and no responses as outcomes rather than removing them from history. An existing position reduction can have different economics from initiating a new long-short trade. Falsification / kill test. Reject results based on invented dealer identities, stale/copy-pasted axes, unrecorded internal transfers or an unknown initial inventory stock. Distinguish a dealer reducing risk because of new issuer information from a purely technical burden. Require incremental performance over observable customer flow and block concessions. If the true inventory data are unavailable in production, downgrade to the proxy model and revalidate from scratch.

14.7 Agent implementation runbook: data discovery to safe production

Operating rule. The agent must advance through auditable gates rather than implement the entire alpha in one pass. If a required field is unavailable, ambiguous, delayed, or not historically point-in-time, stop that branch and record the limitation; never silently substitute a convenient proxy.

- A1. Data discovery.** Inventory candidate sources/tables/files; definitions; keys; units; time zones; economic, publication and receipt timestamps; corrections/cancels; historical coverage; entitlements; delayed-history rules; and whether each field was genuinely observable live. Output schema and coverage diagnostics only.
- A2. Raw profiling.** Report rows, bonds, issuers, days, missingness, duplicates, timestamp inversions, corrections, size distributions and join success by IG/HY, liquidity, sector and period. Do not impute yet.
- A3. Canonical identifiers.** Version the security master and map security IDs to legal obligor, parent, guarantee, seniority, currency, coupon, maturity, option schedule and corporate actions. Quarantine ambiguous/time-varying mappings.
- A4. As-of layer.** Require every query to accept a decision timestamp and return only records then available. Unit-test late records, corrections and cancels. Persist available-at and provenance fields.
- A5. Price layer.** Normalize clean/dirty price, accrued, spread convention, side, size and quote status. Create freshness, executable-size, dispersion and dealer-breadth metadata. Any forward fill must explicitly age and lose confidence.
- A6. Analytics layer.** Produce cash flows, yield/OAS as appropriate, spread duration, convexity, KRD, CS01/DV01, carry and roll from point-in-time terms; reconcile a sample against an independent implementation.
- A7. Targets.** Build 5d/10d efficient-price diagnostic and executable-return labels. Estimate residualisation only on prior data; cost every changed hedge leg, fees, financing and impact. Persist label maturity for purging.
- A8. Signal-specific prerequisites.** Execute the following in order, emitting exclusions, distributions and hand-audited examples after each sub-step:

- a) Discover dealer axes/inventory only where entitled and historically live; document dealer-identity stability and survivorship.
- b) Normalize burden relative to dealer- and bond-specific history; separate firm-wide balance-sheet pressure from bond-specific axes.
- c) Estimate exhaustion from declining offered size, quote normalization, completed flow and inventory relief using contemporaneous data.
- d) Interact burden with independent cheapness and low information toxicity; inventory pressure alone is not a directional alpha.
- e) Model dealer-specific quote availability/fill probability at requested size; never assume an internal axe is externally executable.
- f) Run dealer holdouts, stale-axe decay and dominant-dealer removal; block production if historical fields were not live-observable.

- A9. Literal raw feature.** Translate the documented equation exactly. Assert units, sign, finite values, peer counts, timestamp ordering and boundaries. Hand-reconcile at least 20 events.
- A10. Causal transforms.** Add rolling median/MAD z-score, changes, persistence, acceleration and percentile with past-only windows. Fit curves/betas/normalizers only inside training windows and version them.
- A11. State variables.** Join price confidence, information/toxicity, pressure completion, liquidity regime, volatility, event age and requested size as separate columns before tuning interactions.
- A12. Leakage tests.** Append records arriving after decision time and prove historical feature values do not change. Compare live-available with enhanced-history fields and test shifted timestamps. Any failure blocks modelling.
- A13. Measurement tests.** Recompute with fresh quotes only, trades only, independent prices where possible, alternative stale thresholds and stricter confidence. Reject effects caused only by asynchronous updating.
- A14. Transparent baseline.** Fit horizon-specific linear/robust models in walk-forward folds before nonlinear models; record sign, stability, sample size and incremental information.
- A15. State/convergence models.** Estimate information versus temporary-pressure probabilities and convergence speed where meaningful. Cross-fit state probabilities used as features.
- A16. Execution simulator.** Request a new executable entry after the signal timestamp; apply latency, size, withdrawal, partial-fill and legging rules. No executable observation means unfilled, not a midpoint fill.
- A17. Fill/cost models.** Estimate fill probability and conditional spread/impact/fees/financing by size, protocol and state. Calibrate on held-out fills and cost every changed leg.
- A18. Net outcomes.** Store expected executable return conditional on fill, conservative lower quantile and capacity. Keep gross, mark, executable and fully-net outcomes as distinct audit columns.
- A19. Adversarial validation.** Use chronological purged folds, issuer holdouts, liquidity/rating/regime splits, event exclusions, alternative residualisations, placebo dates and within-family ablations.
- A20. Shadow mode.** Freeze versions and generate live predictions without trading. Log source timestamps, features, quotes, hypothetical orders, predicted fills/costs and 1d/5d/10d mark-outs.
- A21. Canary.** Start with strict size, concentration, liquidity and turnover limits. Kill automatically on timestamp anomalies, data gaps, feature drift, cost underprediction, unexplained residual beta or repeated adverse selection.
- A22. Promotion/rollback.** Promote only a fully reproducible version. Default to no-trade when health checks fail and retain one-command rollback to the prior approved model.

Incremental safe prompts. Use these as separate prompts/commits; each prompt may inspect prior artifacts but must not expand scope silently.

- P1.** “Inventory only the required data, schemas, keys, timestamps, live-history availability, entitlements, coverage and unresolved questions. Do not build a feature.”
- P2.** “Build and unit-test canonical identifiers and as-of joins. Return diagnostics and quarantined ambiguities. Do not calculate alpha.”
- P3.** “Build the price/analytics layer with freshness/confidence and reconcile a hand-audited sample. Stop on unexplained discrepancies.”
- P4.** “Implement only the literal raw signal with assertions and 20 hand-reconciled examples. Do not optimize parameters.”
- P5.** “Add past-only transforms and state variables. Run leakage and measurement-artifact tests before fitting any predictive model.”
- P6.** “Fit the transparent walk-forward baseline for 5d/10d. Report coefficients, uncertainty, ablations and failures; do not tune on the final test period.”
- P7.** “Add execution, fill and transaction-cost simulation. Recompute conditional executable net outcomes and mark unavailable opportunities unfilled.”
- P8.** “Run adversarial validation and within-family ablations. Produce a go/no-go research recommendation based only on predeclared gates.”
- P9.** “Run shadow mode with frozen versions. Do not send orders. Compare predicted versus observable fills, costs and mark-outs.”
- P10.** “Only after explicit human approval, create a tightly limited canary configuration with hard risk limits, health checks and rollback; do not broaden limits automatically.”

14.8 Production row and acceptance gate

Persist raw score, z-score, event age, quote age, peer count/effective sample size, confidence, state probabilities, size/ADV, expected spread/impact, issuer/sector/rating/tenor, KRD/CS01 and source timestamps. The feature graduates only if it adds incremental out-of-sample information after other members of its economic family, survives alternative residualisations and retains a positive conservative executable-net-return statistic at realistic capacity.

15 Signal 14: Dealer quote-revision breadth

15.1 Mathematics and feature construction

The canonical raw score is

$$QB_{i,t} = \sum_d w_d \text{sign}(\Delta q_{d,i,t}) 1\{|\Delta q| > \delta\}. \quad (53)$$

Retain level, one- and three-day change, persistence, acceleration, trailing percentile and robust z-score. Add interactions with $c_{i,t}$, $p_{i,t}^{info}$, $p_{i,t}^{complete}$, liquidity regime, volatility and requested size. Thresholds and normalisation are learned on training data only. Missingness is explicit.

15.2 Point-in-time implementation

Maintain an immutable as-of ledger with economic, publication and receipt timestamps, corrections/cancels, security-master version and entitlements. At each decision time freeze the universe; resolve terms and corporate actions; construct fresh prices; compute peers/factors from earlier data; calculate the feature; obtain or simulate a fresh executable entry; and persist the feature snapshot. A confidence-weighted price can use

$$w_q = e^{-age_q/\tau_a} \min(1, size_q/v_0) w_d e^{-dispersion_q/\tau_\sigma}, \quad P^* = \frac{\sum_q w_q P_q}{\sum_q w_q}.$$

Never use future prints through symmetric smoothing.

15.3 Five- and ten-day forecast

Benchmark each horizon with

$$y_{i,t,H}^{exec} = a_H + b_H z + d_H zc + e_H z p^{info} + f_H z p^{complete} + u_{i,t,H}.$$

For dislocations estimate $D_{t+h} = D_t e^{-\kappa h} + \epsilon$ and expected capture $D_t(1 - e^{-\kappa H})$; for information diffusion use a bounded catch-up fraction. This gives an economic mapping from state to the 5d/10d horizon.

15.4 Execution and costs

Estimate jointly

$$p^{fill} = \mathbb{P}(fill \mid X, size, protocol), \quad (54)$$

$$C = g(size, spread, ADV, depth, vol, flow, toxicity, protocol), \quad (55)$$

$$\mu_H = \mathbb{E}[y_H^{exec} \mid fill, X]. \quad (56)$$

Use a fresh executable entry, never the historical print that revealed the signal. Model quote withdrawal and adverse selection. Relative-value packages pay every leg. Capacity is evaluated by rerunning net P&L at increasing requested size and participation constraints.

15.5 Falsification and monitoring

Kill the feature if it disappears with fresh independent prices, relies on unavailable data, is concentrated in a few issuers/events, or loses its edge under realistic fills and costs. Run leave-one-issuer-out tests, IG/HY and liquidity splits, calm/stress regimes, protocol splits and alternative residualisations. Monitor drift, price confidence, fill calibration, realised versus predicted costs, mark-outs and residual factor exposure.

15.6 Signal-specific mechanism and implementation

Data: Permissioned multi-dealer quote histories | Target: Base and issuer-neutral | 5-day emphasis; test 10 breadth = weighted_consensus(residual_midpoint_revisions across independent dealers) Mechanism and evidence. A broad

migration in fresh dealer valuations may convey more information than a single print or a static quote dispersion. The proposed alpha is persistent consensus revision not yet incorporated in the target's attainable price. Auction/search research motivates analysing dealer information and execution separately; it does not establish this five-day feature's profitability. Feature and implementation. Archive timestamped bid, ask, size, firmness and dealer/source identifiers. Compare each dealer only with its own prior valid quote; synchronise rate and market moves before measuring a credit-specific midpoint revision. Aggregate the signed revisions using reliability, age and executable-size weights, downweighting copied vendor composites and highly correlated dealers. Keep breadth, median revision, disagreement and effective independent dealer count as separate features. A positive price revision from several independent dealers is a candidate continuation feature. A spread widening with unchanged midpoint is a liquidity warning, not the same direction signal. Five-/ten-day forecast and hedge. Fit five-/ten-day residual-return models controlling for the bond's current consensus price and signals 05/07. The question is whether earlier cross-dealer revisions predict future movement after the current executable quote, not whether yesterday's quotes lag today's consensus. Under issuer-neutralisation, retain revisions unique to this bond relative to same-issuer quotes. Estimate whether wide disagreement forecasts slower convergence or simply higher uncertainty; do not assign a universal sign to dispersion. Executable entry and costs. Price the trade at the final executable dealer quote, with allowance for refreshes while RFQs are collected. The apparently stale dealer may update before accepting the trade; count such outcomes as missed opportunities. Simulate the routing rule that would have existed at that time and include dealer-specific rejects, minimum size and fill probability. Use the common conditional-on-fill cost gate. Falsification / kill test. Kill the feature when its apparent breadth comes from one underlying source syndicated to multiple feeds, when quote timestamps are ingestion rather than creation times, or when only nonfirm indications are available. Require value beyond a simple fresh consensus midpoint and sibling lead-lag. A model that predicts quote updates but cannot access a price before those updates is a mark-quality model, not a tradable alpha.

15.7 Agent implementation runbook: data discovery to safe production

Operating rule. The agent must advance through auditable gates rather than implement the entire alpha in one pass. If a required field is unavailable, ambiguous, delayed, or not historically point-in-time, stop that branch and record the limitation; never silently substitute a convenient proxy.

- A1. Data discovery.** Inventory candidate sources/tables/files; definitions; keys; units; time zones; economic, publication and receipt timestamps; corrections/cancels; historical coverage; entitlements; delayed-history rules; and whether each field was genuinely observable live. Output schema and coverage diagnostics only.
- A2. Raw profiling.** Report rows, bonds, issuers, days, missingness, duplicates, timestamp inversions, corrections, size distributions and join success by IG/HY, liquidity, sector and period. Do not impute yet.
- A3. Canonical identifiers.** Version the security master and map security IDs to legal obligor, parent, guarantee, seniority, currency, coupon, maturity, option schedule and corporate actions. Quarantine ambiguous/time-varying mappings.
- A4. As-of layer.** Require every query to accept a decision timestamp and return only records then available. Unit-test late records, corrections and cancels. Persist available-at and provenance fields.
- A5. Price layer.** Normalize clean/dirty price, accrued, spread convention, side, size and quote status. Create freshness, executable-size, dispersion and dealer-breadth metadata. Any forward fill must explicitly age and lose confidence.
- A6. Analytics layer.** Produce cash flows, yield/OAS as appropriate, spread duration, convexity, KRD, CS01/DV01, carry and roll from point-in-time terms; reconcile a sample against an independent implementation.
- A7. Targets.** Build 5d/10d efficient-price diagnostic and executable-return labels. Estimate residualisation only on prior data; cost every changed hedge leg, fees, financing and impact. Persist label maturity for purging.
- A8. Signal-specific prerequisites.** Execute the following in order, emitting exclusions, distributions and hand-audited examples after each sub-step:
 - a) Discover timestamped dealer quotes with side, size, status/cancel events and stable dealer identifiers.
 - b) Create revisions from each dealer own prior live quote; reject crossed/stale/indicative updates and normalize to common spread units.
 - c) Weight dealers by prior-data freshness/reliability; compute signed breadth separately from magnitude.

- d) Classify issuer-wide information versus target catch-up using siblings/CDS/equity and target update age.
- e) Use a fresh executable target quote after breadth is observed; apply explicit latency before assuming fill.
- f) Stress dealer weights, minimum breadth, thresholds and removal of fastest dealer; test duplicate-feed/quote-stuffing robustness.

- A9. Literal raw feature.** Translate the documented equation exactly. Assert units, sign, finite values, peer counts, timestamp ordering and boundaries. Hand-reconcile at least 20 events.
- A10. Causal transforms.** Add rolling median/MAD z-score, changes, persistence, acceleration and percentile with past-only windows. Fit curves/betas/normalizers only inside training windows and version them.
- A11. State variables.** Join price confidence, information/toxicity, pressure completion, liquidity regime, volatility, event age and requested size as separate columns before tuning interactions.
- A12. Leakage tests.** Append records arriving after decision time and prove historical feature values do not change. Compare live-available with enhanced-history fields and test shifted timestamps. Any failure blocks modelling.
- A13. Measurement tests.** Recompute with fresh quotes only, trades only, independent prices where possible, alternative stale thresholds and stricter confidence. Reject effects caused only by asynchronous updating.
- A14. Transparent baseline.** Fit horizon-specific linear/robust models in walk-forward folds before nonlinear models; record sign, stability, sample size and incremental information.
- A15. State/convergence models.** Estimate information versus temporary-pressure probabilities and convergence speed where meaningful. Cross-fit state probabilities used as features.
- A16. Execution simulator.** Request a new executable entry after the signal timestamp; apply latency, size, withdrawal, partial-fill and legging rules. No executable observation means unfilled, not a midpoint fill.
- A17. Fill/cost models.** Estimate fill probability and conditional spread/impact/fees/financing by size, protocol and state. Calibrate on held-out fills and cost every changed leg.
- A18. Net outcomes.** Store expected executable return conditional on fill, conservative lower quantile and capacity. Keep gross, mark, executable and fully-net outcomes as distinct audit columns.
- A19. Adversarial validation.** Use chronological purged folds, issuer holdouts, liquidity/rating/regime splits, event exclusions, alternative residualisations, placebo dates and within-family ablations.
- A20. Shadow mode.** Freeze versions and generate live predictions without trading. Log source timestamps, features, quotes, hypothetical orders, predicted fills/costs and 1d/5d/10d mark-outs.
- A21. Canary.** Start with strict size, concentration, liquidity and turnover limits. Kill automatically on timestamp anomalies, data gaps, feature drift, cost underprediction, unexplained residual beta or repeated adverse selection.
- A22. Promotion/rollback.** Promote only a fully reproducible version. Default to no-trade when health checks fail and retain one-command rollback to the prior approved model.

Incremental safe prompts. Use these as separate prompts/commits; each prompt may inspect prior artifacts but must not expand scope silently.

- P1.** “Inventory only the required data, schemas, keys, timestamps, live-history availability, entitlements, coverage and unresolved questions. Do not build a feature.”
- P2.** “Build and unit-test canonical identifiers and as-of joins. Return diagnostics and quarantined ambiguities. Do not calculate alpha.”
- P3.** “Build the price/analytics layer with freshness/confidence and reconcile a hand-audited sample. Stop on unexplained discrepancies.”
- P4.** “Implement only the literal raw signal with assertions and 20 hand-reconciled examples. Do not optimize parameters.”

- P5.** “Add past-only transforms and state variables. Run leakage and measurement-artifact tests before fitting any predictive model.”
- P6.** “Fit the transparent walk-forward baseline for 5d/10d. Report coefficients, uncertainty, ablations and failures; do not tune on the final test period.”
- P7.** “Add execution, fill and transaction-cost simulation. Recompute conditional executable net outcomes and mark unavailable opportunities unfilled.”
- P8.** “Run adversarial validation and within-family ablations. Produce a go/no-go research recommendation based only on predeclared gates.”
- P9.** “Run shadow mode with frozen versions. Do not send orders. Compare predicted versus observable fills, costs and mark-outs.”
- P10.** “Only after explicit human approval, create a tightly limited canary configuration with hard risk limits, health checks and rollback; do not broaden limits automatically.”

15.8 Production row and acceptance gate

Persist raw score, z-score, event age, quote age, peer count/effective sample size, confidence, state probabilities, size/ADV, expected spread/impact, issuer/sector/rating/tenor, KRD/CS01 and source timestamps. The feature graduates only if it adds incremental out-of-sample information after other members of its economic family, survives alternative residualisations and retains a positive conservative executable-net-return statistic at realistic capacity.

16 Signal 15: Aggregate RFQ demand and response skew

16.1 Mathematics and feature construction

The canonical raw score is

$$RFQ_{i,t} = \omega_1 z(Demand_{i,t}) + \omega_2 z(ResponseSkew_{i,t}) + \omega_3 z(Coverage_{i,t}), \quad (57)$$

Retain level, one- and three-day change, persistence, acceleration, trailing percentile and robust z-score. Add interactions with $c_{i,t}$, $p_{i,t}^{info}$, $p_{i,t}^{complete}$, liquidity regime, volatility and requested size. Thresholds and normalisation are learned on training data only. Missingness is explicit.

16.2 Point-in-time implementation

Maintain an immutable as-of ledger with economic, publication and receipt timestamps, corrections/cancels, security-master version and entitlements. At each decision time freeze the universe; resolve terms and corporate actions; construct fresh prices; compute peers/factors from earlier data; calculate the feature; obtain or simulate a fresh executable entry; and persist the feature snapshot. A confidence-weighted price can use

$$w_q = e^{-age_q/\tau_a} \min(1, size_q/v_0) w_d e^{-dispersion_q/\tau_\sigma}, \quad P^* = \frac{\sum_q w_q P_q}{\sum_q w_q}.$$

Never use future prints through symmetric smoothing.

16.3 Five- and ten-day forecast

Benchmark each horizon with

$$y_{i,t,H}^{exec} = a_H + b_H z + d_H zc + e_H z p^{info} + f_H z p^{complete} + u_{i,t,H}.$$

For dislocations estimate $D_{t+h} = D_t e^{-\kappa h} + \epsilon$ and expected capture $D_t(1 - e^{-\kappa H})$; for information diffusion use a bounded catch-up fraction. This gives an economic mapping from state to the 5d/10d horizon.

16.4 Execution and costs

Estimate jointly

$$p^{fill} = \mathbb{P}(fill \mid X, size, protocol), \quad (58)$$

$$C = g(size, spread, ADV, depth, vol, flow, toxicity, protocol), \quad (59)$$

$$\mu_H = \mathbb{E}[y_H^{exec} \mid fill, X]. \quad (60)$$

Use a fresh executable entry, never the historical print that revealed the signal. Model quote withdrawal and adverse selection. Relative-value packages pay every leg. Capacity is evaluated by rerunning net P&L at increasing requested size and participation constraints.

16.5 Falsification and monitoring

Kill the feature if it disappears with fresh independent prices, relies on unavailable data, is concentrated in a few issuers/events, or loses its edge under realistic fills and costs. Run leave-one-issuer-out tests, IG/HY and liquidity splits, calm/stress regimes, protocol splits and alternative residualisations. Monitor drift, price confidence, fill calibration, realised versus predicted costs, mark-outs and residual factor exposure.

16.6 Signal-specific mechanism and implementation

Data: Permissioned completed RFQs, including failures | Target: Base; relative version possible | 5 and 10 days
 $\text{RFQ_score} = \text{signed_demand_surprise} + \text{fitted_response_skew} + \text{coverage_state}$ Mechanism and evidence. RFQ activity reveals demand and willingness to supply liquidity beyond completed trades, but that information is highly selected and often confidential. Electronic-auction research supports studying the protocol, not the particular alpha formula. This candidate requires a legitimate historical dataset including unsuccessful inquiries. Feature and implementation. For completed or appropriately delayed, aggregated RFQs, record customer-side direction, requested size, recipients, response count, quote dispersion, response latency, winning price and trade/no-trade result. Do not equate a customer buy inquiry with a completed purchase. Model signed demand relative to each bond's normal inquiry intensity and control for repeated shopping of one parent order. Define response skew relative to a contemporaneous fair-price benchmark with an explicit sign convention; learn its relationship to continuation and pressure rather than guessing from raw quote count. Include dealer selection and client mix where permitted and properly anonymised. Five-/ten-day forecast and hedge. Train separate five-/ten-day prediction and fill models. A tentative long state is positive completed-demand surprise corroborated by upward reservation-price revisions; a no-response episode can instead signal deteriorating liquidity. Use only fully matured historical labels for response-quality statistics. A bond-versus-sibling version is needed to survive strict issuer-neutralisation when issuer-wide demand explains most of the effect. Estimate whether this feature adds to signed trades rather than reproducing them. Executable entry and costs. Maintain information barriers and do not trade ahead of live identifiable client orders. The backtest must replay the same inquiry and routing policy as production; changing the number of dealers changes both information and fills. Include information leakage, failed inquiries and adverse selection in cost estimates. Use actual historical request sizes rather than assuming unlimited execution at the best recorded response. Falsification / kill test. The main failure is selection bias: a dataset of winning quotes alone omits the most difficult trades. Veto source uses without appropriate permissions. Require out-of-sample performance across routing-policy changes, leave-dealer-out tests and a comparison with price-only and trade-flow baselines. Reject apparent alpha driven by future RFQ completion status, by repeated counting of one order, or by a fill model that assumes the best quote was always attainable.

16.7 Agent implementation runbook: data discovery to safe production

Operating rule. The agent must advance through auditable gates rather than implement the entire alpha in one pass. If a required field is unavailable, ambiguous, delayed, or not historically point-in-time, stop that branch and record the limitation; never silently substitute a convenient proxy.

- A1. Data discovery.** Inventory candidate sources/tables/files; definitions; keys; units; time zones; economic, publication and receipt timestamps; corrections/cancels; historical coverage; entitlements; delayed-history rules; and whether each field was genuinely observable live. Output schema and coverage diagnostics only.
- A2. Raw profiling.** Report rows, bonds, issuers, days, missingness, duplicates, timestamp inversions, corrections, size distributions and join success by IG/HY, liquidity, sector and period. Do not impute yet.
- A3. Canonical identifiers.** Version the security master and map security IDs to legal obligor, parent, guarantee, seniority, currency, coupon, maturity, option schedule and corporate actions. Quarantine ambiguous/time-varying mappings.
- A4. As-of layer.** Require every query to accept a decision timestamp and return only records then available. Unit-test late records, corrections and cancels. Persist available-at and provenance fields.
- A5. Price layer.** Normalize clean/dirty price, accrued, spread convention, side, size and quote status. Create freshness, executable-size, dispersion and dealer-breadth metadata. Any forward fill must explicitly age and lose confidence.
- A6. Analytics layer.** Produce cash flows, yield/OAS as appropriate, spread duration, convexity, KRD, CS01/DV01, carry and roll from point-in-time terms; reconcile a sample against an independent implementation.
- A7. Targets.** Build 5d/10d efficient-price diagnostic and executable-return labels. Estimate residualisation only on prior data; cost every changed hedge leg, fees, financing and impact. Persist label maturity for purging.
- A8. Signal-specific prerequisites.** Execute the following in order, emitting exclusions, distributions and hand-audited examples after each sub-step:

- a) Discover RFQ inquiries/responses, side, requested size, dealer count, response times, cover/dispersion and completion status.
- b) Define signed demand, response skew and coverage; normalize by bond/liquidity/size bucket on prior data.
- c) Estimate component weights inside training folds or expose components separately; do not tune equal-weight composites on test outcomes.
- d) Separate information-rich from liquidity-seeking RFQs using dispersion, cross-market confirmation and quote revisions.
- e) Model winner quote, no-trade RFQs, partial size, response latency and selection into completed trades.
- f) Ablate dealer count, size, protocol and completion; verify every response field was observable before the trading decision.

- A9. Literal raw feature.** Translate the documented equation exactly. Assert units, sign, finite values, peer counts, timestamp ordering and boundaries. Hand-reconcile at least 20 events.
- A10. Causal transforms.** Add rolling median/MAD z-score, changes, persistence, acceleration and percentile with past-only windows. Fit curves/betas/normalizers only inside training windows and version them.
- A11. State variables.** Join price confidence, information/toxicity, pressure completion, liquidity regime, volatility, event age and requested size as separate columns before tuning interactions.
- A12. Leakage tests.** Append records arriving after decision time and prove historical feature values do not change. Compare live-available with enhanced-history fields and test shifted timestamps. Any failure blocks modelling.
- A13. Measurement tests.** Recompute with fresh quotes only, trades only, independent prices where possible, alternative stale thresholds and stricter confidence. Reject effects caused only by asynchronous updating.
- A14. Transparent baseline.** Fit horizon-specific linear/robust models in walk-forward folds before nonlinear models; record sign, stability, sample size and incremental information.
- A15. State/convergence models.** Estimate information versus temporary-pressure probabilities and convergence speed where meaningful. Cross-fit state probabilities used as features.
- A16. Execution simulator.** Request a new executable entry after the signal timestamp; apply latency, size, withdrawal, partial-fill and legging rules. No executable observation means unfilled, not a midpoint fill.
- A17. Fill/cost models.** Estimate fill probability and conditional spread/impact/fees/financing by size, protocol and state. Calibrate on held-out fills and cost every changed leg.
- A18. Net outcomes.** Store expected executable return conditional on fill, conservative lower quantile and capacity. Keep gross, mark, executable and fully-net outcomes as distinct audit columns.
- A19. Adversarial validation.** Use chronological purged folds, issuer holdouts, liquidity/rating/regime splits, event exclusions, alternative residualisations, placebo dates and within-family ablations.
- A20. Shadow mode.** Freeze versions and generate live predictions without trading. Log source timestamps, features, quotes, hypothetical orders, predicted fills/costs and 1d/5d/10d mark-outs.
- A21. Canary.** Start with strict size, concentration, liquidity and turnover limits. Kill automatically on timestamp anomalies, data gaps, feature drift, cost underprediction, unexplained residual beta or repeated adverse selection.
- A22. Promotion/rollback.** Promote only a fully reproducible version. Default to no-trade when health checks fail and retain one-command rollback to the prior approved model.

Incremental safe prompts. Use these as separate prompts/commits; each prompt may inspect prior artifacts but must not expand scope silently.

- P1.** “Inventory only the required data, schemas, keys, timestamps, live-history availability, entitlements, coverage and unresolved questions. Do not build a feature.”
- P2.** “Build and unit-test canonical identifiers and as-of joins. Return diagnostics and quarantined ambiguities. Do not calculate alpha.”

- P3.** “Build the price/analytics layer with freshness/confidence and reconcile a hand-audited sample. Stop on unexplained discrepancies.”
- P4.** “Implement only the literal raw signal with assertions and 20 hand-reconciled examples. Do not optimize parameters.”
- P5.** “Add past-only transforms and state variables. Run leakage and measurement-artifact tests before fitting any predictive model.”
- P6.** “Fit the transparent walk-forward baseline for 5d/10d. Report coefficients, uncertainty, ablations and failures; do not tune on the final test period.”
- P7.** “Add execution, fill and transaction-cost simulation. Recompute conditional executable net outcomes and mark unavailable opportunities unfilled.”
- P8.** “Run adversarial validation and within-family ablations. Produce a go/no-go research recommendation based only on predeclared gates.”
- P9.** “Run shadow mode with frozen versions. Do not send orders. Compare predicted versus observable fills, costs and mark-outs.”
- P10.** “Only after explicit human approval, create a tightly limited canary configuration with hard risk limits, health checks and rollback; do not broaden limits automatically.”

16.8 Production row and acceptance gate

Persist raw score, z-score, event age, quote age, peer count/effective sample size, confidence, state probabilities, size/ADV, expected spread/impact, issuer/sector/rating/tenor, KRD/CS01 and source timestamps. The feature graduates only if it adds incremental out-of-sample information after other members of its economic family, survives alternative residualisations and retains a positive conservative executable-net-return statistic at realistic capacity.

17 Signal 16: ETF basket-specific pressure and completion

17.1 Mathematics and feature construction

The canonical raw score is

$$BP_{i,t} = BasketWeight_{i,t}, Flow_t, Scarcity_{i,t}, p_{i,t}^{complete}, \quad (61)$$

Retain level, one- and three-day change, persistence, acceleration, trailing percentile and robust z-score. Add interactions with $c_{i,t}$, $p_{i,t}^{info}$, $p_{i,t}^{complete}$, liquidity regime, volatility and requested size. Thresholds and normalisation are learned on training data only. Missingness is explicit.

17.2 Point-in-time implementation

Maintain an immutable as-of ledger with economic, publication and receipt timestamps, corrections/cancels, security-master version and entitlements. At each decision time freeze the universe; resolve terms and corporate actions; construct fresh prices; compute peers/factors from earlier data; calculate the feature; obtain or simulate a fresh executable entry; and persist the feature snapshot. A confidence-weighted price can use

$$w_q = e^{-age_q/\tau_a} \min(1, size_q/v_0) w_d e^{-dispersion_q/\tau_\sigma}, \quad P^* = \frac{\sum_q w_q P_q}{\sum_q w_q}.$$

Never use future prints through symmetric smoothing.

17.3 Five- and ten-day forecast

Benchmark each horizon with

$$y_{i,t,H}^{exec} = a_H + b_H z + d_H zc + e_H zp^{info} + f_H zp^{complete} + u_{i,t,H}.$$

For dislocations estimate $D_{t+h} = D_t e^{-\kappa h} + \epsilon$ and expected capture $D_t(1 - e^{-\kappa H})$; for information diffusion use a bounded catch-up fraction. This gives an economic mapping from state to the 5d/10d horizon.

17.4 Execution and costs

Estimate jointly

$$p^{fill} = \mathbb{P}(fill \mid X, size, protocol), \quad (62)$$

$$C = g(size, spread, ADV, depth, vol, flow, toxicity, protocol), \quad (63)$$

$$\mu_H = \mathbb{E}[y_H^{exec} \mid fill, X]. \quad (64)$$

Use a fresh executable entry, never the historical print that revealed the signal. Model quote withdrawal and adverse selection. Relative-value packages pay every leg. Capacity is evaluated by rerunning net P&L at increasing requested size and participation constraints.

17.5 Falsification and monitoring

Kill the feature if it disappears with fresh independent prices, relies on unavailable data, is concentrated in a few issuers/events, or loses its edge under realistic fills and costs. Run leave-one-issuer-out tests, IG/HY and liquidity splits, calm/stress regimes, protocol splits and alternative residualisations. Monitor drift, price confidence, fill calibration, realised versus predicted costs, mark-outs and residual factor exposure.

17.6 Signal-specific mechanism and implementation

Data: Historical actual baskets + published flows | Target: Base and issuer-relative | 5 and 10 days pressure_i = sum_f(published_redemption_units_f * actual_basket_quantity_f, i) Mechanism and evidence. ETF creation/redemption activity can redistribute specific bonds, but the actual baskets need not resemble pro-rata holdings.

The relevant hypothesis is temporary pressure in bonds actually delivered or received, followed by observable completion. The BIS study supports the basket distinction, not a universal sign or a ten-day reversal rule. Feature and implementation. Archive the actual historical creation and redemption baskets, cash substitutions, share-unit sizes and publication times for each fund. Combine only as-of-known flow/creation information with basket quantities, then normalise estimated bond supply by issue size and realistic trading capacity. Separate delivery to an authorised participant from a verified sale by that participant: redemptions alone do not establish immediate market selling. Use completed prints, dealer interest and quote concessions as confirmation. Where negotiated custom baskets are unobserved, carry an explicit missing-exposure flag rather than substituting holdings weights as truth. Five-/ten-day forecast and hedge. Estimate five-/ten-day responses after a confirmed basket-related supply event and after a separately defined completion signal. Persistent redemption waves may cause continued pressure instead of reversal. Fit creation and redemption effects independently, with liquidity-regime interactions. Remove broad ETF/credit moves and compare affected bonds with matched same-issuer or same-sector controls; distinguish the basket-pressure signal from ETF price discovery in signal 08. Executable entry and costs. Require a remaining executable discount in the affected bond after the basket and flow information actually became available. Count all hedge/round-trip costs and the possibility that an authorised participant warehouses the bond rather than selling it. Ordinary strategy returns must not include primary-market ETF creation privileges, negotiated baskets or financing unavailable to the strategy. Use a smaller eligible universe rather than silently inventing historical basket data. Falsification / kill test. Kill any version driven by pro-rata-holdings assumptions, publication-time leakage or a presumed sale for every redeemed bond. Compare known-basket exposure with placebo basket assignments and unexposed siblings. Reject a rebound model that only works after choosing the event's final redemption day with hindsight. Data uncertainty and event clustering make this a specialist candidate, not a first-line public-data alpha.

17.7 Agent implementation runbook: data discovery to safe production

Operating rule. The agent must advance through auditable gates rather than implement the entire alpha in one pass. If a required field is unavailable, ambiguous, delayed, or not historically point-in-time, stop that branch and record the limitation; never silently substitute a convenient proxy.

- A1. Data discovery.** Inventory candidate sources/tables/files; definitions; keys; units; time zones; economic, publication and receipt timestamps; corrections/cancels; historical coverage; entitlements; delayed-history rules; and whether each field was genuinely observable live. Output schema and coverage diagnostics only.
- A2. Raw profiling.** Report rows, bonds, issuers, days, missingness, duplicates, timestamp inversions, corrections, size distributions and join success by IG/HY, liquidity, sector and period. Do not impute yet.
- A3. Canonical identifiers.** Version the security master and map security IDs to legal obligor, parent, guarantee, seniority, currency, coupon, maturity, option schedule and corporate actions. Quarantine ambiguous/time-varying mappings.
- A4. As-of layer.** Require every query to accept a decision timestamp and return only records then available. Unit-test late records, corrections and cancels. Persist available-at and provenance fields.
- A5. Price layer.** Normalize clean/dirty price, accrued, spread convention, side, size and quote status. Create freshness, executable-size, dispersion and dealer-breadth metadata. Any forward fill must explicitly age and lose confidence.
- A6. Analytics layer.** Produce cash flows, yield/OAS as appropriate, spread duration, convexity, KRD, CS01/DV01, carry and roll from point-in-time terms; reconcile a sample against an independent implementation.
- A7. Targets.** Build 5d/10d efficient-price diagnostic and executable-return labels. Estimate residualisation only on prior data; cost every changed hedge leg, fees, financing and impact. Persist label maturity for purging.
- A8. Signal-specific prerequisites.** Execute the following in order, emitting exclusions, distributions and hand-audited examples after each sub-step:
 - a) Discover ETF primary creations/redemptions, actual/custom baskets, basket publication times, holdings and cash-bond prices.
 - b) Map each bond to the exact basket version effective at decision time; distinguish creation/redemption and in-kind/cash substitutions.

- c) Estimate expected bond-level mechanical flow from basket weight and primary flow, scaled by scarcity/liquidity/outstanding amount.
- d) Estimate completion from information available at the time; never use end-of-day flow totals before publication.
- e) Refresh the bond executable quote after basket/flow information and model AP sourcing/delivery frictions.
- f) Stress basket substitutions, premium/discount, flow timing and holdings-proxy error; require actual-basket versions to beat naive pro-rata proxies.

- A9. Literal raw feature.** Translate the documented equation exactly. Assert units, sign, finite values, peer counts, timestamp ordering and boundaries. Hand-reconcile at least 20 events.
- A10. Causal transforms.** Add rolling median/MAD z-score, changes, persistence, acceleration and percentile with past-only windows. Fit curves/betas/normalizers only inside training windows and version them.
- A11. State variables.** Join price confidence, information/toxicity, pressure completion, liquidity regime, volatility, event age and requested size as separate columns before tuning interactions.
- A12. Leakage tests.** Append records arriving after decision time and prove historical feature values do not change. Compare live-available with enhanced-history fields and test shifted timestamps. Any failure blocks modelling.
- A13. Measurement tests.** Recompute with fresh quotes only, trades only, independent prices where possible, alternative stale thresholds and stricter confidence. Reject effects caused only by asynchronous updating.
- A14. Transparent baseline.** Fit horizon-specific linear/robust models in walk-forward folds before nonlinear models; record sign, stability, sample size and incremental information.
- A15. State/convergence models.** Estimate information versus temporary-pressure probabilities and convergence speed where meaningful. Cross-fit state probabilities used as features.
- A16. Execution simulator.** Request a new executable entry after the signal timestamp; apply latency, size, withdrawal, partial-fill and legging rules. No executable observation means unfilled, not a midpoint fill.
- A17. Fill/cost models.** Estimate fill probability and conditional spread/impact/fees/financing by size, protocol and state. Calibrate on held-out fills and cost every changed leg.
- A18. Net outcomes.** Store expected executable return conditional on fill, conservative lower quantile and capacity. Keep gross, mark, executable and fully-net outcomes as distinct audit columns.
- A19. Adversarial validation.** Use chronological purged folds, issuer holdouts, liquidity/rating/regime splits, event exclusions, alternative residualisations, placebo dates and within-family ablations.
- A20. Shadow mode.** Freeze versions and generate live predictions without trading. Log source timestamps, features, quotes, hypothetical orders, predicted fills/costs and 1d/5d/10d mark-outs.
- A21. Canary.** Start with strict size, concentration, liquidity and turnover limits. Kill automatically on timestamp anomalies, data gaps, feature drift, cost underprediction, unexplained residual beta or repeated adverse selection.
- A22. Promotion/rollback.** Promote only a fully reproducible version. Default to no-trade when health checks fail and retain one-command rollback to the prior approved model.

Incremental safe prompts. Use these as separate prompts/commits; each prompt may inspect prior artifacts but must not expand scope silently.

- P1.** “Inventory only the required data, schemas, keys, timestamps, live-history availability, entitlements, coverage and unresolved questions. Do not build a feature.”
- P2.** “Build and unit-test canonical identifiers and as-of joins. Return diagnostics and quarantined ambiguities. Do not calculate alpha.”
- P3.** “Build the price/analytics layer with freshness/confidence and reconcile a hand-audited sample. Stop on unexplained discrepancies.”

- P4.** “Implement only the literal raw signal with assertions and 20 hand-reconciled examples. Do not optimize parameters.”
- P5.** “Add past-only transforms and state variables. Run leakage and measurement-artifact tests before fitting any predictive model.”
- P6.** “Fit the transparent walk-forward baseline for 5d/10d. Report coefficients, uncertainty, ablations and failures; do not tune on the final test period.”
- P7.** “Add execution, fill and transaction-cost simulation. Recompute conditional executable net outcomes and mark unavailable opportunities unfilled.”
- P8.** “Run adversarial validation and within-family ablations. Produce a go/no-go research recommendation based only on predeclared gates.”
- P9.** “Run shadow mode with frozen versions. Do not send orders. Compare predicted versus observable fills, costs and mark-outs.”
- P10.** “Only after explicit human approval, create a tightly limited canary configuration with hard risk limits, health checks and rollback; do not broaden limits automatically.”

17.8 Production row and acceptance gate

Persist raw score, z-score, event age, quote age, peer count/effective sample size, confidence, state probabilities, size/ADV, expected spread/impact, issuer/sector/rating/tenor, KRD/CS01 and source timestamps. The feature graduates only if it adds incremental out-of-sample information after other members of its economic family, survives alternative residualisations and retains a positive conservative executable-net-return statistic at realistic capacity.

18 Signal 17: Fund outflow pressure after stabilisation

18.1 Mathematics and feature construction

The canonical raw score is

$$FP_{i,t} = Exposure_{i,t}, OutflowShock_t, SalePropensity_{i,t}, p_{i,t}^{stabilise}, \quad (65)$$

Retain level, one- and three-day change, persistence, acceleration, trailing percentile and robust z-score. Add interactions with $c_{i,t}$, $p_{i,t}^{info}$, $p_{i,t}^{complete}$, liquidity regime, volatility and requested size. Thresholds and normalisation are learned on training data only. Missingness is explicit.

18.2 Point-in-time implementation

Maintain an immutable as-of ledger with economic, publication and receipt timestamps, corrections/cancels, security-master version and entitlements. At each decision time freeze the universe; resolve terms and corporate actions; construct fresh prices; compute peers/factors from earlier data; calculate the feature; obtain or simulate a fresh executable entry; and persist the feature snapshot. A confidence-weighted price can use

$$w_q = e^{-age_q/\tau_a} \min(1, size_q/v_0) w_d e^{-dispersion_q/\tau_\sigma}, \quad P^* = \frac{\sum_q w_q P_q}{\sum_q w_q}.$$

Never use future prints through symmetric smoothing.

18.3 Five- and ten-day forecast

Benchmark each horizon with

$$y_{i,t,H}^{exec} = a_H + b_H z + d_H zc + e_H z p^{info} + f_H z p^{complete} + u_{i,t,H}.$$

For dislocations estimate $D_{t+h} = D_t e^{-\kappa h} + \epsilon$ and expected capture $D_t(1 - e^{-\kappa H})$; for information diffusion use a bounded catch-up fraction. This gives an economic mapping from state to the 5d/10d horizon.

18.4 Execution and costs

Estimate jointly

$$p^{fill} = \mathbb{P}(fill \mid X, size, protocol), \quad (66)$$

$$C = g(size, spread, ADV, depth, vol, flow, toxicity, protocol), \quad (67)$$

$$\mu_H = \mathbb{E}[y_H^{exec} \mid fill, X]. \quad (68)$$

Use a fresh executable entry, never the historical print that revealed the signal. Model quote withdrawal and adverse selection. Relative-value packages pay every leg. Capacity is evaluated by rerunning net P&L at increasing requested size and participation constraints.

18.5 Falsification and monitoring

Kill the feature if it disappears with fresh independent prices, relies on unavailable data, is concentrated in a few issuers/events, or loses its edge under realistic fills and costs. Run leave-one-issuer-out tests, IG/HY and liquidity splits, calm/stress regimes, protocol splits and alternative residualisations. Monitor drift, price confidence, fill calibration, realised versus predicted costs, mark-outs and residual factor exposure.

18.6 Signal-specific mechanism and implementation

Data: Point-in-time holdings + timely licensed flows | Target: Base and issuer-relative | 10-day emphasis; test 5 forced_supply_i = sum_f(last_published_holding_f,i * current_known_outflow_fraction_f) Mechanism and evidence. Corporate-bond fund-flow research supports fragility and forced-selling mechanisms. It does not establish that publicly delayed holdings and fund reports can produce a timely five-day reversal strategy. This candidate needs unusually good as-of data and an explicit distinction between continued selling and completed pressure. Feature and implementation. Use the last holdings snapshot actually published by each fund, not a later reconstructed portfolio. Estimate outflow-driven dollar demand from flow data available before the decision; reconcile flow-frequency and publication lag. Weight exposures by fund liquidity buffer, concentration and likelihood of selling the bond, while acknowledging that managers choose what to sell. Divide predicted supply by plausible bond trading capacity. Separate predicted pressure from confirmation in actual sales and executable spreads. Preserve holdings age and prediction uncertainty; stale holdings must reduce confidence rather than be treated as precise current ownership. Five-/ten-day forecast and hedge. The reversal proposal buys a remaining discount only after observable outflow intensity subsides and two-sided liquidity begins returning. A continuation-risk score should veto entry while outflows accelerate. Fit five- and ten-day returns from that decision, not from the best retrospective event trough. A ten-day emphasis is an economic hypothesis, not a claim of an established half-life. Remove common sector/credit moves and, where possible, compare similarly exposed siblings with different fund ownership. Executable entry and costs. Count the cost of taking the side opposite forced sellers and the realistic exit if outflows resume. Do not assume the strategy can access a fund's sale price after a report is published. Model a finite executable position size relative to trading capacity and include correlated exposure across many bonds held by the same funds. A market-wide outflow episode can concentrate nearly every nominally diversified trade in one liquidity factor. Falsification / kill test. Kill the signal if daily flow estimates are backfilled, if future holdings reveal which bonds were sold, or if public reporting delays consume the full forecast horizon. Perform placebo tests on publication dates versus economic period-end dates. Require value beyond observed bond flow and reversal. Without timely licensed inputs this is better used as a slow risk-control variable than advertised as a live short-horizon alpha.

18.7 Agent implementation runbook: data discovery to safe production

Operating rule. The agent must advance through auditable gates rather than implement the entire alpha in one pass. If a required field is unavailable, ambiguous, delayed, or not historically point-in-time, stop that branch and record the limitation; never silently substitute a convenient proxy.

- A1. Data discovery.** Inventory candidate sources/tables/files; definitions; keys; units; time zones; economic, publication and receipt timestamps; corrections/cancels; historical coverage; entitlements; delayed-history rules; and whether each field was genuinely observable live. Output schema and coverage diagnostics only.
- A2. Raw profiling.** Report rows, bonds, issuers, days, missingness, duplicates, timestamp inversions, corrections, size distributions and join success by IG/HY, liquidity, sector and period. Do not impute yet.
- A3. Canonical identifiers.** Version the security master and map security IDs to legal obligor, parent, guarantee, seniority, currency, coupon, maturity, option schedule and corporate actions. Quarantine ambiguous/time-varying mappings.
- A4. As-of layer.** Require every query to accept a decision timestamp and return only records then available. Unit-test late records, corrections and cancels. Persist available-at and provenance fields.
- A5. Price layer.** Normalize clean/dirty price, accrued, spread convention, side, size and quote status. Create freshness, executable-size, dispersion and dealer-breadth metadata. Any forward fill must explicitly age and lose confidence.
- A6. Analytics layer.** Produce cash flows, yield/OAS as appropriate, spread duration, convexity, KRD, CS01/DV01, carry and roll from point-in-time terms; reconcile a sample against an independent implementation.
- A7. Targets.** Build 5d/10d efficient-price diagnostic and executable-return labels. Estimate residualisation only on prior data; cost every changed hedge leg, fees, financing and impact. Persist label maturity for purging.
- A8. Signal-specific prerequisites.** Execute the following in order, emitting exclusions, distributions and hand-audited examples after each sub-step:

- a) Discover fund flows, holdings, reporting lags, liquidity buffers and bond prices; record when each holdings snapshot became public/available.
- b) Construct exposure only from then-known holdings and model sale propensity rather than assuming pro-rata liquidation.
- c) Aggregate predicted selling pressure across funds while controlling issuer-time information and liquidity management.
- d) Estimate stabilization from flow deceleration and liquidity recovery; do not buy mechanically during continuing information shocks.
- e) Use fresh executable prices and separate delayed-public-data implementations from timely proprietary holdings implementations.
- f) Run same-issuer controls and zero-fire-sale nulls; require strong validation because published evidence finds little average fire-sale pressure after issuer-time controls.

- A9. Literal raw feature.** Translate the documented equation exactly. Assert units, sign, finite values, peer counts, timestamp ordering and boundaries. Hand-reconcile at least 20 events.
- A10. Causal transforms.** Add rolling median/MAD z-score, changes, persistence, acceleration and percentile with past-only windows. Fit curves/betas/normalizers only inside training windows and version them.
- A11. State variables.** Join price confidence, information/toxicity, pressure completion, liquidity regime, volatility, event age and requested size as separate columns before tuning interactions.
- A12. Leakage tests.** Append records arriving after decision time and prove historical feature values do not change. Compare live-available with enhanced-history fields and test shifted timestamps. Any failure blocks modelling.
- A13. Measurement tests.** Recompute with fresh quotes only, trades only, independent prices where possible, alternative stale thresholds and stricter confidence. Reject effects caused only by asynchronous updating.
- A14. Transparent baseline.** Fit horizon-specific linear/robust models in walk-forward folds before nonlinear models; record sign, stability, sample size and incremental information.
- A15. State/convergence models.** Estimate information versus temporary-pressure probabilities and convergence speed where meaningful. Cross-fit state probabilities used as features.
- A16. Execution simulator.** Request a new executable entry after the signal timestamp; apply latency, size, withdrawal, partial-fill and legging rules. No executable observation means unfilled, not a midpoint fill.
- A17. Fill/cost models.** Estimate fill probability and conditional spread/impact/fees/financing by size, protocol and state. Calibrate on held-out fills and cost every changed leg.
- A18. Net outcomes.** Store expected executable return conditional on fill, conservative lower quantile and capacity. Keep gross, mark, executable and fully-net outcomes as distinct audit columns.
- A19. Adversarial validation.** Use chronological purged folds, issuer holdouts, liquidity/rating/regime splits, event exclusions, alternative residualisations, placebo dates and within-family ablations.
- A20. Shadow mode.** Freeze versions and generate live predictions without trading. Log source timestamps, features, quotes, hypothetical orders, predicted fills/costs and 1d/5d/10d mark-outs.
- A21. Canary.** Start with strict size, concentration, liquidity and turnover limits. Kill automatically on timestamp anomalies, data gaps, feature drift, cost underprediction, unexplained residual beta or repeated adverse selection.
- A22. Promotion/rollback.** Promote only a fully reproducible version. Default to no-trade when health checks fail and retain one-command rollback to the prior approved model.

Incremental safe prompts. Use these as separate prompts/commits; each prompt may inspect prior artifacts but must not expand scope silently.

- P1.** “Inventory only the required data, schemas, keys, timestamps, live-history availability, entitlements, coverage and unresolved questions. Do not build a feature.”
- P2.** “Build and unit-test canonical identifiers and as-of joins. Return diagnostics and quarantined ambiguities. Do not calculate alpha.”
- P3.** “Build the price/analytics layer with freshness/confidence and reconcile a hand-audited sample. Stop on unexplained discrepancies.”
- P4.** “Implement only the literal raw signal with assertions and 20 hand-reconciled examples. Do not optimize parameters.”
- P5.** “Add past-only transforms and state variables. Run leakage and measurement-artifact tests before fitting any predictive model.”
- P6.** “Fit the transparent walk-forward baseline for 5d/10d. Report coefficients, uncertainty, ablations and failures; do not tune on the final test period.”
- P7.** “Add execution, fill and transaction-cost simulation. Recompute conditional executable net outcomes and mark unavailable opportunities unfilled.”
- P8.** “Run adversarial validation and within-family ablations. Produce a go/no-go research recommendation based only on predeclared gates.”
- P9.** “Run shadow mode with frozen versions. Do not send orders. Compare predicted versus observable fills, costs and mark-outs.”
- P10.** “Only after explicit human approval, create a tightly limited canary configuration with hard risk limits, health checks and rollback; do not broaden limits automatically.”

18.8 Production row and acceptance gate

Persist raw score, z-score, event age, quote age, peer count/effective sample size, confidence, state probabilities, size/ADV, expected spread/impact, issuer/sector/rating/tenor, KRD/CS01 and source timestamps. The feature graduates only if it adds incremental out-of-sample information after other members of its economic family, survives alternative residualisations and retains a positive conservative executable-net-return statistic at realistic capacity.

19 Signal 18: Remaining new-issue concession / seasoning

19.1 Mathematics and feature construction

The canonical raw score is

$$NIC_{i,t} = s_{i,t}^{new} - \hat{s}_{issuer,t}^{peer}(T_i) - Adj_{terms,liq}. \quad (69)$$

Retain level, one- and three-day change, persistence, acceleration, trailing percentile and robust z-score. Add interactions with $c_{i,t}$, $p_{i,t}^{info}$, $p_{i,t}^{complete}$, liquidity regime, volatility and requested size. Thresholds and normalisation are learned on training data only. Missingness is explicit.

19.2 Point-in-time implementation

Maintain an immutable as-of ledger with economic, publication and receipt timestamps, corrections/cancels, security-master version and entitlements. At each decision time freeze the universe; resolve terms and corporate actions; construct fresh prices; compute peers/factors from earlier data; calculate the feature; obtain or simulate a fresh executable entry; and persist the feature snapshot. A confidence-weighted price can use

$$w_q = e^{-age_q/\tau_a} \min(1, size_q/v_0) w_d e^{-dispersion_q/\tau_\sigma}, \quad P^* = \frac{\sum_q w_q P_q}{\sum_q w_q}.$$

Never use future prints through symmetric smoothing.

19.3 Five- and ten-day forecast

Benchmark each horizon with

$$y_{i,t,H}^{exec} = a_H + b_H z + d_H zc + e_H zp^{info} + f_H zp^{complete} + u_{i,t,H}.$$

For dislocations estimate $D_{t+h} = D_t e^{-\kappa h} + \epsilon$ and expected capture $D_t(1 - e^{-\kappa H})$; for information diffusion use a bounded catch-up fraction. This gives an economic mapping from state to the 5d/10d horizon.

19.4 Execution and costs

Estimate jointly

$$p^{fill} = \mathbb{P}(fill \mid X, size, protocol), \quad (70)$$

$$C = g(size, spread, ADV, depth, vol, flow, toxicity, protocol), \quad (71)$$

$$\mu_H = \mathbb{E}[y_H^{exec} \mid fill, X]. \quad (72)$$

Use a fresh executable entry, never the historical print that revealed the signal. Model quote withdrawal and adverse selection. Relative-value packages pay every leg. Capacity is evaluated by rerunning net P&L at increasing requested size and participation constraints.

19.5 Falsification and monitoring

Kill the feature if it disappears with fresh independent prices, relies on unavailable data, is concentrated in a few issuers/events, or loses its edge under realistic fills and costs. Run leave-one-issuer-out tests, IG/HY and liquidity splits, calm/stress regimes, protocol splits and alternative residualisations. Monitor drift, price confidence, fill calibration, realised versus predicted costs, mark-outs and residual factor exposure.

19.6 Signal-specific mechanism and implementation

Data: Issuance terms + allocation/fill + peer curve | Target: Base and issuer-neutral | 5 and 10 days NIC_remaining = executable_new_issue_spread - adjusted_secondary_fair_spread Mechanism and evidence. Primary-market

intermediation can require a concession and inventory capacity affects offering economics. The attainable alpha is any concession remaining at an actual allocation or secondary-market fill, after adjusting for the new bond's structure and liquidity. Published underwriter and large-issue research supports the mechanism, not universal access to offering-price gains. Feature and implementation. Archive announcement, initial guidance, revisions, final pricing, allocation, issue/settlement and secondary-quote timestamps. Build the issuer curve before including the new bond, mapping its maturity, coupon, legal claim and optionality correctly. Adjust for the new issue's potential benchmark liquidity advantage rather than assuming it should trade exactly on an older sibling curve. For primary participation, use only the allocation actually received and the price/fees actually paid. For secondary entry, recompute concession at the available ask after final terms were known. Record deal size relative to issuer debt, syndicate distribution indicators and market conditions without using confidential order-book information improperly. Five-/ten-day forecast and hedge. Fit seasoning from actual fill over five and ten trading days, controlling broad rates/credit and contemporaneous issuer news. Separate refinancing from incremental leverage or acquisition financing. The issuer-neutral target is attractive when hedged against comparable seasoned siblings, but a new capital-structure fact can change all bonds' fair value. Estimate remaining convergence conditional on initial secondary trading and flow, not an unconditional calendar-age premium. Executable entry and costs. Count allocation uncertainty, secondary bid-ask costs, funding from settlement, hedge costs and any delay before the bond is shortable or liquid. Do not apply both an offering-price fill and an already-converged secondary mark to a strategy that received no allocation. Use a no-trade band when expected remaining seasoning is smaller than the round trip. Exit under a pretested policy rather than assuming issuance automatically guarantees liquidity at day ten. Falsification / kill test. Kill the model if gains depend on receiving every attractive allocation, on terms published after the decision or on excluding poorly performing new issues. Model deal cancellations and execution failures. Distinguish true concession from higher leverage, structural subordination, adverse use of proceeds or permanent liquidity differences. Compare incremental net results with issuer-curve value; the event timing, not merely the wide spread, must justify a separate signal.

19.7 Agent implementation runbook: data discovery to safe production

Operating rule. The agent must advance through auditable gates rather than implement the entire alpha in one pass. If a required field is unavailable, ambiguous, delayed, or not historically point-in-time, stop that branch and record the limitation; never silently substitute a convenient proxy.

- A1. Data discovery.** Inventory candidate sources/tables/files; definitions; keys; units; time zones; economic, publication and receipt timestamps; corrections/cancels; historical coverage; entitlements; delayed-history rules; and whether each field was genuinely observable live. Output schema and coverage diagnostics only.
- A2. Raw profiling.** Report rows, bonds, issuers, days, missingness, duplicates, timestamp inversions, corrections, size distributions and join success by IG/HY, liquidity, sector and period. Do not impute yet.
- A3. Canonical identifiers.** Version the security master and map security IDs to legal obligor, parent, guarantee, seniority, currency, coupon, maturity, option schedule and corporate actions. Quarantine ambiguous/time-varying mappings.
- A4. As-of layer.** Require every query to accept a decision timestamp and return only records then available. Unit-test late records, corrections and cancels. Persist available-at and provenance fields.
- A5. Price layer.** Normalize clean/dirty price, accrued, spread convention, side, size and quote status. Create freshness, executable-size, dispersion and dealer-breadth metadata. Any forward fill must explicitly age and lose confidence.
- A6. Analytics layer.** Produce cash flows, yield/OAS as appropriate, spread duration, convexity, KRD, CS01/DV01, carry and roll from point-in-time terms; reconcile a sample against an independent implementation.
- A7. Targets.** Build 5d/10d efficient-price diagnostic and executable-return labels. Estimate residualisation only on prior data; cost every changed hedge leg, fees, financing and impact. Persist label maturity for purging.
- A8. Signal-specific prerequisites.** Execute the following in order, emitting exclusions, distributions and hand-audited examples after each sub-step:
 - a) Discover new-issue terms, IPT/reoffer/final pricing, allocations if entitled, and secondary quotes.
 - b) Create an exact event timeline from announcement through pricing, allocation and secondary seasoning.
 - c) Estimate fair spread from the pre-existing issuer curve with contractual/liquidity adjustments.

- d) Measure remaining concession at attainable allocation or fresh secondary price, not unavailable reoffer.
- e) Model concession decay by deal size, market state and time since pricing.
- f) Evaluate primary-allocation and secondary-entry versions separately because selection and capacity differ.

- A9. Literal raw feature.** Translate the documented equation exactly. Assert units, sign, finite values, peer counts, timestamp ordering and boundaries. Hand-reconcile at least 20 events.
- A10. Causal transforms.** Add rolling median/MAD z-score, changes, persistence, acceleration and percentile with past-only windows. Fit curves/betas/normalizers only inside training windows and version them.
- A11. State variables.** Join price confidence, information/toxicity, pressure completion, liquidity regime, volatility, event age and requested size as separate columns before tuning interactions.
- A12. Leakage tests.** Append records arriving after decision time and prove historical feature values do not change. Compare live-available with enhanced-history fields and test shifted timestamps. Any failure blocks modelling.
- A13. Measurement tests.** Recompute with fresh quotes only, trades only, independent prices where possible, alternative stale thresholds and stricter confidence. Reject effects caused only by asynchronous updating.
- A14. Transparent baseline.** Fit horizon-specific linear/robust models in walk-forward folds before nonlinear models; record sign, stability, sample size and incremental information.
- A15. State/convergence models.** Estimate information versus temporary-pressure probabilities and convergence speed where meaningful. Cross-fit state probabilities used as features.
- A16. Execution simulator.** Request a new executable entry after the signal timestamp; apply latency, size, withdrawal, partial-fill and legging rules. No executable observation means unfilled, not a midpoint fill.
- A17. Fill/cost models.** Estimate fill probability and conditional spread/impact/fees/financing by size, protocol and state. Calibrate on held-out fills and cost every changed leg.
- A18. Net outcomes.** Store expected executable return conditional on fill, conservative lower quantile and capacity. Keep gross, mark, executable and fully-net outcomes as distinct audit columns.
- A19. Adversarial validation.** Use chronological purged folds, issuer holdouts, liquidity/rating/regime splits, event exclusions, alternative residualisations, placebo dates and within-family ablations.
- A20. Shadow mode.** Freeze versions and generate live predictions without trading. Log source timestamps, features, quotes, hypothetical orders, predicted fills/costs and 1d/5d/10d mark-outs.
- A21. Canary.** Start with strict size, concentration, liquidity and turnover limits. Kill automatically on timestamp anomalies, data gaps, feature drift, cost underprediction, unexplained residual beta or repeated adverse selection.
- A22. Promotion/rollback.** Promote only a fully reproducible version. Default to no-trade when health checks fail and retain one-command rollback to the prior approved model.

Incremental safe prompts. Use these as separate prompts/commits; each prompt may inspect prior artifacts but must not expand scope silently.

- P1.** “Inventory only the required data, schemas, keys, timestamps, live-history availability, entitlements, coverage and unresolved questions. Do not build a feature.”
- P2.** “Build and unit-test canonical identifiers and as-of joins. Return diagnostics and quarantined ambiguities. Do not calculate alpha.”
- P3.** “Build the price/analytics layer with freshness/confidence and reconcile a hand-audited sample. Stop on unexplained discrepancies.”
- P4.** “Implement only the literal raw signal with assertions and 20 hand-reconciled examples. Do not optimize parameters.”
- P5.** “Add past-only transforms and state variables. Run leakage and measurement-artifact tests before fitting any predictive model.”

- P6.** “Fit the transparent walk-forward baseline for 5d/10d. Report coefficients, uncertainty, ablations and failures; do not tune on the final test period.”
- P7.** “Add execution, fill and transaction-cost simulation. Recompute conditional executable net outcomes and mark unavailable opportunities unfilled.”
- P8.** “Run adversarial validation and within-family ablations. Produce a go/no-go research recommendation based only on predeclared gates.”
- P9.** “Run shadow mode with frozen versions. Do not send orders. Compare predicted versus observable fills, costs and mark-outs.”
- P10.** “Only after explicit human approval, create a tightly limited canary configuration with hard risk limits, health checks and rollback; do not broaden limits automatically.”

19.8 Production row and acceptance gate

Persist raw score, z-score, event age, quote age, peer count/effective sample size, confidence, state probabilities, size/ADV, expected spread/impact, issuer/sector/rating/tenor, KRD/CS01 and source timestamps. The feature graduates only if it adds incremental out-of-sample information after other members of its economic family, survives alternative residualisations and retains a positive conservative executable-net-return statistic at realistic capacity.

20 Signal 19: Existing-bond issuance spillover recovery

20.1 Mathematics and feature construction

The canonical raw score is

$$ISP_{i,t} = \Delta s_{i,event}^{old} - \widehat{\Delta s_{i,event}}^{cf}. \quad (73)$$

Retain level, one- and three-day change, persistence, acceleration, trailing percentile and robust z-score. Add interactions with $c_{i,t}$, $p_{i,t}^{info}$, $p_{i,t}^{complete}$, liquidity regime, volatility and requested size. Thresholds and normalisation are learned on training data only. Missingness is explicit.

20.2 Point-in-time implementation

Maintain an immutable as-of ledger with economic, publication and receipt timestamps, corrections/cancels, security-master version and entitlements. At each decision time freeze the universe; resolve terms and corporate actions; construct fresh prices; compute peers/factors from earlier data; calculate the feature; obtain or simulate a fresh executable entry; and persist the feature snapshot. A confidence-weighted price can use

$$w_q = e^{-age_q/\tau_a} \min(1, size_q/v_0) w_d e^{-dispersion_q/\tau_\sigma}, \quad P^* = \frac{\sum_q w_q P_q}{\sum_q w_q}.$$

Never use future prints through symmetric smoothing.

20.3 Five- and ten-day forecast

Benchmark each horizon with

$$y_{i,t,H}^{exec} = a_H + b_H z + d_H zc + e_H z p^{info} + f_H z p^{complete} + u_{i,t,H}.$$

For dislocations estimate $D_{t+h} = D_t e^{-\kappa h} + \epsilon$ and expected capture $D_t(1 - e^{-\kappa H})$; for information diffusion use a bounded catch-up fraction. This gives an economic mapping from state to the 5d/10d horizon.

20.4 Execution and costs

Estimate jointly

$$p^{fill} = \mathbb{P}(fill \mid X, size, protocol), \quad (74)$$

$$C = g(size, spread, ADV, depth, vol, flow, toxicity, protocol), \quad (75)$$

$$\mu_H = \mathbb{E}[y_H^{exec} \mid fill, X]. \quad (76)$$

Use a fresh executable entry, never the historical print that revealed the signal. Model quote withdrawal and adverse selection. Relative-value packages pay every leg. Capacity is evaluated by rerunning net P&L at increasing requested size and participation constraints.

20.5 Falsification and monitoring

Kill the feature if it disappears with fresh independent prices, relies on unavailable data, is concentrated in a few issuers/events, or loses its edge under realistic fills and costs. Run leave-one-issuer-out tests, IG/HY and liquidity splits, calm/stress regimes, protocol splits and alternative residualisations. Monitor drift, price confidence, fill calibration, realised versus predicted costs, mark-outs and residual factor exposure.

20.6 Signal-specific mechanism and implementation

Data: Public issuance events + sibling quotes / flow | Target: Base and issuer-neutral | 5 and 10 days spillover = $current_spread_gap - predeal_gap - fundamental_repricing$ Mechanism and evidence. A large new issue can place

price pressure on outstanding bonds of the same issuer. Published mega-bond research finds effects consistent with pressure in both new and existing bonds. The separate proposal here is a tradable recovery in seasoned bonds after distribution, not an assumption that all announcement-driven widening is temporary. Feature and implementation. Create a public event clock for announcement, final pricing and known distribution milestones. Estimate each outstanding bond's abnormal price/spread move relative to its pre-event issuer curve and matched unaffected issuers, correcting for rates and broad credit. Measure supply intensity relative to existing issuer debt and focus on maturities likely to substitute for the new bond. Confirm selling with available completed flow and dealer quotes. Define the score as a positive remaining cheapness caused by the event after removing estimated fundamental repricing. Keep the actual new-issue bond out of this seasoned-bond universe to separate signal 18. Five-/ten-day forecast and hedge. Start the five-/ten-day clock at an observable distribution/completion trigger and feasible fill, not at the retrospectively lowest price. Fit announcement-pressure and post-pricing-recovery phases separately. Under strict issuer-neutralisation, compare affected maturity buckets with less affected siblings; a common issuer-wide rebound will be removed. Use public financing-purpose information to distinguish simple refinancing from transactions that materially alter leverage, recovery or guarantees. Executable entry and costs. Buy only at a currently available discounted price and hedge the genuinely common issuer/rate exposures. Count the possibility of further tranches, deferred distribution and weak dealer balance-sheet capacity. Since the attractive old bond may be less liquid than the new issue, compare the cost-adjusted opportunity in both rather than assuming the seasoned bond is better. Limit overlapping events and issuer-level gross exposure. Falsification / kill test. A leveraged acquisition, tender, security pledge or structural subordination can permanently worsen old bonds. Reject a model that mechanically fades these changes. Use non-issuing matched controls and placebo announcement dates, and test with realistic news timestamps. Require incremental return beyond signal 01 and broad reversal. A pressure mechanism is evidenced; its completion timing and net five-/ten-day recovery are still hypotheses to be falsified.

20.7 Agent implementation runbook: data discovery to safe production

Operating rule. The agent must advance through auditable gates rather than implement the entire alpha in one pass. If a required field is unavailable, ambiguous, delayed, or not historically point-in-time, stop that branch and record the limitation; never silently substitute a convenient proxy.

- A1. Data discovery.** Inventory candidate sources/tables/files; definitions; keys; units; time zones; economic, publication and receipt timestamps; corrections/cancels; historical coverage; entitlements; delayed-history rules; and whether each field was genuinely observable live. Output schema and coverage diagnostics only.
- A2. Raw profiling.** Report rows, bonds, issuers, days, missingness, duplicates, timestamp inversions, corrections, size distributions and join success by IG/HY, liquidity, sector and period. Do not impute yet.
- A3. Canonical identifiers.** Version the security master and map security IDs to legal obligor, parent, guarantee, seniority, currency, coupon, maturity, option schedule and corporate actions. Quarantine ambiguous/time-varying mappings.
- A4. As-of layer.** Require every query to accept a decision timestamp and return only records then available. Unit-test late records, corrections and cancels. Persist available-at and provenance fields.
- A5. Price layer.** Normalize clean/dirty price, accrued, spread convention, side, size and quote status. Create freshness, executable-size, dispersion and dealer-breadth metadata. Any forward fill must explicitly age and lose confidence.
- A6. Analytics layer.** Produce cash flows, yield/OAS as appropriate, spread duration, convexity, KRD, CS01/DV01, carry and roll from point-in-time terms; reconcile a sample against an independent implementation.
- A7. Targets.** Build 5d/10d efficient-price diagnostic and executable-return labels. Estimate residualisation only on prior data; cost every changed hedge leg, fees, financing and impact. Persist label maturity for purging.
- A8. Signal-specific prerequisites.** Execute the following in order, emitting exclusions, distributions and hand-audited examples after each sub-step:
 - a) Discover issuance announcements, IPT/guidance/final pricing, size, use of proceeds, tender/exchange links and existing-bond prices.
 - b) Build event clock and classify refinancing, incremental leverage, secured/subordinated issuance and liability-management links.

- c) Estimate old-bond counterfactual moves from issuer curve, market/sector credit and rates using only pre-event parameters.
- d) Measure residual old-bond cheapening and classify fundamental versus technical components using use-of-proceeds and cross-market confirmation.
- e) Enter only at a fresh executable old-bond quote; hedge rate/issuer risks and cost simultaneous capital-structure repricing.
- f) Ablate refinancing, acquisition financing, tender-linked deals, size and event window; reject effects explained by fundamental leverage changes.

- A9. Literal raw feature.** Translate the documented equation exactly. Assert units, sign, finite values, peer counts, timestamp ordering and boundaries. Hand-reconcile at least 20 events.
- A10. Causal transforms.** Add rolling median/MAD z-score, changes, persistence, acceleration and percentile with past-only windows. Fit curves/betas/normalizers only inside training windows and version them.
- A11. State variables.** Join price confidence, information/toxicity, pressure completion, liquidity regime, volatility, event age and requested size as separate columns before tuning interactions.
- A12. Leakage tests.** Append records arriving after decision time and prove historical feature values do not change. Compare live-available with enhanced-history fields and test shifted timestamps. Any failure blocks modelling.
- A13. Measurement tests.** Recompute with fresh quotes only, trades only, independent prices where possible, alternative stale thresholds and stricter confidence. Reject effects caused only by asynchronous updating.
- A14. Transparent baseline.** Fit horizon-specific linear/robust models in walk-forward folds before nonlinear models; record sign, stability, sample size and incremental information.
- A15. State/convergence models.** Estimate information versus temporary-pressure probabilities and convergence speed where meaningful. Cross-fit state probabilities used as features.
- A16. Execution simulator.** Request a new executable entry after the signal timestamp; apply latency, size, withdrawal, partial-fill and legging rules. No executable observation means unfilled, not a midpoint fill.
- A17. Fill/cost models.** Estimate fill probability and conditional spread/impact/fees/financing by size, protocol and state. Calibrate on held-out fills and cost every changed leg.
- A18. Net outcomes.** Store expected executable return conditional on fill, conservative lower quantile and capacity. Keep gross, mark, executable and fully-net outcomes as distinct audit columns.
- A19. Adversarial validation.** Use chronological purged folds, issuer holdouts, liquidity/rating/regime splits, event exclusions, alternative residualisations, placebo dates and within-family ablations.
- A20. Shadow mode.** Freeze versions and generate live predictions without trading. Log source timestamps, features, quotes, hypothetical orders, predicted fills/costs and 1d/5d/10d mark-outs.
- A21. Canary.** Start with strict size, concentration, liquidity and turnover limits. Kill automatically on timestamp anomalies, data gaps, feature drift, cost underprediction, unexplained residual beta or repeated adverse selection.
- A22. Promotion/rollback.** Promote only a fully reproducible version. Default to no-trade when health checks fail and retain one-command rollback to the prior approved model.

Incremental safe prompts. Use these as separate prompts/commits; each prompt may inspect prior artifacts but must not expand scope silently.

- P1.** “Inventory only the required data, schemas, keys, timestamps, live-history availability, entitlements, coverage and unresolved questions. Do not build a feature.”
- P2.** “Build and unit-test canonical identifiers and as-of joins. Return diagnostics and quarantined ambiguities. Do not calculate alpha.”
- P3.** “Build the price/analytics layer with freshness/confidence and reconcile a hand-audited sample. Stop on unexplained discrepancies.”

- P4.** “Implement only the literal raw signal with assertions and 20 hand-reconciled examples. Do not optimize parameters.”
- P5.** “Add past-only transforms and state variables. Run leakage and measurement-artifact tests before fitting any predictive model.”
- P6.** “Fit the transparent walk-forward baseline for 5d/10d. Report coefficients, uncertainty, ablations and failures; do not tune on the final test period.”
- P7.** “Add execution, fill and transaction-cost simulation. Recompute conditional executable net outcomes and mark unavailable opportunities unfilled.”
- P8.** “Run adversarial validation and within-family ablations. Produce a go/no-go research recommendation based only on predeclared gates.”
- P9.** “Run shadow mode with frozen versions. Do not send orders. Compare predicted versus observable fills, costs and mark-outs.”
- P10.** “Only after explicit human approval, create a tightly limited canary configuration with hard risk limits, health checks and rollback; do not broaden limits automatically.”

20.8 Production row and acceptance gate

Persist raw score, z-score, event age, quote age, peer count/effective sample size, confidence, state probabilities, size/ADV, expected spread/impact, issuer/sector/rating/tenor, KRD/CS01 and source timestamps. The feature graduates only if it adds incremental out-of-sample information after other members of its economic family, survives alternative residualisations and retains a positive conservative executable-net-return statistic at realistic capacity.

21 Signal 20: Mechanical index-pressure completion

21.1 Mathematics and feature construction

The canonical raw score is

$$IDX_{i,t} = ExpectedPassiveFlow_{i,t}, PricePressure_{i,t}, p_{i,t}^{complete}. \quad (77)$$

Retain level, one- and three-day change, persistence, acceleration, trailing percentile and robust z-score. Add interactions with $c_{i,t}$, $p_{i,t}^{info}$, $p_{i,t}^{complete}$, liquidity regime, volatility and requested size. Thresholds and normalisation are learned on training data only. Missingness is explicit.

21.2 Point-in-time implementation

Maintain an immutable as-of ledger with economic, publication and receipt timestamps, corrections/cancels, security-master version and entitlements. At each decision time freeze the universe; resolve terms and corporate actions; construct fresh prices; compute peers/factors from earlier data; calculate the feature; obtain or simulate a fresh executable entry; and persist the feature snapshot. A confidence-weighted price can use

$$w_q = e^{-age_q/\tau_a} \min(1, size_q/v_0) w_d e^{-dispersion_q/\tau_\sigma}, \quad P^* = \frac{\sum_q w_q P_q}{\sum_q w_q}.$$

Never use future prints through symmetric smoothing.

21.3 Five- and ten-day forecast

Benchmark each horizon with

$$y_{i,t,H}^{exec} = a_H + b_H z + d_H zc + e_H z p^{info} + f_H z p^{complete} + u_{i,t,H}.$$

For dislocations estimate $D_{t+h} = D_t e^{-\kappa h} + \epsilon$ and expected capture $D_t(1 - e^{-\kappa H})$; for information diffusion use a bounded catch-up fraction. This gives an economic mapping from state to the 5d/10d horizon.

21.4 Execution and costs

Estimate jointly

$$p^{fill} = \mathbb{P}(fill \mid X, size, protocol), \quad (78)$$

$$C = g(size, spread, ADV, depth, vol, flow, toxicity, protocol), \quad (79)$$

$$\mu_H = \mathbb{E}[y_H^{exec} \mid fill, X]. \quad (80)$$

Use a fresh executable entry, never the historical print that revealed the signal. Model quote withdrawal and adverse selection. Relative-value packages pay every leg. Capacity is evaluated by rerunning net P&L at increasing requested size and participation constraints.

21.5 Falsification and monitoring

Kill the feature if it disappears with fresh independent prices, relies on unavailable data, is concentrated in a few issuers/events, or loses its edge under realistic fills and costs. Run leave-one-issuer-out tests, IG/HY and liquidity splits, calm/stress regimes, protocol splits and alternative residualisations. Monitor drift, price confidence, fill calibration, realised versus predicted costs, mark-outs and residual factor exposure.

21.6 Signal-specific mechanism and implementation

Data: Dated index rules + notices + flow / quotes | Target: Base and issuer-neutral | 5 and 10 days around completion event
 $\text{event_edge} = \text{signed_mechanical_demand} * \text{remaining_concession} * \text{completion_state}$ Mechanism and evidence. Mechanical index changes can create demand for immediacy independent of new issuer information. Research exploiting index exclusions demonstrates real intermediation costs around such events. The proposed alpha takes the opposite side only where a remaining concession pays for those costs; inclusion/deletion is not automatically a free reversal. Feature and implementation. Archive the methodology version, eligibility data, announced membership changes and effective dates known at each decision. Prefer genuinely mechanical maturity or rule-driven exclusions in the initial test; separate credit-rating transitions and other information-rich changes. Estimate affected passive demand from point-in-time tracked assets and holdings, with wide uncertainty for anticipatory trading. Measure the remaining bond-specific price concession against fresh comparable bonds and observed flow. Define event completion from the published effective time plus observed stabilisation, never from a hindsight-selected minimum. Five-/ten-day forecast and hedge. Test both pre-event pressure and post-event recovery, but choose one locked trading policy per experiment. Measure five-/ten-day returns from the actual entry. A fallen angel can remain under selling pressure much longer and carries new default information; do not import a long-horizon fallen-angel premium into this mechanical-event signal. Use issuer-neutral peers when a maturity bucket is affected, and base residuals for the broader event portfolio. Estimate event clustering and month-end common liquidity effects. Executable entry and costs. Use available size and executable prices, allowing for crowded pre-positioning, event-day spread widening and imperfect hedges. The day of maximum predicted flow need not offer the best net entry. Include all transaction, funding and short-borrow costs, as well as realistic fills around the effective time. Compare against simply delaying an otherwise necessary portfolio rebalance: execution savings and standalone alpha are distinct outcomes. Falsification / kill test. Kill a result based on today's index rules applied retrospectively, final membership before announcement, or future knowledge of affected passive holdings. Placebo-test unaffected bonds and adjacent non-event dates. Reject a pure reversal rule whose returns are explained by rating news or common month-end credit beta. Require an incremental net contribution after issuance, fund-flow and simple-reversal controls, not merely an attractive gross event-study chart. 08 / COMBINE ONLY INCREMENTAL INFORMATION

21.7 Agent implementation runbook: data discovery to safe production

Operating rule. The agent must advance through auditable gates rather than implement the entire alpha in one pass. If a required field is unavailable, ambiguous, delayed, or not historically point-in-time, stop that branch and record the limitation; never silently substitute a convenient proxy.

- A1. Data discovery.** Inventory candidate sources/tables/files; definitions; keys; units; time zones; economic, publication and receipt timestamps; corrections/cancels; historical coverage; entitlements; delayed-history rules; and whether each field was genuinely observable live. Output schema and coverage diagnostics only.
- A2. Raw profiling.** Report rows, bonds, issuers, days, missingness, duplicates, timestamp inversions, corrections, size distributions and join success by IG/HY, liquidity, sector and period. Do not impute yet.
- A3. Canonical identifiers.** Version the security master and map security IDs to legal obligor, parent, guarantee, seniority, currency, coupon, maturity, option schedule and corporate actions. Quarantine ambiguous/time-varying mappings.
- A4. As-of layer.** Require every query to accept a decision timestamp and return only records then available. Unit-test late records, corrections and cancels. Persist available-at and provenance fields.
- A5. Price layer.** Normalize clean/dirty price, accrued, spread convention, side, size and quote status. Create freshness, executable-size, dispersion and dealer-breadth metadata. Any forward fill must explicitly age and lose confidence.
- A6. Analytics layer.** Produce cash flows, yield/OAS as appropriate, spread duration, convexity, KRD, CS01/DV01, carry and roll from point-in-time terms; reconcile a sample against an independent implementation.
- A7. Targets.** Build 5d/10d efficient-price diagnostic and executable-return labels. Estimate residualisation only on prior data; cost every changed hedge leg, fees, financing and impact. Persist label maturity for purging.
- A8. Signal-specific prerequisites.** Execute the following in order, emitting exclusions, distributions and hand-audited examples after each sub-step:

- a) Discover index methodology, eligibility rules, rebalance calendars, constituent files, outstanding amounts and publication timestamps.
- b) Reconstruct membership point-in-time from rules; never use final constituent lists before their announcement/effective availability.
- c) Estimate expected passive notional from index weights and tracking assets under explicit uncertainty; separate additions, deletions and weight changes.
- d) Measure abnormal pre/post-event price pressure after rates/credit controls and estimate completion from event stage, not future realized flow.
- e) Use fresh executable prices and include crowding/front-running, borrow and unwind costs; avoid assuming trade at official close.
- f) Run placebo rebalance dates, alternative AUM assumptions, surprise versus anticipated events and crowding splits; require decay after completion.

- A9. Literal raw feature.** Translate the documented equation exactly. Assert units, sign, finite values, peer counts, timestamp ordering and boundaries. Hand-reconcile at least 20 events.
- A10. Causal transforms.** Add rolling median/MAD z-score, changes, persistence, acceleration and percentile with past-only windows. Fit curves/betas/normalizers only inside training windows and version them.
- A11. State variables.** Join price confidence, information/toxicity, pressure completion, liquidity regime, volatility, event age and requested size as separate columns before tuning interactions.
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- A14. Transparent baseline.** Fit horizon-specific linear/robust models in walk-forward folds before nonlinear models; record sign, stability, sample size and incremental information.
- A15. State/convergence models.** Estimate information versus temporary-pressure probabilities and convergence speed where meaningful. Cross-fit state probabilities used as features.
- A16. Execution simulator.** Request a new executable entry after the signal timestamp; apply latency, size, withdrawal, partial-fill and legging rules. No executable observation means unfilled, not a midpoint fill.
- A17. Fill/cost models.** Estimate fill probability and conditional spread/impact/fees/financing by size, protocol and state. Calibrate on held-out fills and cost every changed leg.
- A18. Net outcomes.** Store expected executable return conditional on fill, conservative lower quantile and capacity. Keep gross, mark, executable and fully-net outcomes as distinct audit columns.
- A19. Adversarial validation.** Use chronological purged folds, issuer holdouts, liquidity/rating/regime splits, event exclusions, alternative residualisations, placebo dates and within-family ablations.
- A20. Shadow mode.** Freeze versions and generate live predictions without trading. Log source timestamps, features, quotes, hypothetical orders, predicted fills/costs and 1d/5d/10d mark-outs.
- A21. Canary.** Start with strict size, concentration, liquidity and turnover limits. Kill automatically on timestamp anomalies, data gaps, feature drift, cost underprediction, unexplained residual beta or repeated adverse selection.
- A22. Promotion/rollback.** Promote only a fully reproducible version. Default to no-trade when health checks fail and retain one-command rollback to the prior approved model.

Incremental safe prompts. Use these as separate prompts/commits; each prompt may inspect prior artifacts but must not expand scope silently.

- P1.** “Inventory only the required data, schemas, keys, timestamps, live-history availability, entitlements, coverage and unresolved questions. Do not build a feature.”
- P2.** “Build and unit-test canonical identifiers and as-of joins. Return diagnostics and quarantined ambiguities. Do not calculate alpha.”
- P3.** “Build the price/analytics layer with freshness/confidence and reconcile a hand-audited sample. Stop on unexplained discrepancies.”
- P4.** “Implement only the literal raw signal with assertions and 20 hand-reconciled examples. Do not optimize parameters.”
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- P6.** “Fit the transparent walk-forward baseline for 5d/10d. Report coefficients, uncertainty, ablations and failures; do not tune on the final test period.”
- P7.** “Add execution, fill and transaction-cost simulation. Recompute conditional executable net outcomes and mark unavailable opportunities unfilled.”
- P8.** “Run adversarial validation and within-family ablations. Produce a go/no-go research recommendation based only on predeclared gates.”
- P9.** “Run shadow mode with frozen versions. Do not send orders. Compare predicted versus observable fills, costs and mark-outs.”
- P10.** “Only after explicit human approval, create a tightly limited canary configuration with hard risk limits, health checks and rollback; do not broaden limits automatically.”

21.8 Production row and acceptance gate

Persist raw score, z-score, event age, quote age, peer count/effective sample size, confidence, state probabilities, size/ADV, expected spread/impact, issuer/sector/rating/tenor, KRD/CS01 and source timestamps. The feature graduates only if it adds incremental out-of-sample information after other members of its economic family, survives alternative residualisations and retains a positive conservative executable-net-return statistic at realistic capacity.

22 Combining signals with XGBoost or another nonlinear model

The twenty features are correlated views of five latent engines: relative-value dislocation A , information diffusion B , temporary liquidity pressure C , slow institutional mechanics D , and state/toxicity E . Preserve raw features and add cross-fitted engine summaries:

$$X = [x_{1:20}, A, B, C, D, E, c, p^{info}, p^{complete}, liquidity, size, protocol, risk].$$

Fit separate models

$$\hat{p}^{fill} = f_{fill}(X), \quad \hat{p}^{info} = f_{info}(X), \quad \hat{C} = f_{cost}(X, size, protocol), \quad \hat{\mu}_H = f_{ret,H}(X \mid fill).$$

For XGBoost use shallow trees, row/column subsampling, minimum-child constraints, L_1/L_2 regularisation and early stopping. Tune inside rolling training/validation windows only. Prevent issuer memorisation with held-out-issuer validation and economically meaningful identifiers. Keep elastic net/GAM benchmarks; consider LightGBM/CatBoost only as robustness alternatives.

22.1 State gating

The crucial interaction is pressure versus information:

$$\alpha^{REV} = -Dislocation \times Pressure \times (1 - p^{info}) \times c, \\ \alpha^{CONT} = Dislocation \times InformationConfirmation \times p^{info} \times c.$$

The model must be able to abstain. Do not let the same observed sale mechanically contribute to both continuation and reversal.

22.2 Distributional decision rule

Fit a conditional lower quantile $\hat{q}_{\alpha,H}$ of executable net return. Trade only if

$$\hat{q}_{\alpha,H}(y^{exec} \mid fill, X) > r_{min}, \quad p^{fill} > p_{min}, \quad OperationalChecks = 1.$$

Choose α, r_{min}, p_{min} without touching the final test set. Calibrate by liquidity bucket and protocol. Evaluate probability calibration for fill/state models and empirical coverage for return quantiles.

22.3 Walk-forward validation and portfolio layer

Purge overlapping 5d/10d labels and embargo clustered events where necessary. Re-estimate peer curves, graphs, scalars, feature selection and hyperparameters inside each fold. Repeat with held-out issuers and calendar regimes. Translate forecasts into risk with

$$\max_w \hat{\mu}^\top w - \lambda w^\top \hat{\Sigma} w - \eta TC(w - w_{old})$$

subject to issuer, sector, rating, duration, CS01, liquidity, turnover, locate and concentration limits. Rebalance only when expected improvement exceeds the cost of changing the existing book.

23 Extending each signal into a family

A scalar feature should become a causal family

$$\mathcal{F}(x) = \{x, \Delta_1 x, \Delta_3 x, \textit{persistence}, \textit{acceleration}, \textit{rank}, \textit{age}, x \times c, x \times p^{\textit{info}}, x \times p^{\textit{complete}}\}.$$

For relative-value signals create IG/HY, sector, seniority, currency, tenor, issue-size, liquidity and callable/non-callable variants with hierarchical shrinkage. For information diffusion vary leader source, innovation window, confirmation breadth and event class. For flow/pressure signals vary side, size bucket, parent-order persistence, completion definition and execution protocol. For institutional mechanics vary event stage, expected mechanical flow, crowding and time-to-completion.

Family expansion must not become feature mining. Pre-register transformations, cap effective degrees of freedom, cluster highly correlated variants, and require incremental value at the family level before inspecting individual members. A family is useful when it generalises a mechanism across bonds; it is dangerous when it merely creates hundreds of correlated thresholds around the same historical episodes.

23.1 Hierarchical representation

Let $x_{i,t,k}$ be variant k of a family. A partial-pooling benchmark is

$$y_{i,t,H} = \alpha_H + \sum_k (\bar{\beta}_{H,f} + u_{H,k}) x_{i,t,k} + \epsilon, \quad u_{H,k} \sim N(0, \tau_f^2).$$

In trees, approximate this with shared base features plus categorical bucket interactions and strong regularisation. Family summaries should be cross-fitted so the downstream combiner never sees an in-sample fitted value.

24 How the combined model should be fitted

Use nested walk-forward estimation. First define an untouched final test period. Within the remaining history, create rolling outer folds. In each outer training sample: rebuild the point-in-time universe; fit efficient-price and residualisation models; construct peers and leader graphs; fit state/fill/cost models; generate cross-fitted engine summaries; tune the final return/quantile model on an inner validation window; freeze all parameters; then score the outer validation window. Concatenate only out-of-fold predictions to choose the trading gate and portfolio hyperparameters.

The final model is refit once on all pre-test data using frozen hyperparameter rules and evaluated once on the untouched test. Report results by 5d/10d horizon, IG/HY, liquidity, size, protocol, issuer holdout, event/non-event state and capacity. The primary metric is conservative executable net return conditional on fill with turnover/capacity; secondary metrics include rank IC, calibration, hit rate, drawdown and residual exposures. A production candidate must remain useful after realistic latency, costs, failed RFQs and quote withdrawals.

25 Selected references

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